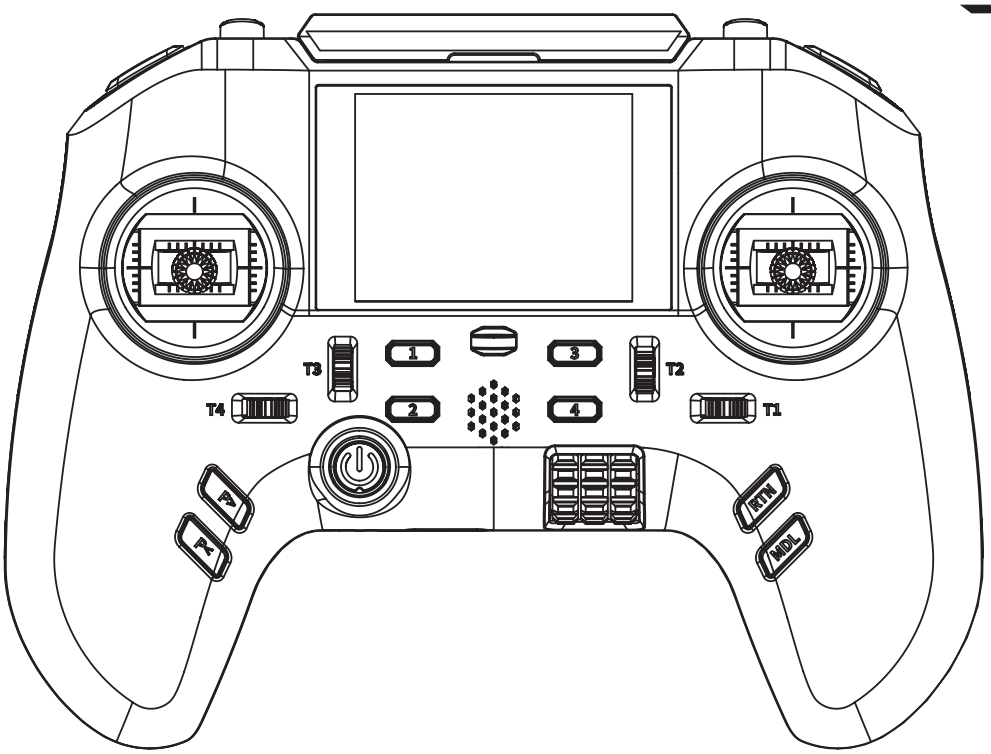
Scan for info.

ProArt PA01

User Manual

2.4GHz AFHDS 3



FLYSKY

Touching Infinity

Copyright ©2025-2026 Flysky Technology Co., Ltd.

15+

WARNING:

This product is only for 15 years old or above.

微信公众号 Bilibili Website Facebook

Thank you for purchasing our products.

Read the manual carefully to ensure your personal safety as well as the safety of your equipment.

If you encounter any problems during using, please refer to this manual first. If the problem is still not resolved, please contact the local dealer directly or contact the customer service staff via the website below:

http://www.flysky-cn.com

Table of Contents

1.Safety....5
1.1 Safety Symbols....5
1.2 Safety Guide....5
2.Battery Safety Instructions....6
3. Product Description....7
3.1 System Features....7
3.2 Transmitter Overview....8
3.2.1 Controls/Interfaces Introduction....11
3.2.2 LED Indicators for Battery and Charging....11
3.3 FS-DZ02 Base (Optional Accessory)....12
3.3.1 Using the FS-DZ02 Base....14
3.3.2 How to Use the FS-DZ02 Base....14
3.4 TMr Receiver Overview....15
3.4.1 **Receiver LED**....**15**
3.4.2 **Antenna**....**15**
3.5 PR01 Receiver Overview....16
3.5.1 **Receiver LED**....**16**
3.5.2 **Antenna**....**16**
4.Before Use....17
4.1 Transmitter Battery Installation & Removal....17
4.1.1 Battery Installation....17
4.1.2 Battery Removal....18
4.2 Receiver Installation....18
4.3 RF Module Installation....19
4.4 Gimbal Assembly Adjustment Instructions....20
4.5 Stick Height Adjustment....21
4.6 Antenna Adjustment....21
5.**Operation Guide**....**22**
5.1 Basic Setting Guidelines....22
5.2 Language Selection....22
5.3 Power On....24
5.4 Binding....25
5.5 Power Off....28
5.6 Upgrading Transmitter Firmware....28
6. System Interface....29
7. Function Settings....30
7.1 Manage Models....31
7.1.1 New Model....31
7.1.2 Create New Label....32
7.2 Channel Monitor....33
7.3 Model Settings....34
7.3.1 Model Name....34
7.3.2 Labels....34
7.3.3 Model Image....34
7.3.4 Internal RF....34
7.3.5 External RF....39
7.3.6 Trainer....40
7.3.7 Timer1....41
7.3.8 Timer2....42
7.3.9 Timer3....42
7.3.10 Pre-start Checks....43
7.3.11 Trims....44
7.3.12 Throttle....45
7.3.13 Enabled Features....45

7.3.14 USB Joystick 46
7.3.15 Customizable Switches 48
7.3.16 Other 49
7.3.17 HELI Setup 49
7.3.18 Flight Modes 50
7.3.19 Inputs 51
7.3.20 Mixes 53
7.3.21 Outputs 56
7.3.22 Curves 57
7.3.23 Global Variables 58
7.3.24 Logical Switches 59
7.3.25 Special Functions 60
7.3.26 Telemetry 62
7.4 Radio Settings 65
7.4.1 Tools 65
7.4.2 SD Card 65
7.4.3 Radio Setup 66
7.4.4 Themes 71
7.4.5 Global Functions 71
7.4.6 Trainer 71
7.4.7 Hardware 72
7.4.8 Version 75
7.5 Screen Settings 76
7.6 Reset Telemetry 77
7.7 Statistics 77
7.8 About EdgeTX 77
8. Product Specifications 78
8.1 Transmitter Specifications 78
9. Package Contents 79
10. Certifications 80
10.1 DoC 80
10.2 CE Warning 80
10.3 FCC Statement 80
10.4 Environmentally friendly disposal 81

1.Safety

1.1 Safety Symbols

Pay close attention to the following symbols and their meanings. Failure to follow these warnings could cause damage, injury or death.

Danger	• **Not following these instructions may lead to serious injuries or death.**
Warning	• **Not following these instructions may lead to major injuries.**
Attention	• **Not following these instructions may lead to minor injuries.**

1.2 Safety Guide

Prohibited **Mandatory**

- **Do not use the product at night or in bad weather like rain or thunderstorm. It can cause erratic operation or loss of control.**
- **Do not use the product when visibility is limited.**
- **Do not use the product on rain or snow days. Any exposure to moisture (water or snow) may cause erratic operation or loss of control.**
- **Interference may cause loss of control. To ensure the safety of you and others, do not operate in the following places:**
  - Near any site where other radio control activity may occur
  - Near power lines or communication broadcasting antennas
  - Near people or roads
  - On any body of water when passenger boats are present
- **Do not use this product when you are tired, uncomfortable, or under the influence of alcohol or drugs. Doing so may cause serious injury to yourself or others.**
- **The 2.4GHz radio band is limited to line of sight. Always keep your model in sight as a large object can block the RF signal and lead to loss of control.**
- **Do not touch any part of the model that may generate heat during operation, or immediately after use. The engine, motor or speed control, may be very hot and can cause serious burns.**

- **Misuse of this product may lead to serious injury or death. To ensure the safety of you and your equipment, read this manual and follow the instructions.**
- **Make sure the product is properly installed in your model. Failure to do so may result in serious injury.**
- **Make sure to disconnect the receiver battery before turning off the transmitter. Failure to do so may lead to unintended operation and cause an accident.**
- **Ensure that all motors operate in the correct direction. If not, adjust the direction first.**
- **Make sure the model stays within the systems maximum range to prevent loss of control.**
- The PA01 system uses EdgeTX open source firmware. Official updates will be posted on our official website from time to time. If you modify or use unofficial software, the system may encounter out-of-control or error. Our company will not accept legal responsibility caused by unofficial software!

2.Battery Safety Instructions



Do not use batteries that are damaged, aged or have abnormalities such as leaks.

Do not short circuit the battery.

Do not repair, modify, dissaemble, crush or punture the battery.

Do not allow contact with water.

Do not charge or store the battery in high temperatures or in direct sun light.

Store the battery in a cool, dry environment.

Do not charge under the following conditions:

- Covered or without ventalation.
- In extremely cold or hot conditions.

Do not use around flammable gases or liquids.

Do not bend, stretch or pull on the power cord.

- The power cord may be damaged by fire, heat, electric shock etc.



The battery should be stored in a location that is not accessible to infants and toddlers.

Be sure to charge the battery before flying.

- If a battery leak accidentally comes in contact with skin, clothing, etc., wash it immediately with water. If it comes in contact with eyes wash immedietly. Seek medical attention if necessary.

When not in use for a long time, remove the battery from the transmitter and store it in a dry environment.

Do not throw used batteries into the garbage can, etc., and dispose of them according to the treatment methods specified in each area.

Use only the battery specified by the manufacturer.

Take care to prevent the battery from falling, bumping or vibrating.

If the battery is damaged, stop using it immediately.

3. Product Description

The PA01 transmitter is equipped with a 2.4-inch, full dot-matrix, color, non-touch IPS display. It features an RF system with built-in support for both AFHDS 3 and ELRS, as well as an external RF option. When the built-in RF is set to AFHDS 3, it supports 18-channel output; when it is set to ELRS, it supports 16-channel output. The external RF module, the FRM303, supports expansion up to 32 channels (with a default maximum output of 18 channels). It operates on the EdgeTX OS and forms an FPV-optimized system when bound to TMr receivers.

3.1 System Features

AFHDS 3

AFHDS 3 (third-generation automatic frequency hopping digital system) is a newly developed digital wireless system with independent intellectual property rights by FLYSKY. It is compatible with one-way and two-way real-time data packet transmission and transparent data stream transmission.In other words, this system has advantages of both AFHDS 2A and WS2A wireless system. It equips with a brand-new 2.4G chip, stable and reliable connection, good real-time performance, and supports various configurations. Bring you the optimal configuration for multi-scenario application performance.

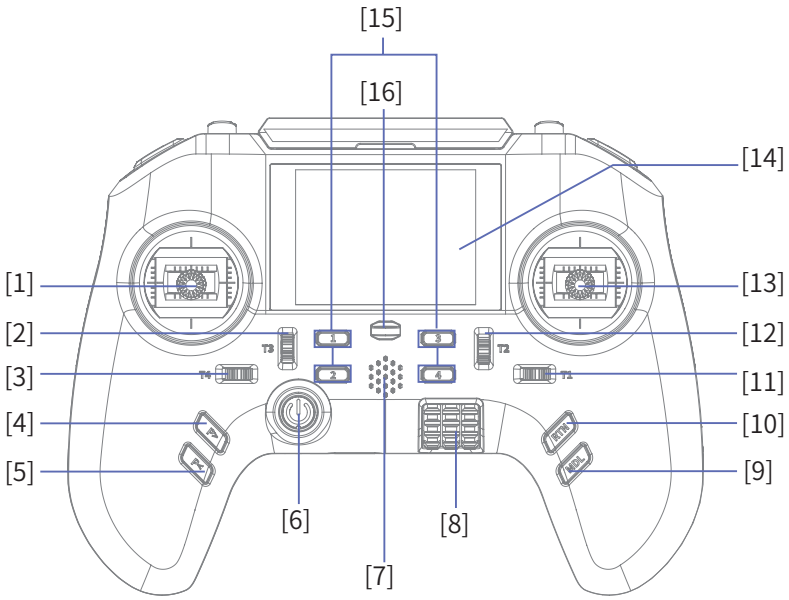
Compatible with One-way/two-way Real-time Data Transmission	The system supports one-way and two-way connections. When the transmitter is working in one-way transmission way, the receiver can receive data from the transmitter. When the transmitter is working in two-way transmission way. The receiver can receive data from the transmitter and the transmitter can also receive data from the receiver, as well as the information cross from the temperature and speed sensor modules.
Data Transparent Transmission	The independent data transparent transmission module is built into RF system, which can realize data transmission via transmitter and transparent transmission. It can be used for data transmission of flight control.
Intelligent RF Configuration	To set the parameters which affect RF transmission distance, speed and anti-interference, such as numbers of channels, resolution, bandwidth and receiver sensitivity. The system can be set according to different application requirements to obtain the most suitable performance.
Multi-channel Frequency Hopping	This systems bandwidth ranges from 2.402GHz to 2.480GHz. Set intelligent RF configurations accordding to your required, it can avoid or reduce the interference from other transmitters with the same frequency via different configurations, different time of powering on the transmitter, various patterns to the hopping frequency and various using frequency spots.
Unique ID Recognition System	Each transmitter and receiver has it's own unique ID. Once the transmitter and receiver have been bound, they will save the each other's ID and only connect with each other. When the syetem is working,if the IDs are matched with each other, then the connection will be connected, otherwise, there is no connection between transmitter and receiver. This unique ID recognition syetem resists the interference so as to make the system stabler and more reliable.
Low Power Consumption	It is built using highly sensitive, low power consumption components. And it works in the way of interval data transmission to improve transmitting efficient effectively and extend the working time of the battery distinctly, while it consumes as little as one tenth the power of a standard FM system.

EdgeTX

EdgeTX is an open source operating system for transmitter. EdgeTX is very powerful. Users can configure the transmitter according to their own needs, making transmitter more comprehensive and higher performance.

3.2 Transmitter Overview

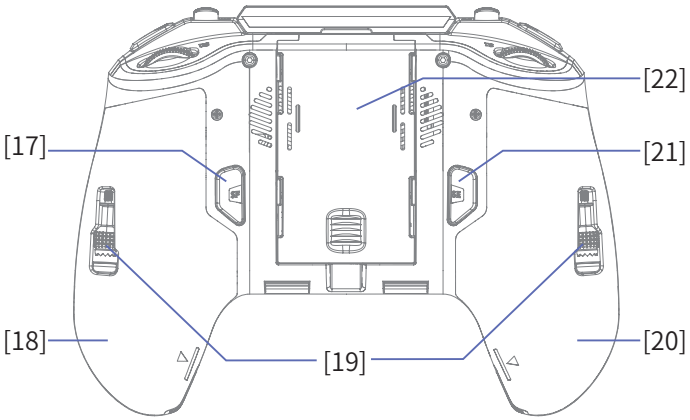
Front View



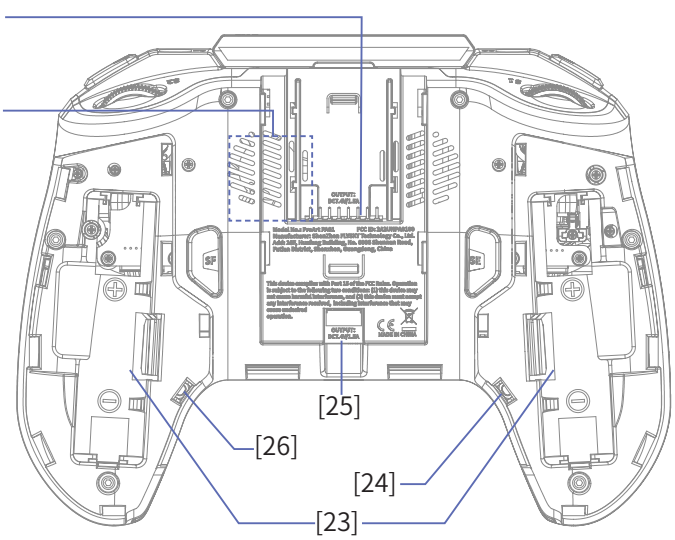
[1] Left Stick
[2] T3 (Trim Button)
[3] T4 (Trim Button)
[4] P> (Right Page Turn)
[5] P< (Left Page Turn)
[6] Power Switch(Built-in LED)
[7] Speaker
[8] Scroll Wheel
[9] MDL (Model Settings)
[10] RTN (Return Button)
[11] T1 (Trim Button)
[12] T2 (Trim Button)
[13] Right Stick
[14] Screen
[15] Customizable Illuminated Buttons (SW1~SW4)
[16] Neck Strap Hook

微信公众号 Bilibili Website Facebook

Back View



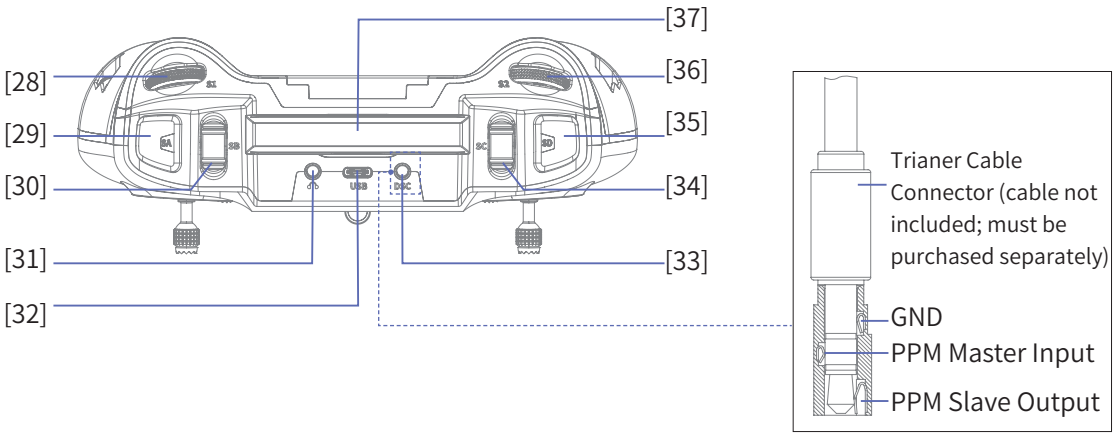
[17] SF Button
[18] Right Grip
[19] Gimbal Stick Holder
[20] Left Grip
[21] SE Button
[22] RF Module Installation Cover

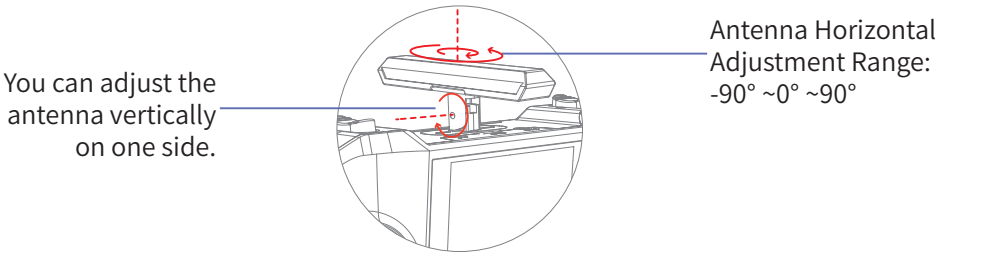


Note: The fan control is exclusively available when the ELRS protocol is selected. You can enable it and set a power activation threshold (TX Power ≥ Fan Thresh) in the TX power. The fan speed will vary automatically based on the current transmission power. When using the AFHDS 3 protocol, the fan does not activate due to its low power consumption.

[23] Battery Compartment
[24] Reset Button
[25] FRM303 Module Connector
[26] Transimitter MCU-DFU Button
[27] Nano Module Connector

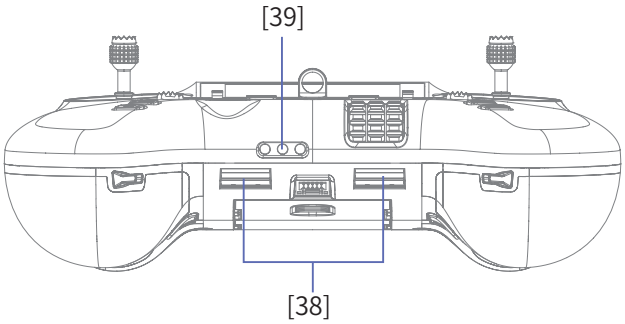
Top View





[28] S1 (Non-returning middle dial)
[29] SA (self-locking button)
[30] SB (Three-position Switch)
[31] 3.5mm Audio Jack
[32] USB Type-C Port
[33] DSC 3.5mm Trainer Connector
[34] SC (Three-position Switch)
[35] SD Button
[36] S2 (Non-Self-Centering Dial)
[37] Antenna

Bottom View



[38] Magnet
[39] Base Contact Points

微信公众号 Bilibili Website Facebook

3.2.1 Controls/Interfaces Introduction

Scroll Wheel

- On the Main interface, short press the scroll wheel to enter the function menu;
- When not on the Main interface, rotate the scroll wheel to navigate options, short press to enter the selected function.

RTN (Return Button)

When not on the Main interface, short press the RTN button to return to the previous interface or exit edit mode.

MDL (Model Settings)

Short press the MDL button in any interface to enter the Model Settings menu.

P>/P< (Right/(Left Page Turn)

On the function menu, short press P> (or P<) to turn pages right (or left).

Reset Button and Transimitter MCU-DFU Button

Before using these two buttons, you need to first remove the left and right grips. You can use a slender tool to press these two buttons.

- Reset Button: When the power button fails to turn off the transmitter, this button is needed to reset the transmitter.
- Transimitter MCU-DFU Button: Used for updating the transmitter's firmware.

USB Type-C Port

The device supports USB charging, firmware updates, and multi-mode USB communication (including HID joystick, mass storage, and serial port functionality).

DSC 3.5mm Trainer Connector

For connecting a trainer cable.

3.5mm Audio Jack

You can connect headphones to listen to the audio.

3.2.2 LED Indicators for Battery and Charging

When the transmitter is powered off, the power switch light, grip light bar, and SW1~SW4 button lights can indicate the battery level and charging status.

When not connected to a charger: Pressing the power switch will activate the indicator lights to display the battery level. The lights will turn off after 6 seconds. The indications are as follows:

Voltage Range	Grip Light Bar&Power Switch Light	SW1	SW2	SW3	SW4	Color
< 6.5V	Solid ON	OFF	OFF	OFF	OFF	Red
6.5V - 7.2V	Solid ON	Solid ON	OFF	OFF	OFF	Purple
7.2V - 7.6V	Solid ON	Solid ON	Solid ON	OFF	OFF	Yellow
7.6V - 8.0V	Solid ON	Solid ON	Solid ON	Solid ON	OFF	Green
⩾ 8.0V	Solid ON	Solid ON	Solid ON	Solid ON	Solid ON	Green

While connected to a charger, the indicator activates for 6 seconds upon plugging in, unplugging, or pressing the power switch to display the battery level and charging status. The indications are as follows:

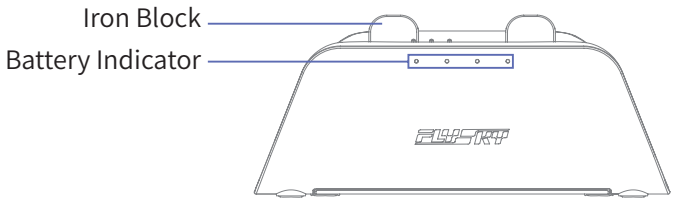
Voltage Range	Grip Light Bar&power Switch Light	SW1	SW2	SW3	SW4	Color
< 6.5V	Breathing	Breathing	OFF	OFF	OFF	Red
6.5V - 7.2V	Breathing	Breathing	OFF	OFF	OFF	Purple
7.2V - 7.6V	Breathing	Solid ON	Breathing	OFF	OFF	Yellow
7.6V - 8.0V	Breathing	Solid ON	Solid ON	Breathing	OFF	Green
8.0V - 8.23V	Breathing	Solid ON	Solid ON	Solid ON	Breathing	Green
⩾ 8.23V	Breathing	Solid ON	Solid ON	Solid ON	Solid ON	Green

Note: Press and hold the power switch for 1.5 seconds to power on the transmitter. After power-on, the power switch light indicates the receiver connection status and alarms through different colors, whereas the SW1~SW4 button lights and grip light bar will illuminate according to the user's custom settings.

3.3 FS-DZ02 Base (Optional Accessory)

The FS-DZ02 base can not only charge the transmitter but also directly power it.

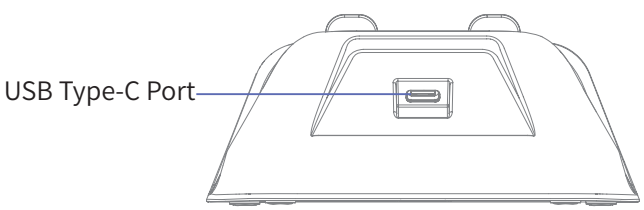
Front View:



Battery Indicator: This base is equipped with 4 LED indicators to show the battery power status and charging status inside the base.
Charging Status: When the four LED indicators are flashing, the device is charging; when they are solid on, the battery is fully charged.
Standby/Power Supply Status: The LED indicators display the current battery level, representing 25%, 50%, 75%, and 100% from left to right.

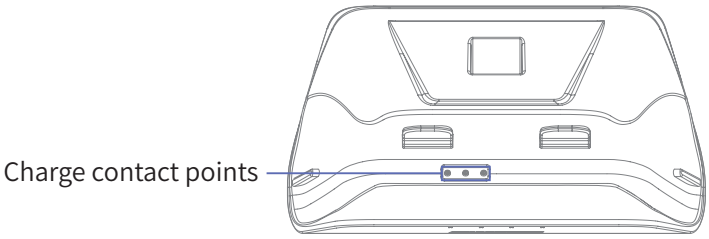
Note: The FS-DZ02 base features a voltage self-balancing function.When there is a significant voltage difference between the two installed batteries, the higher-voltage battery will discharge at 50mA until both voltages equalize. Therefore, always use a matched pair of 18650 batteries with the same voltage. Mixing batteries of different voltages may lead to a significant voltage drop after a short period of storage due to this balancing function.

Back View:

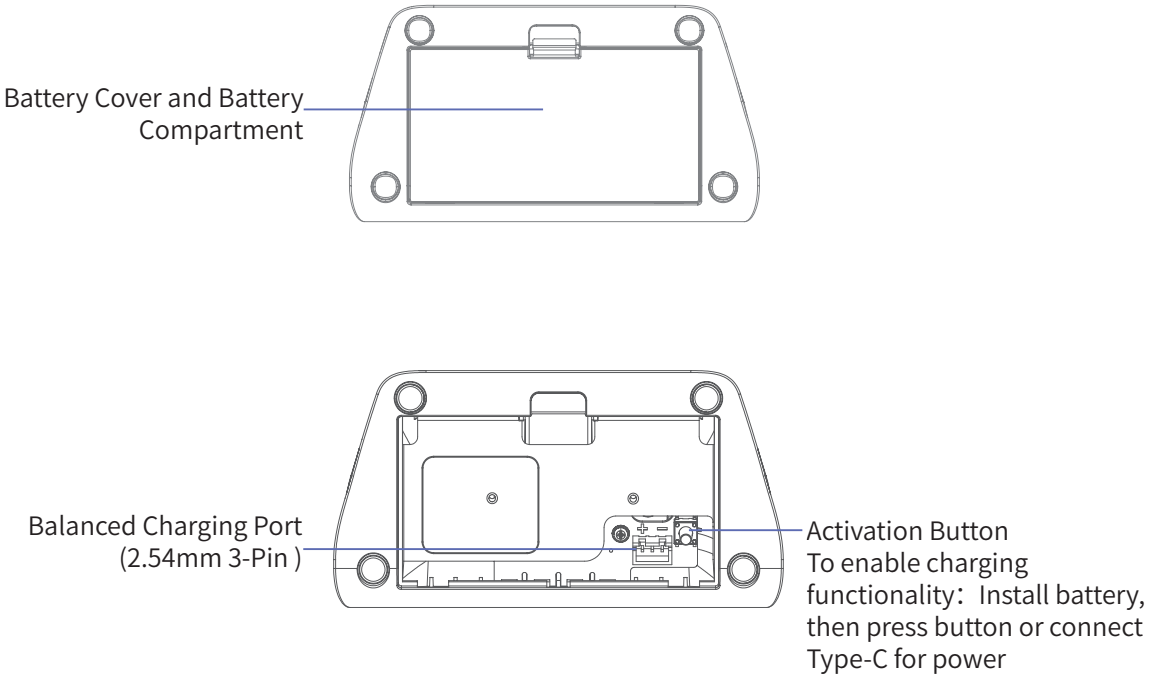


微信公众号 Bilibili Website Facebook

Top View:



Bottom View:



Warning: Only recharge 7.4V 2S LiPo batteries.

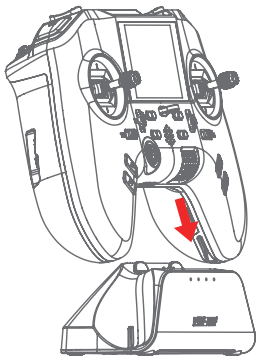
Type-C Port Input Voltage and Current Support:

- When the input voltage is 5V, the maximum allowed current is 3A.
- When the input voltage is 9V, the maximum allowed current is 2A.
- When the input voltage is 12V, the maximum allowed current is 1.5A.

Type-C Port Output Voltage and Current Support:

- When the output voltage is 5V, the maximum allowed current is 3A.
- When the output voltage is 9V, the maximum allowed current is 2A.
- When the output voltage is 12V, the maximum allowed current is 1.5A.

3.3.1 Using the FS-DZ02 Base



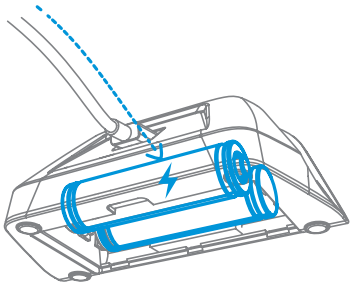
For proper operation during charging:

- The magnetic interface of the transmitter must be correctly aligned and attached to the corresponding iron block on the base.
- The electrical contact points on the transmitter must establish proper electrical contact with the charging terminals on the base.

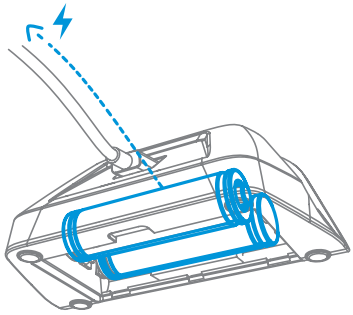
3.3.2 How to Use the FS-DZ02 Base

The FS-DZ02 charging base can be used in the following five modes:

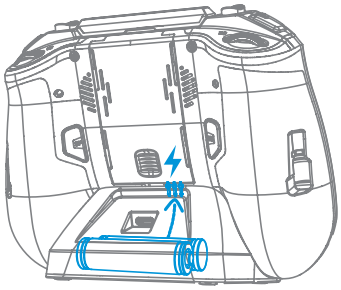
Mode 1: The batteries inside the base can be charged via the Type-C port.



Mode 2: Power or charge external devices via the Type-C port using its internal batteries.

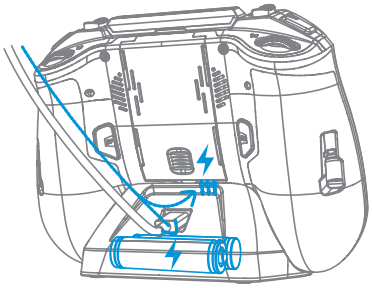


Mode 3: The internal batteries powers the transmitter (and charges it if rechargeable).



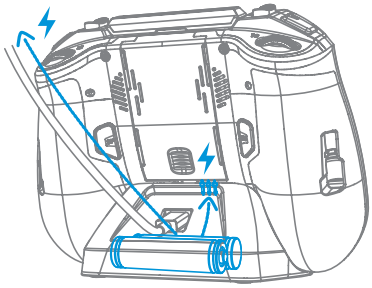
Mode 4: Through the Type-C port, you can:

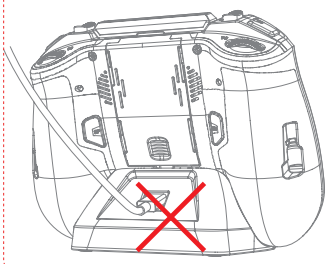
- Charge the internal batteries
- Power/charge the transmitter simultaneously



Mode 5: The internal batteries provides:

- Primary power to the transmitter
- Type-C power output for external devices
- Charging capability for rechargeable devices

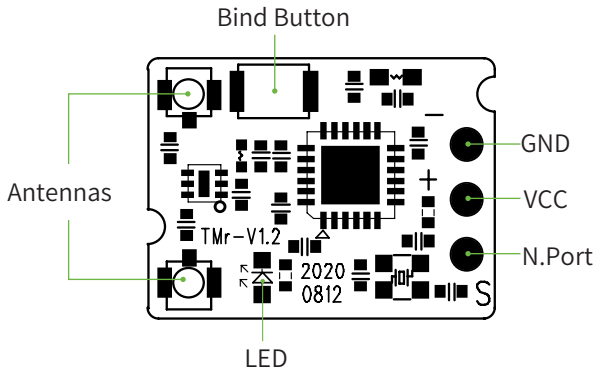




The charging base cannot charge unless a battery is connected.

微信公众号 Bilibili Website Facebook

3.4 TMr Receiver Overview



3.4.1 Receiver LED

The LED status indicates the power supply state of the receiver and its working state.

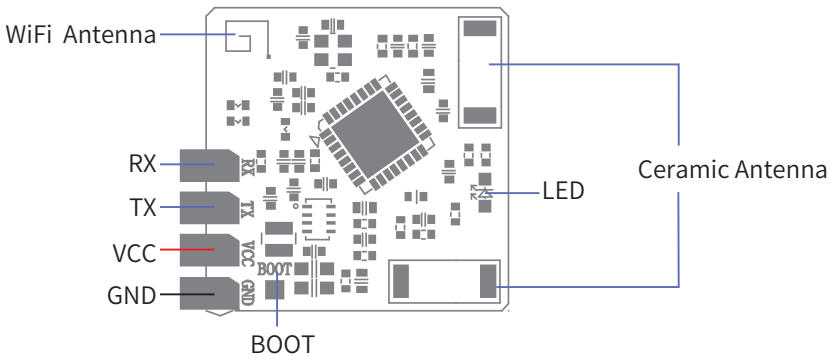
- Off: The receiver is not powered on.
- Solid ON: The receiver works normally.
- Fast Flashing: The receiver is in the binding mode.
- Slow Flashing: The transmitter bound is powered off, or it has been not bound with a transmitter, or the receiver does not receive any signal.
- Three-flash-one-off: The receiver is in the forced update mode.

3.4.2 Antenna

It features dual external antennas.

Warning	**Do not hold the antenna during use in order to prevent signal quality loss.**
Attention	**To ensure signal quality, the antenna should be perpendicular to the model body.**
Attention	**Do not pull the antenna of the receiver. Do not tie the antenna and the servo cable together.**

3.5 PR01 Receiver Overview



3.5.1 Receiver LED

The LED status indicates the power supply state of the receiver and its working state.

- Off: The receiver is not powered on.
- Solid ON: The receiver is connected to the transmitter.
- Two-flash-one-off: The receiver is in the binding mode.
- Three-flash-one-off: The receiver is connected to transmitter but has mismatched model-match configuration.
- Slow flashing: The receiver is attempting to establish connection with the transmitter.
- Medium-speed flashing: The receiver detects no RF chip.
- Fast flashing: The receiver enters WiFi mode.

3.5.2 Antenna

It features dual ceramic antennas.

⚠ **Warning**	**Do not hold the antenna during use in order to prevent signal quality loss.**

微信公众号 Bilibili Website Facebook

4.Before Use

Before operation, please follow the instructions in this chapter to install the battery and connect the device.

⚠ Danger	• Use only the battery specified by the manufacturer.
⚠ Danger	• Do not open, disassemble, or repair the battery yourself.
⚠ Danger	• Do not squeeze, puncture, or touch the metal terminals of the battery.
⚠ Danger	• Do not expose the battery to high temperatures or liquids.
⚠ Danger	• Take care to prevent the battery from dropping, bumping or vibrating.
⚠ Danger	• Store the battery in a dry, cool environment.
⚠ Danger	• If the battery is damaged, stop using it immediately.

4.1 Transmitter Battery Installation & Removal

4.1.1 Battery Installation

⚠ **Warning**	• Only the 18650 battery with dimensions of 18×65mm as illustrated must be used. Using an improperly sized battery may deform the battery compartment, which can lead to poor contact or other malfunctions.

Figure 1

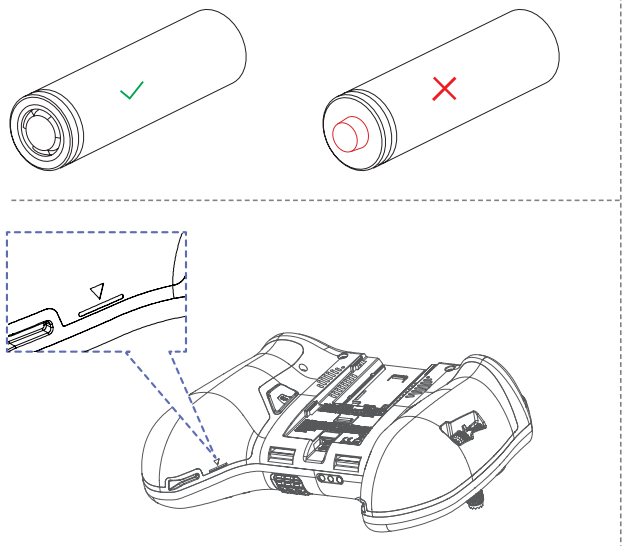
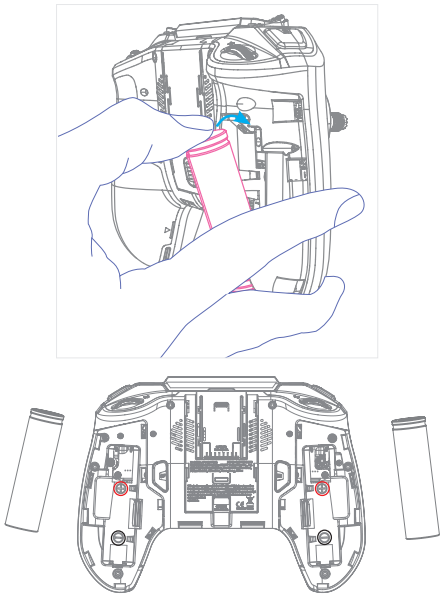


Figure 2



Follow the steps below to install the batteries:

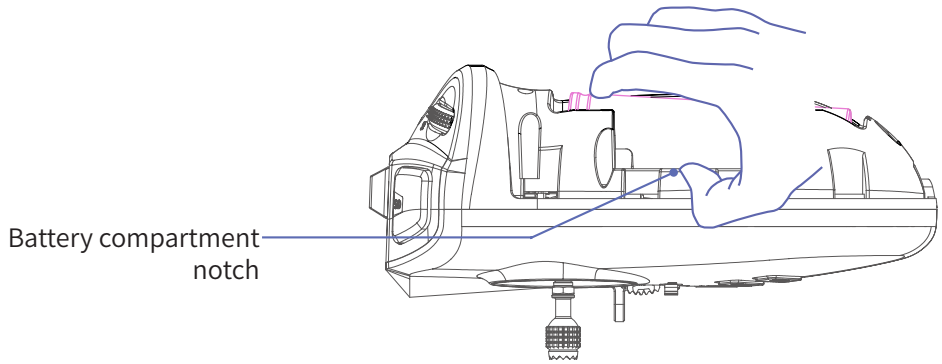
1. As shown in Figure 1, locate the icon ▽ , and then carefully remove the left and right hand grips of the transmitter starting from that position;
2. As shown in Figure 2, first, insert the end of the battery marked with '-' into the corresponding position in the battery compartment. Then press the battery in, ensuring it is fully seated, and make sure the '+' and '-' orientation matches the markings inside the compartment.

微信公众号 Bilibili Website Facebook

- To ensure the normal use of the transmitter, please use dual-battery power supply.

3. Refer to Step 2 to install the other 18650 battery.
4. Reinstall the left and right grips.

4.1.2 Battery Removal



Follow the steps below to remove the batteries:

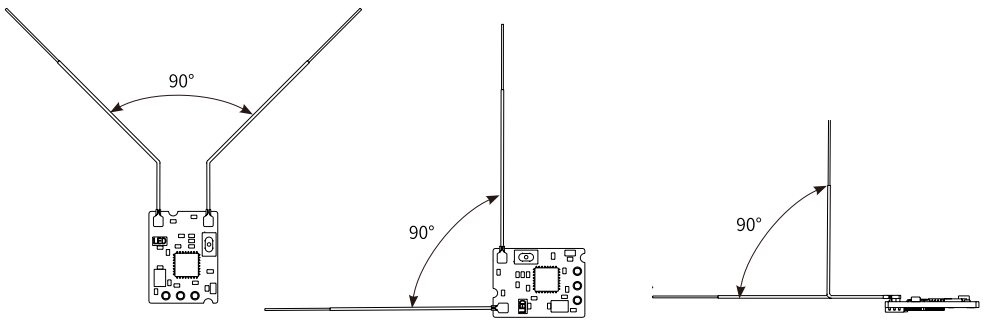
1. Remove left and right hand grips;
2. As shown in the diagram, locate the notch position (Battery compartment notch) on the side of the battery compartment;
3. Apply moderate pressure to disengage and remove the battery;
4. Follow steps 2-3 to remove the second battery.

4.2 Receiver Installation

Make sure that the receiver is mounted in an appropriate location within the model, to ensure a stable signal, maximum range and to mitigate external interference, follow these guidelines:

Pay attention to the following when installing the receiver:

1. Make sure the receiver is not installed near ESCs or other sources of electrical noise.
2. Keep the receiver's antennas away from conductive materials such as carbon or metal. To ensure normal function make sure there is a gap of at least 1cm between the antenna and the conductive material.
3. Make sure that the two antennas of the receiver are fixed at 90 degrees to each other (as shown below).



⚠ Attention	• Do not connect the receiver to the power supply during these steps to avoid damage.

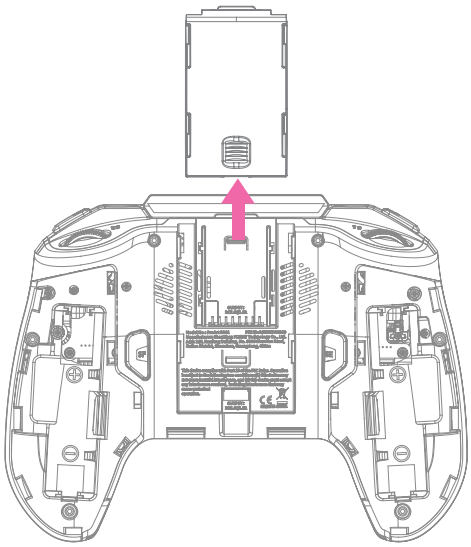
微信公众号 Bilibili Website Facebook

4.3 RF Module Installation

This transmitter comes with an FS-XC506 RF cable for use with the FRM303 RF module. Please select the appropriate RF connector for installation based on the actual RF module.

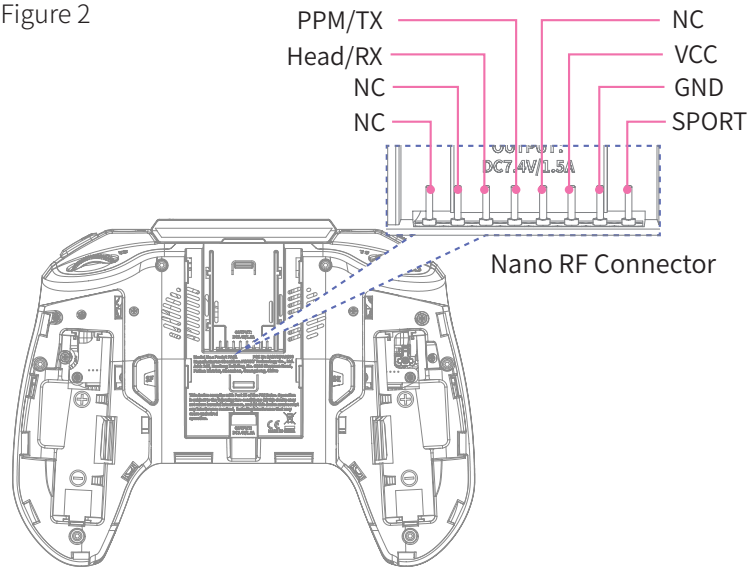
Here are the steps for installing the Nano RF module:

Figure 1



1. As shown in Figure 1, remove the RF module installation cover.

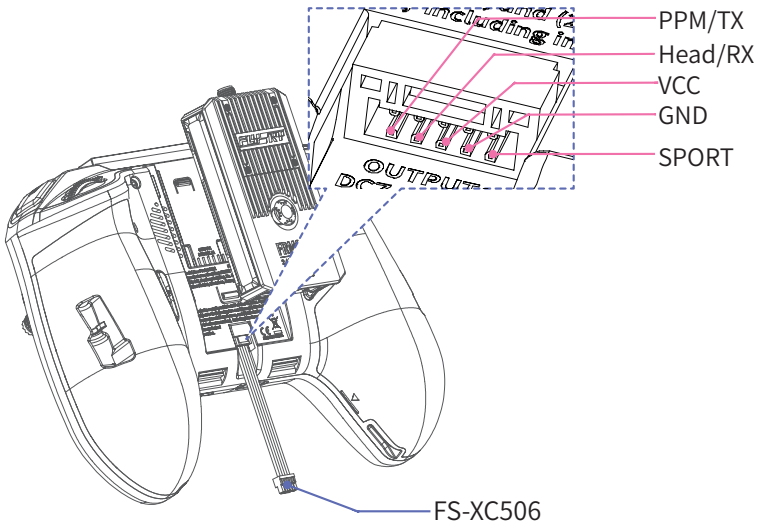
Figure 2



2. As shown in Figure 2, simply install the Nano RF module onto the transmitter's Nano RF connector.

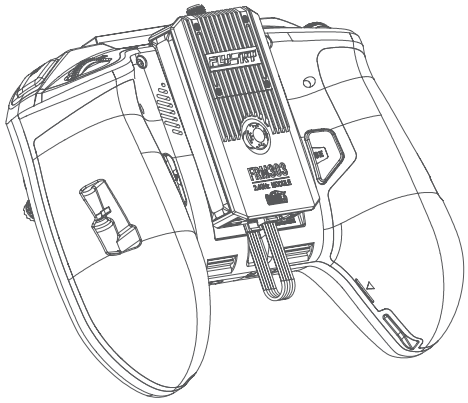
Here are the steps for installing the FRM303 RF module:

Figure 1



1. As shown in Figure 1, remove the RF module installation cover. Then, connect one end of the FS-XC506 RF cable to the FRM303 RF connector on the transmitter, and the other end to the FRM303 RF module itself.

Figure 2



2. As shown in Figure 2, mount the RF module securely onto the transmitter.

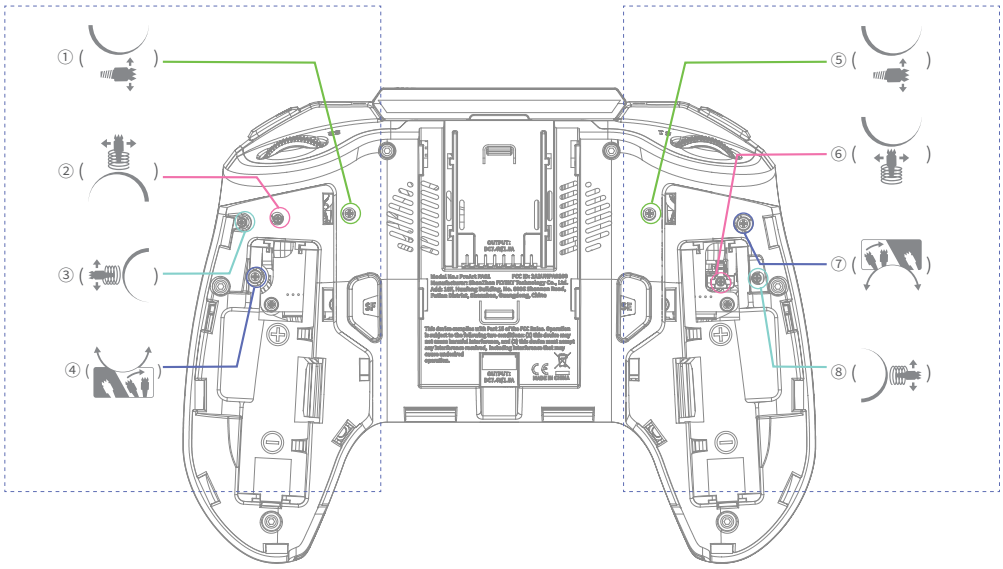
Note: For detailed information on the FRM303 RF module and the Nano RF module, please refer to the respective user manuals.

微信公众号 Bilibili Website Facebook

4.4 Gimbal Assembly Adjustment Instructions

For Right Gimbal Stick Adjustment:

For Left Gimbal Stick Adjustment:



As shown above, by adjusting the screws which are located in the screw holes in the back of the transmitter, the gimbal stick can be set to either self-centering or non-self-centering and the friction in case of non-self-centering, as well as changing stick tension in case of self-centering (Remove the grips to find the relevant screw holes and screws). Screw description is as following:

① . ⑤	Gimbal stick vertical friction adjustment	② . ⑥	Gimbal stick horizontal tension adjustment
③ . ⑧	Gimbal stick vertical tension adjustment	④ . ⑦	Gimbal stick self-centering/non-self-centering adjustment

Notes:

1. During the adjustment process, please be careful of the amount of force you apply. If you loosen the screws too much, they may fall out. If you tighten them too much, you may damage the spring.
2. When performing the following steps, you can adjust the screws while simultaneously moving the stick to achieve the desired adjustment.

Take right gimbal stick as example:

Non-Self-Centering to Self-Centering

1. Use the screwdriver to adjust the screw ④ (shown on previous diagram) counterclockwise until the gimbal stick changes to self-centering.
2. Adjust the screw ① counterclockwise to adjust the frictional force.
3. If you need to adjust the vertical centering force or horizontal centering force, adjust the corresponding screw ③ or ② accordingly. The force increases clockwise, and decreases counterclockwise.

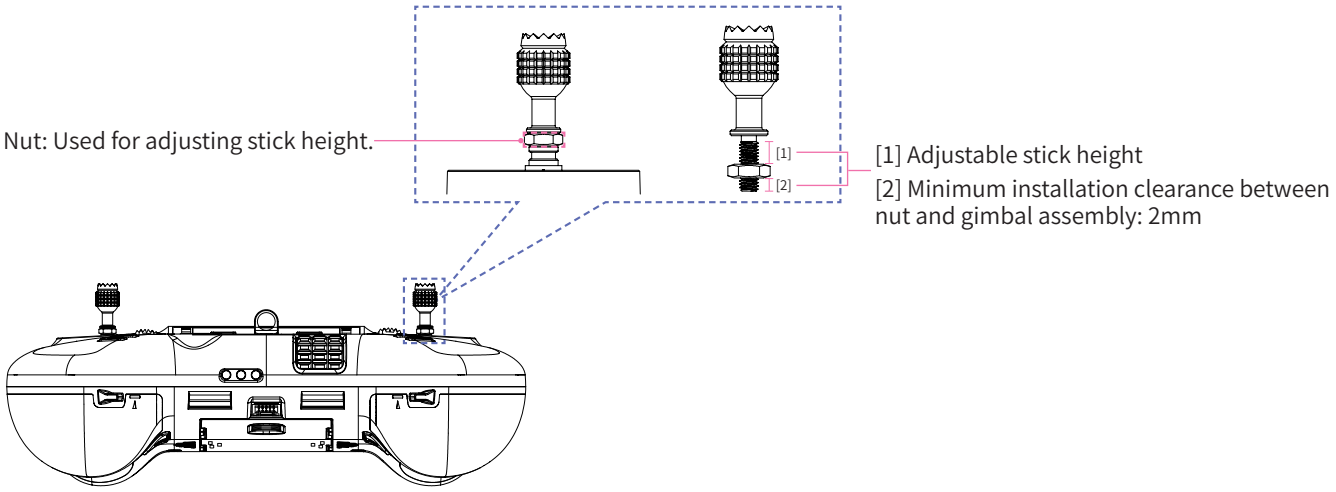
Self-Centering to Non-Self-Centering

1. Use the screwdriver to adjust the screw ④ clockwise until it is tightened so that the gimbal stick changes to non-self-centering.
2. Adjust the screw ① clockwise to strengthen the frictional force.
3. If you need to adjust the horizontal centering force, adjust the screw ② accordingly. The force increases clockwise, and decreases counterclockwise.

微信公众号 Bilibili Website Facebook

4.5 Stick Height Adjustment

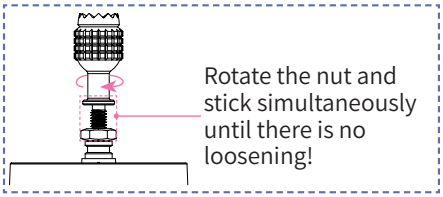
This transmitter features adjustable stick height, allowing users to customize the stick position according to their operating preferences.



Adjustment Procedure:

1. Rotate the joystick counterclockwise (do not fully unscrew it from the nut);
2. Adjust the stick height to your preference, then rotate the nut counterclockwise or clockwise as needed. Finally, rotate both the stick and nut clockwise simultaneously (in the direction of the arrow shown) until fully tightened.

Note: Apply moderate force during adjustment to avoid damaging the nut or stick.

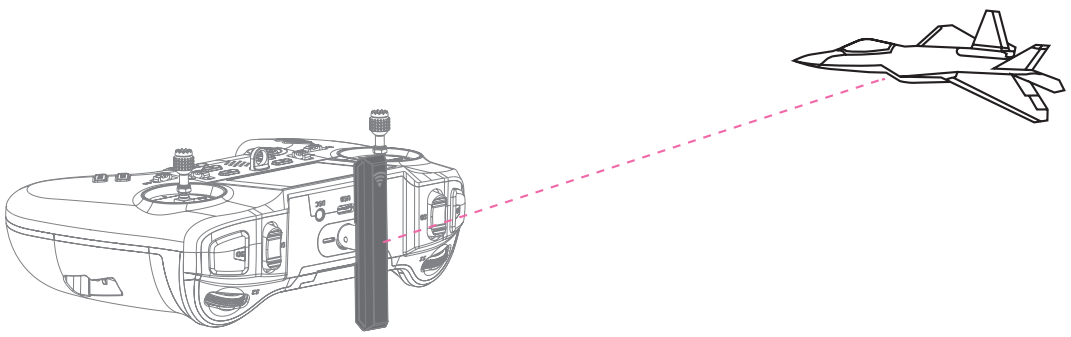


4.6 Antenna Adjustment

Pull out the remote control antenna and rotate it to the appropriate position, as different antenna positions will receive varying signal strengths.

Based on the relative position of the remote control and the model, adjust the direction of the remote control's external antenna and align the antenna indicator towards the model to ensure the best signal quality between the transmitter and the model.

The suggested antenna angles for operating a model with a transmitter are shown in the figure below:



5.Operation Guide

After the preparation is complete, you can follow the instructions in this chapter to get started.

5.1 Basic Setting Guidelines

If you use the open-source system for the first time, you can set up the transmitter according to the following basic setting guidelines:

Language Selection → Mode Selection → Stick Mode → Assign Channel → Stick Calibration → Bind → Protocol Setting → Failsafe.

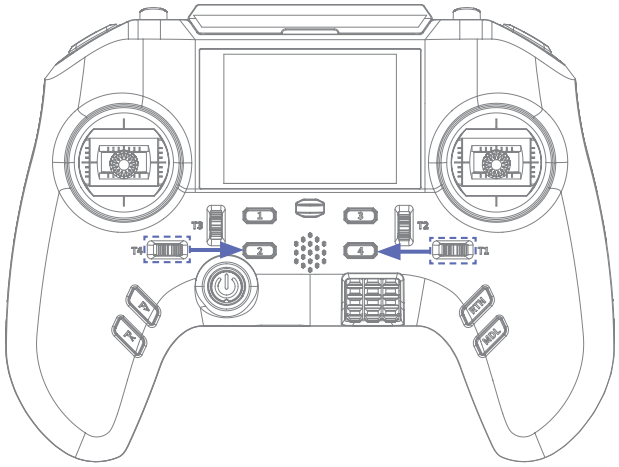
- Language Selection: Refer to "5.2 Language Selection".
- Model Selection: Refer to "7.1 Manage Models".
- Stick Mode: Select the appropriate stick mode according to your habit. Refer to "7.4.3 Radio Setup → Default Channel Order and Mode".
- Assign Channel: Set the channel-related functions. Refer to "7.3.19 Inputs and 7.3.20 Mixes".
- Stick Calibration: Refer to "7.4.7 Hardware → Inputs → Calibration".
- Bind: Bind the transmitter with the receiver. Refer to "5.4 Binding"
- Protocol Setting: Set the output signal type of the receiver interface. Refer to "7.3.4 Internal RF →Type → Module Option".
- Failsafe: Set the failsafe. Refer to "7.3.4 Internal RF → Failsafe Mode".

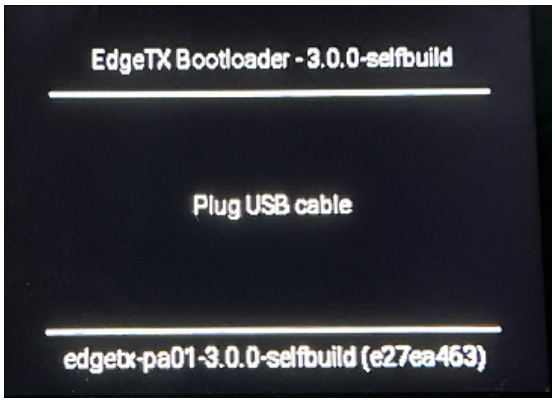
5.2 Language Selection

The transmitter's SD card file contains the default firmware in both Chinese and English versions. If you need to change the language, you must update the firmware corresponding to the desired language.

Operating Steps:

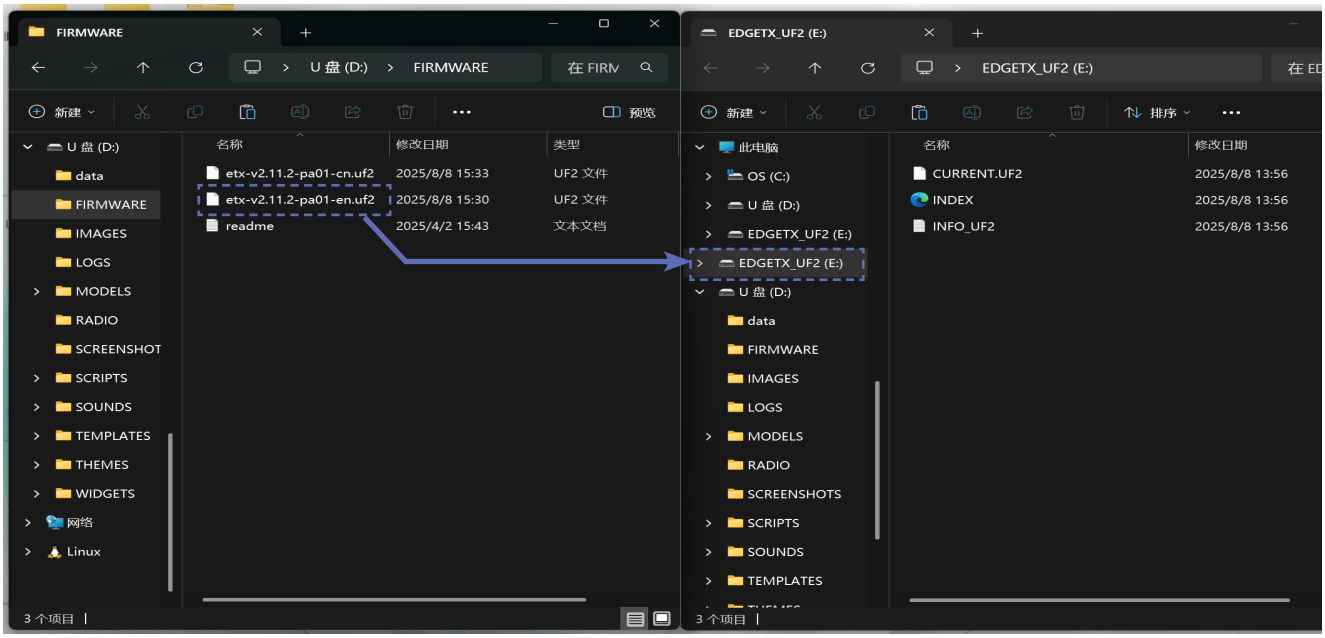
1. While powered off, simultaneously push T1 and T4 trim buttons in the arrow directions, short press the power button for 2 seconds, then release all three controls to enter firmware update mode when the screen lights up.



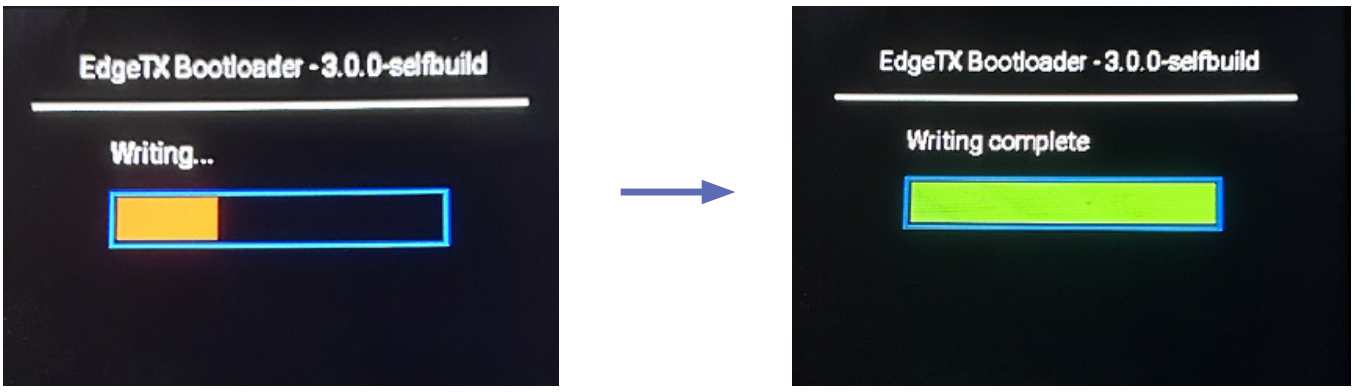


微信公众号 Bilibili Website Facebook

2. Connect the transmitter to your PC via USB Type-C cable .
3. On the PC, the U Disk and the EDGETX_UF2 Disk will automatically pop up.
   - Select [U Disk] > [FIRMWARE], and choose the English version firmware (etx-v2.11.2-pa01-en.uf2) or the Chinese version firmware (etx-v2.11.2-pa01-cn.uf2) according to your needs, then copy it to the EDGETX_UF2 Disk in the direction indicated by the arrow in the diagram.



   - The screen will display update progress.



4. After the update is completed, unplug the USB Type-C cable, and the transmitter will restart automatically.

5.3 Power On

Follow these steps to power on the device:

1. Press and hold the power switch for at least 1.5 seconds. Release it when the transmitter screen turns on.
2. Follow the on-screen prompts to complete the startup process successfully.
   - Check the throttle position and switch positions according to the throttle warning and switch position warning, and move them to the correct positions as prompted, or press any key to skip.
   - If the failsafe is not set for the current model, a failsafe warning will be displayed. You can simply press any key to skip it.

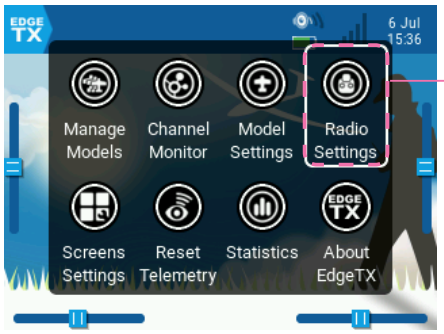
⚠ Warning	• **Do not use the USB Type-C cable to power on the RF module when the battery is not installed, to avoid loss of control due to unstable power supply!**
⚠ Warning	• **Operate with caution in order to avoid damage or injury.**
⚠ Warning	• **For your safety, before powering on, please set the transmitter switch to the highest position and move the throttle to the lowest or safety position.**
⚠ Danger	• **When in use make sure not to fly beyond the maximum range.**
⚠ Attention	• **External signal sources may affect signal quality.**
⚠ Danger	• **Before connecting, ensure all channels are in safe positions to prevent aircraft damage, unintended activation, or personal injury.**

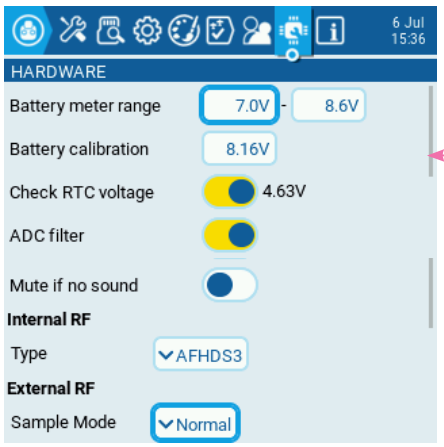
5.4 Binding

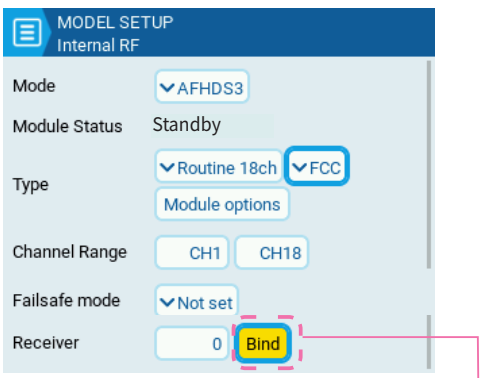
The TMr receiver supports binding in two-way or one-way mode, follow the steps below to finish the binding in two-way mode.

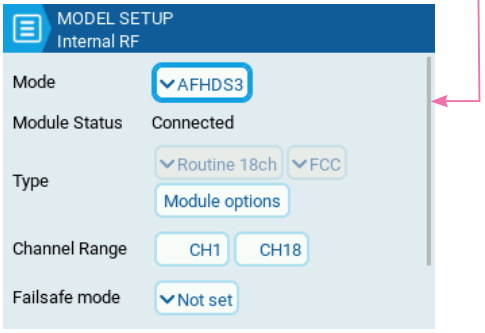
1. Enter the Radio Setttings interface, short press P> or P< to switch to the [Hardware] menu, select [Internal RF], and set the internal RF type to AFHDS3.
2. Enter the Model Setttings menu, then select suitable RF to enter the submenu. After setting the corresponding mode, ie AFHDS 3, and suitable type(Classic 18ch, C-Fast 10ch, Routine 18ch, Fast 8ch or Lora 12ch), select the Bind to put the transmitter into bind state.
   - TMr enhanced receiver requires selecting either Routine 18ch, Fast 8ch, or Lora 12ch mode for proper binding.
3. Press and hold the BIND button of the receiver while powering on the receiver (Alternatively, power on the receiver first, then press and hold the binding button for 3 seconds), the LED of the receiver should be flashing, indicating that the receiver is in bind mode.
4. When the receiver LED is solid on, it indicates successful binding.
5. Check whether the transmitter and receiver are operating properly. For re-binding, please repeat the above steps.

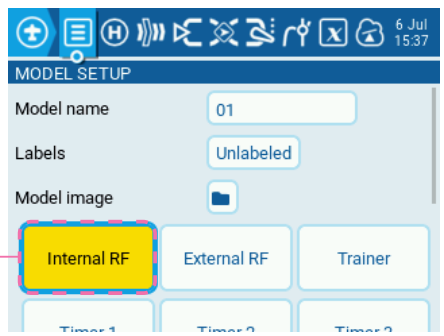
Note: For one-way mode, after the receiver LED becomes slow flashing, then put the transmitter to exit the binding state. At this time, the receiver LED is solid on, indicating that the binding is successful.

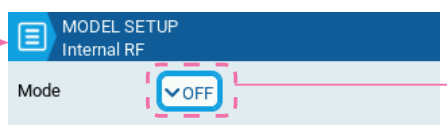


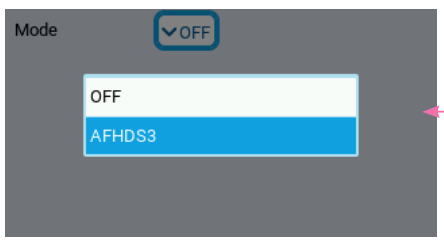










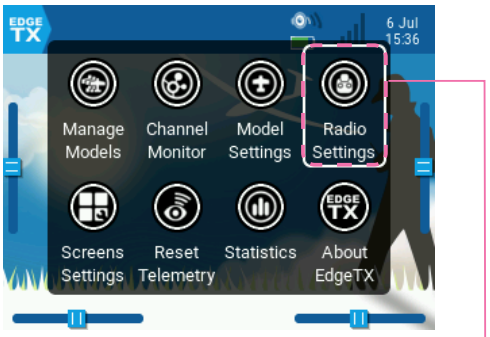


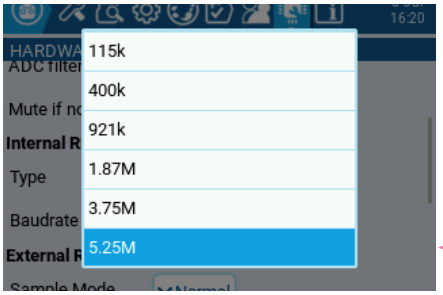
This transmitter is preloaded with the AFHDS 3 built-in RF firmware by default. To bind with ELRS receivers, flash the ELRS built-in RF firmware instead.

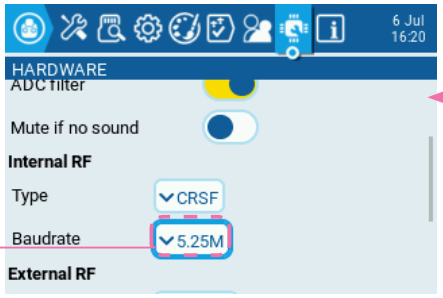
Please await the official ELRS firmware release. Reference: https://expresslrs.github.io/web-flasher/

The PR01 receiver supports binding in two-way or one-way mode, follow the steps below to finish the binding in two-way mode.

1. Enter the Radio Setttings interface, short press P> or P< to switch to the [Hardware] menu.
2. Select [Internal RF], and set the internal RF type to CRSF, and set the Baudrate to 5.25M.

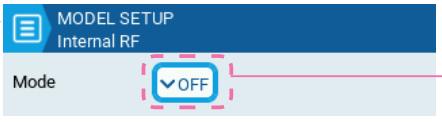


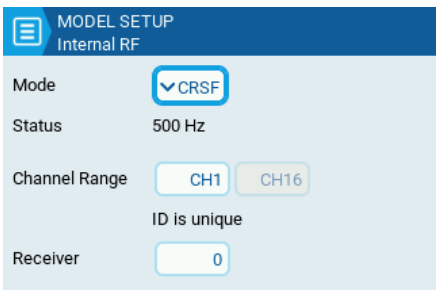


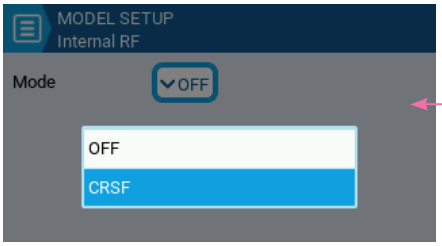


3. Enter the Model Setttings menu, then select suitable RF to enter the submenu. After setting the corresponding mode, ie CRSF.



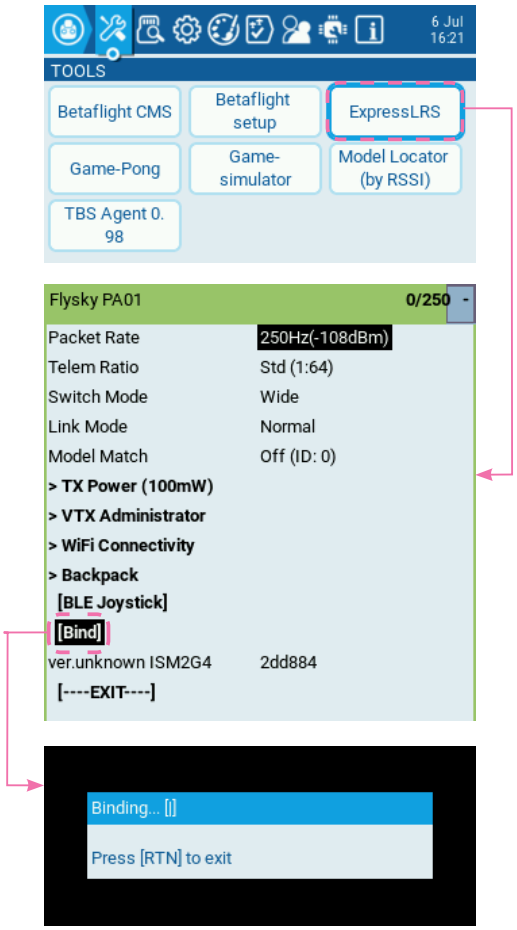






微信公众号 Bilibili Website Facebook

4. Select [Radio Settings]>[TOOLS]>[ExpressLRS] to enter the next interface.
5. After the receiver is powered on three times consecutively, the LED of the receiver should be in two-flash-one-off state, indicating that the receiver is in bind mode.
6. Transmitter Side: In the ExpressLRS interface, select [Bind] to activate the transmitter's binding mode.
7. When the receiver LED is solid on, it indicates successful binding.
8. Check whether the transmitter and receiver are operating properly. For re-binding, please repeat the above steps.



- **Binding methods vary by receiver. Please visit the FLYSKY official website to access the receiver manual or related documentation for specific instructions.**
- **For binding methods with other ELRS receivers, please refer to the corresponding ELRS receiver manual.**
- **As the product is constantly updated, please visit the FLYSKY website for the latest transmitter and receiver compatibility documents.**

5.5 Power Off

Follow the steps below to turn off the transmitter:

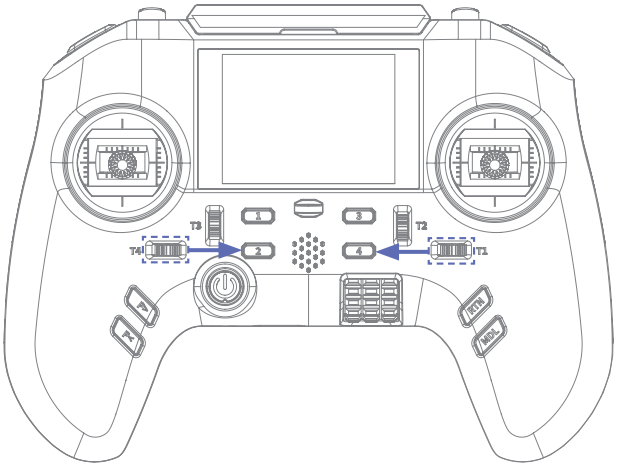
1. Turn off the receiver.
2. Press and hold the Power Switch until the screen turns off.

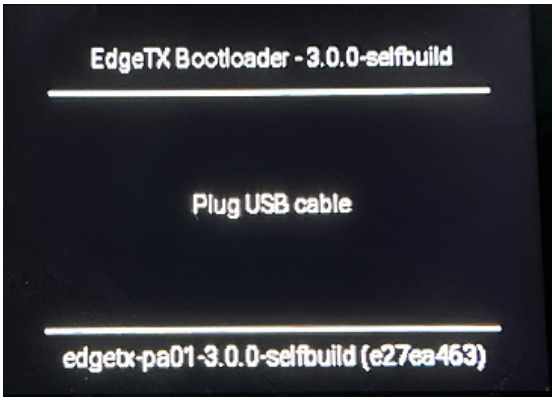
⚠ Danger	• **For safety always power off the receiver before the transmitter.**

5.6 Upgrading Transmitter Firmware

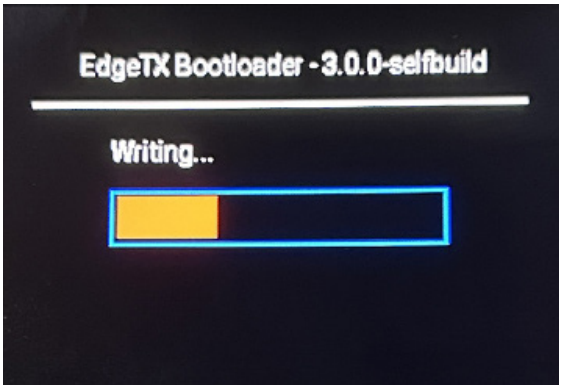
Follow the steps below to update the firmware of the transmitter:

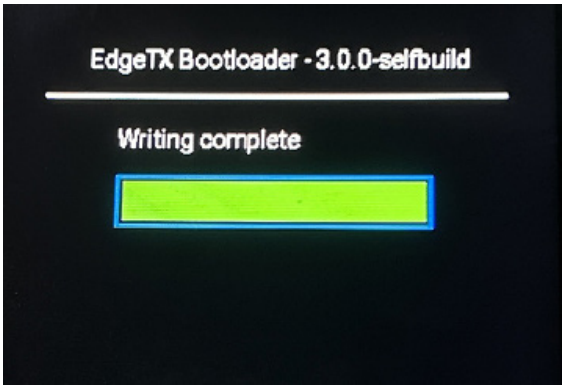
1. Access the PC website: https://github.com/EdgeTX/edgetx/releases to download the latest transmitter firmware.
2. While powered off, simultaneously push T1 and T4 trim buttons in the arrow directions, short press the power button for 2 seconds, then release all three controls to enter firmware update mode when the screen lights up.





3. Connect the transmitter to your PC via USB Type-C cable and select "EDGETX_UF2".
4. Copy the downloaded firmware (in .uf2 format) to the EDGETX_UF2 drive - the screen will display update progress.





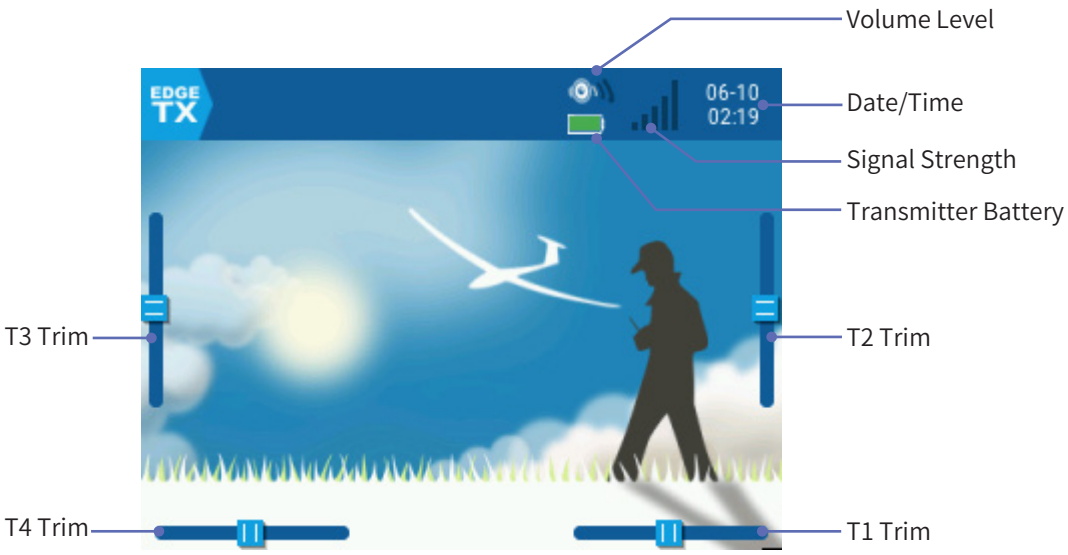
5. After the update is completed, unplug the USB Type-C cable, and the transmitter will restart automatically.

微信公众号 Bilibili Website Facebook

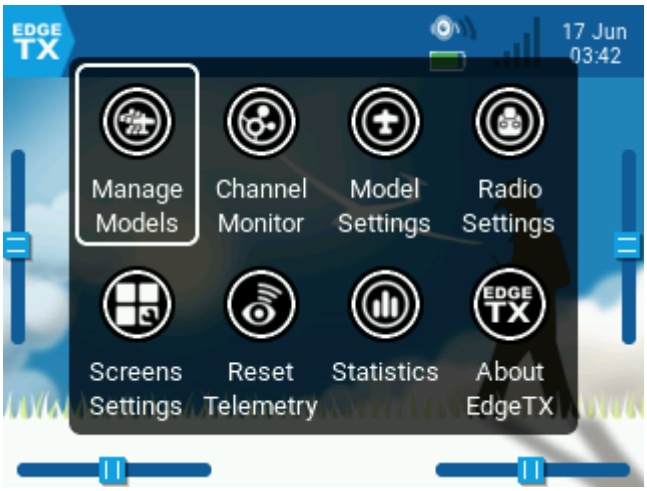
6. System Interface

This is an introduction about the transmitter's main interface.

The main interface includes the function icons, such as Manage Models, Channel Monitor, Model Settings, Radio Settings, Screens Settings, Reset Telemetry, Statistics and About EdgeTX. Select the corresponding icon to access the function interface.



Press the scroll wheel briefly to display the function menu.

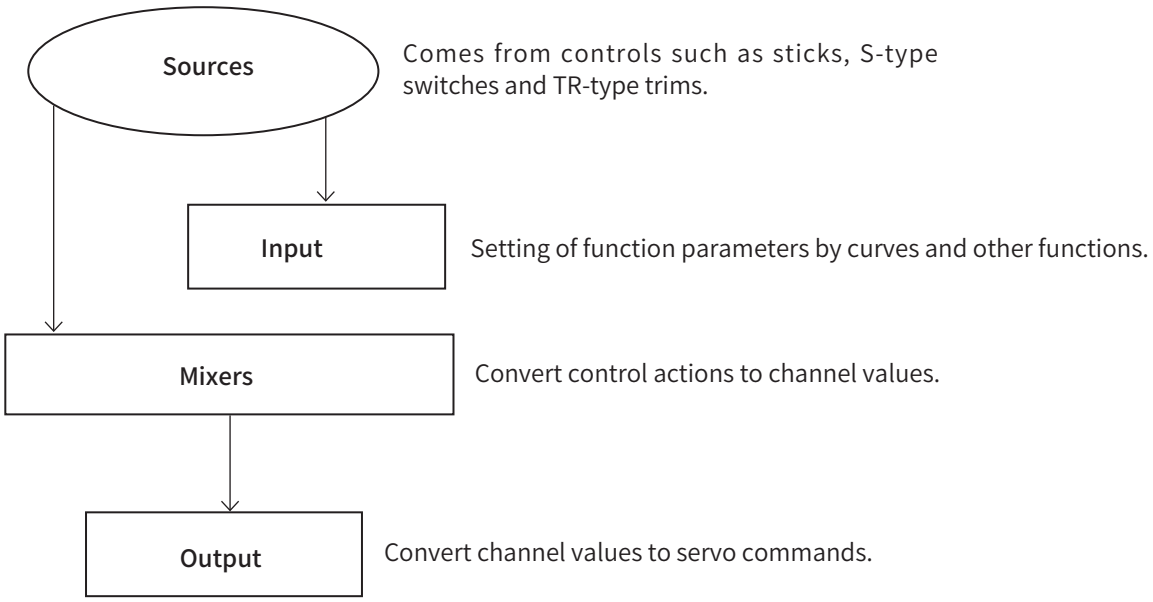


7. Function Settings

This chapter introduces the related settings of functions.

Note that open source operating systems are very different from branded operating systems as far as functionality is concerned.

The Edge TX system functional arithmetic logic is shown below:



微信公众号 Bilibili Website Facebook

7.1 Manage Models

Used to create new models, select which model is active, create and apply model labels, and create model templates.

7.1.1 New Model

Create a new model.

Select [New] → [New Model] to generate a new model.

Under each model, you can "Select model", " Duplicate model", "Save as template" and "Label model", and under the non-current model, you can also choose "Delete model".

Select Model

Selects this model as the active model.

Dulicate Model

With the duplicate function, you can create a new model by using the data of the currently selected model. You can use this function to copy the existing model data when set up a new model, and then modify different contents. The repeated setup can be avoided.

Label Mode

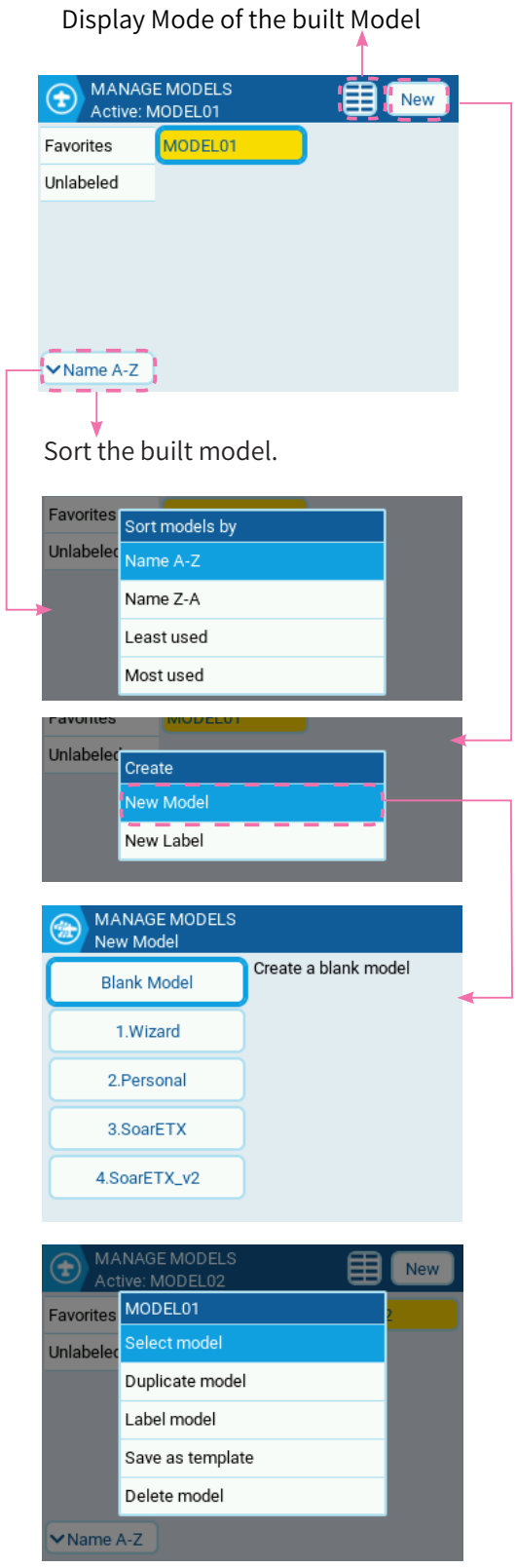
Add one or more tags for each model.

Save as template

Save the selected model as a template. When creating a new model, you can select the saved template from "Personal" to create a new model.

Delete Model

Moves the model to the deleted folder on the SD Card. Only models that are not active can be deleted.



7.1.2 Create New Label

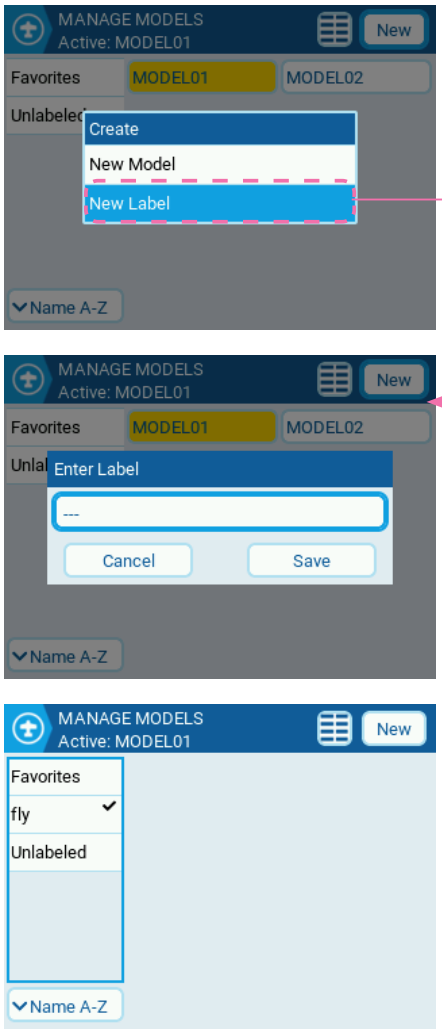
Create a new model group. You can change the name of the model group; you can also organize existing models into groups. Additionally, you can customize the position of the model groups.

Select [New] → [New Label] to generate a new model group.

Edit Label

Select model → [Label model] → Select Group. When a tick appears to the right of a grouping, the interface displays the models belonging to the label, and other models will be hidden.

If the new model is not yet labeled, "unlabeled" will appear.



微信公众号 Bilibili Website Facebook

7.2 Channel Monitor

By using this function, you can view the output of each channel and the status of the logic switch.

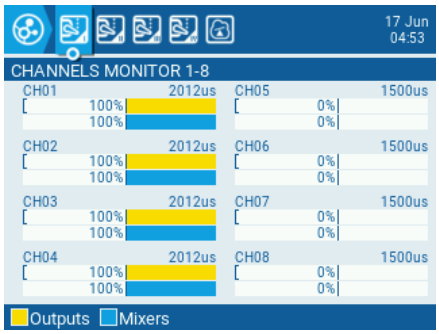
Channel Monitor

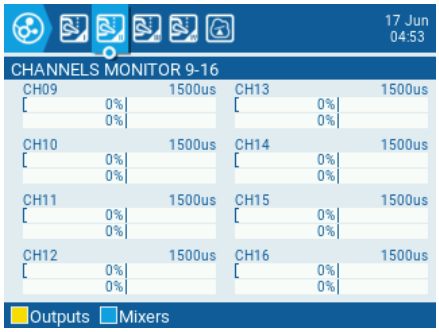
View the output of each channel. EdgeTX can output up to 32 channels.

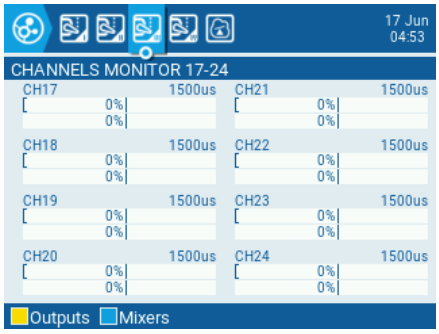
Each channel displays two travel bars in total. One is "Output view", indicating the actual output of the channel, and the other is "Mixes view", indicating the output calculated by superimposing various setup functions such as curve, mixes, etc. The output of "Mixes view" is not the same as "Output view".

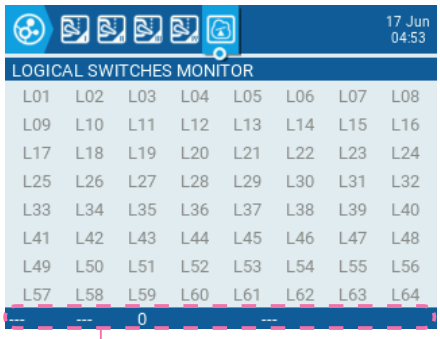
Logical Switches

View details of each logical switch, including its components and logical relationships.

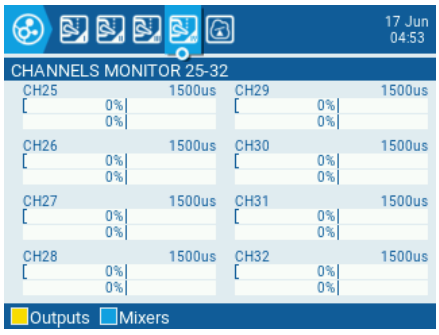








Display Logic Switch Information



7.3 Model Settings

7.3.1 Model Name

It is used to change the model name. You can view it on the model interface after the change is completed. You can enter a maximum of 15 characters.

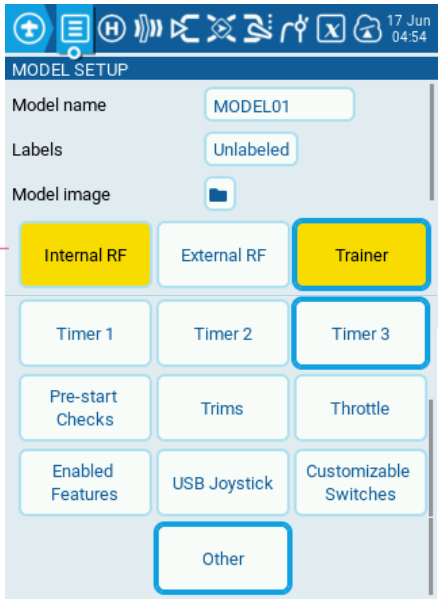
7.3.2 Labels

Assign a label from your defined label list. By default, the model label will be Unlabeled.

7.3.3 Model Image

Modify the picture corresponding to the model. The picture will be displayed in the "Model selection" after the modification is completed. The picture is also displayed when the model is selected as a widget in the main interface.

The format of the picture is recommended to be: PNG or JPG format, with a resolution of 320x480.

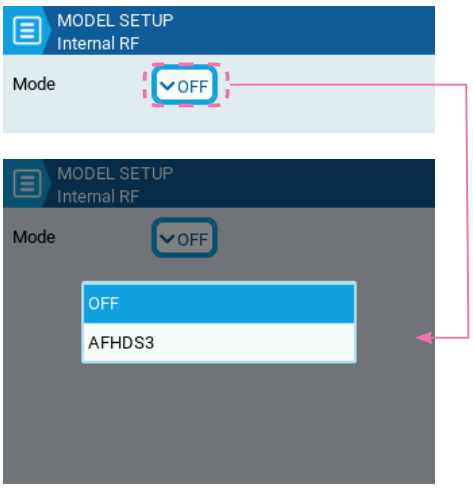


The bounding box background turns yellow when the feature is enabled.

7.3.4 Internal RF

This function interface varies according to the internal RF type set in the radio settings.

⚠ Warning	• **Do not use the USB Type-C cable to power on the RF module when the battery is not installed, to avoid loss of control due to unstable power supply!**



For AFHDS 3 protocol:

Mode

With the built-in radio frequency Flysky AFHDS 3 (Third Gen Automatic Frequency Hopping Digital System) protocol, PA01 is compatible with Flysky enhanced and classic edition receivers.

It is used to set the communication protocol of the transmitter. Select "OFF" to disable the internal radio frequency module. Select "AFHDS 3" to enable the internal radio frequency module. This can be used to bind Flysky AFHDS 3 receivers (TMr/FTr16S/Tr8B /FTr8B). For more information about it, visit the official website of Flysky Technology:https://www.flyskytech.com/

微信公众号 Bilibili Website Facebook

Module Status

Display the current connection status of transmitter and receiver. "Standby" means the transmitter's built-in RF module is in normal communication; "Disconnected" means the transmitter is not connected to the receiver, or the receiver is not detected. "Connected" means the transmitter and receiver are connected. "Ready" indicates that the transmitter and receiver are in a binding state.

Type

Select the transmitter and receiver connection parameters. There are classic and enhanced receivers in Flysky 3rd generation receivers. For the two types of receivers, you need to select the appropriate type.

FCC or CE Set the transmit power of the current transmitter. Legal regulations vary with countries/regions. Please choose the appropriate transmitter power according to the relevant legal information.

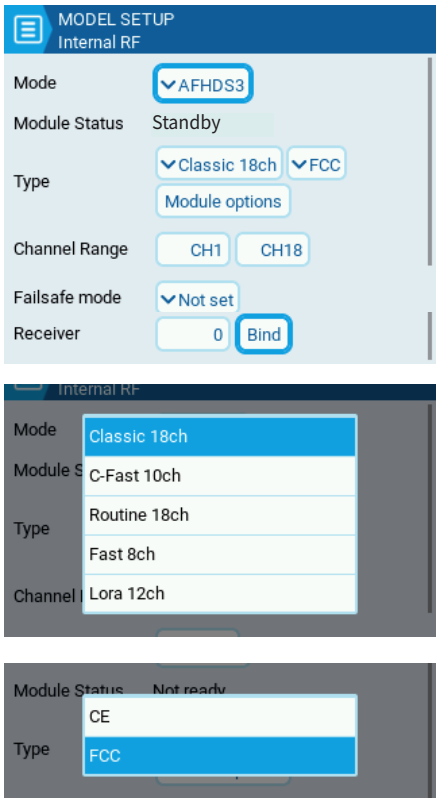
Classic 18ch Adapt to AFHDS 3 Classic receivers, such as FTr10/ FTr4/FTr16S. There are 18 channels for communications.

C-Fast 10ch Adapt to AFHDS 3 Classic receivers. There are 10 channels for communications.

Routine 18ch Adapt to AFHDS 3 Enhanced receivers, such as FGr4B/ Tr8B/TMr, with a moderate communication distance. There are 18 channels for communications.

Fast 8ch Bind the Enhanced receivers, with fast communication speed in a short distance. There are 8 channels for communications.

Lora 12ch Bind the Enhanced receivers, with super anti-interference and moderate communication distance. There are 12 channels for communications.



Module options

Use to set the frequency of the receiver output PWM signal, and the receiver interface output protocol, as well as the receiver signal strength output.

PWM Frequency: Theoretically, the higher the frequency, the faster the signal is refreshed, and the faster the servo responds to the signal change. However, some servos may not support PWM signals with excessively high frequency. You may need to consider the servo's performance capabilities when making these settings.

⚠ ATTENTION	• **The conventional PWM frequency is 50-400 Hz. When SR (PWM frequency: 833 Hz) and SFR (PWM frequency: 1000 Hz) are selected, the overall system delay will be decreased, but the pulse range of PWM signals is changed. Please make sure the servo supports the frequency and the setting is correct. Otherwise the servo may not work properly, or even get damaged.**

This function interface varies according to the Type Settings:

For enhanced version receiver:

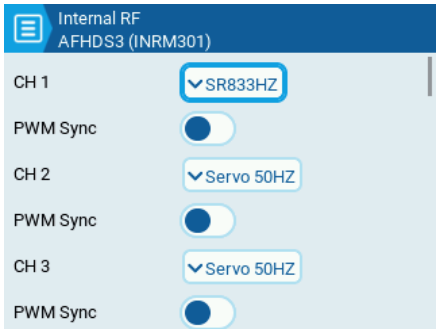
Supports separate PWM frequency setting for each channel, options include Servo 50HZ, Servo 333HZ, SFR 1000HZ or Custom.

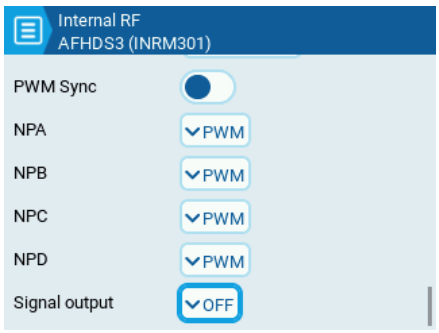
PWM Sync: The PWM output is synchronized with the timing of the (RF) radio signal reception.

Interface Protocol: NewPort(NPA, NPB, NPC and NPD), used to set the signal type of newport to one of the PWM, PPM, SBUS, iBUS IN or iBUS OUT.

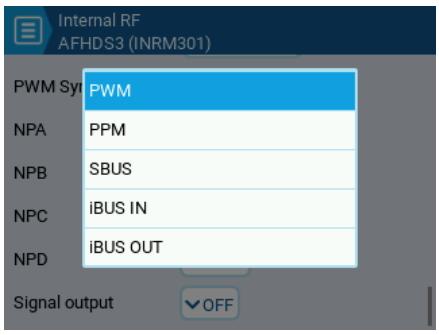
Signal output: Through this function, you can select a channel to output the signal strength value of the receiver. After the function is enabled, the selected channel does not perform the output of transmitter's corresponding channel function, but outputs the receiver's signal strength value. This function is necessary for users who use FPV glasses in the operations of traversers. We recommend you to select the CH14 or any auxiliary channel. You can make corresponding adaptations in the flight control settings to view the signal strength information on FPV glasses.

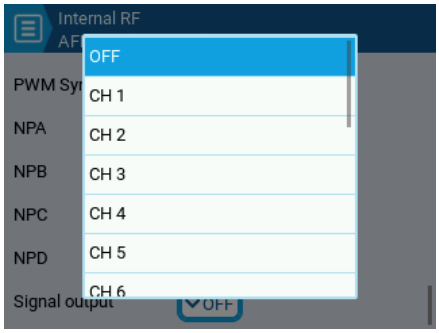
Module options- Enhanced version receiver











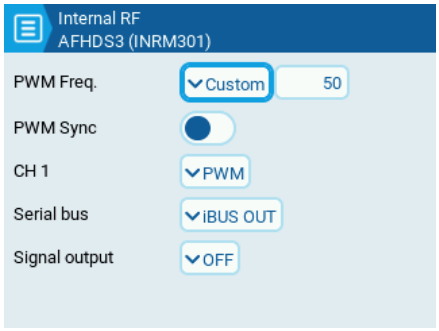
微信公众号 Bilibili Website Facebook

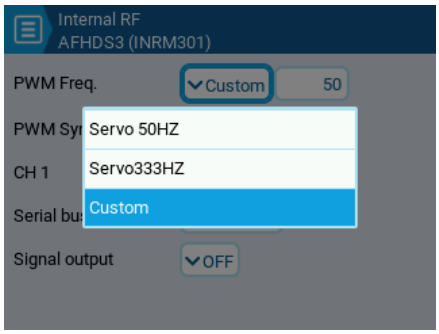
For classic version receiver:

CH1: Sets the interface protocol for the CH1 channel. PWM or PPM is optional.

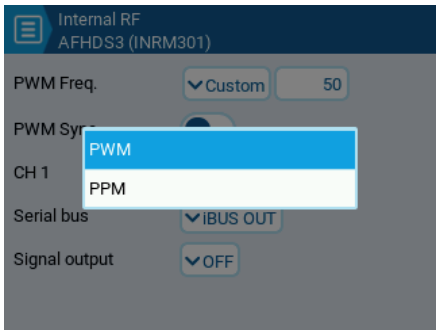
Serial Bus: Set the interface protocol of another interface. iBUS OUT, iBUS IN, or S.BUS is optional.

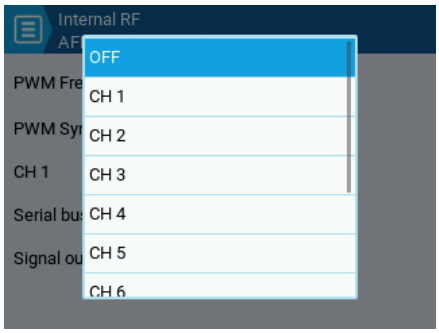
Module options- Classic version receiver





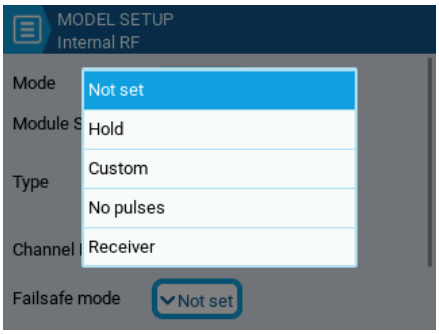






Channel Range

You can set the number of receiver output channels. By default, it is CH1-CH18. You can set the number of channels required by the model.

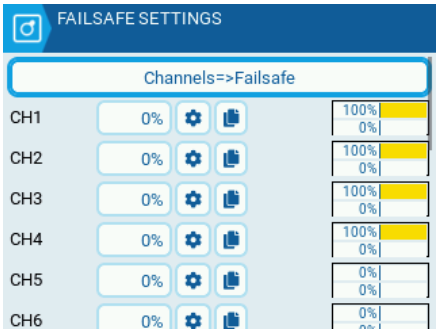


Failsafe Mode

Used to set the output channel value of the receiver after the receiver loses the signals. Five states can be set: "Not set", "Hold", "Custom", "No pulses", and "Receiver".

"Not set", "No pulses" and "Receiver" indicate no signal output in case of out-of-control. "Hold" means that the last signal is output in case of out-of-control. "Custom" , users can set the failsafe preset output value according to the failsafe output requirement of the model. On the setup interface, the dark travel bar indicates the output value of the corresponding position of the current control, and the red travel bar indicates the output value of the failsafe already set.

To set a separate channel by selecting , it can be set to "Hold", "None" or "0.0%" .Set the current value of this channel to the failsafe value by selecting .



Receiver

Set the receiver corresponding to the current model.

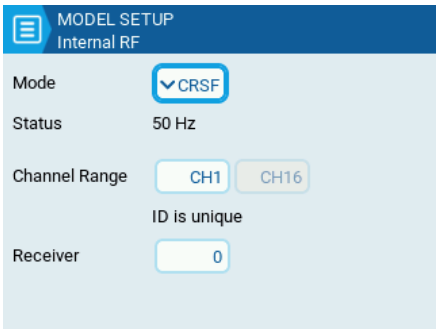
Bind

Sclect "Bind" to make the transmitter enter the binding state.

For CRSF protocol:

Mode

Compatible with corresponding receivers when the internal RF protocol is set to CRSF.



Channel Range

You can set the number of receiver output channels. By default, it is CH1-CH18. You can set the number of channels required by the model.

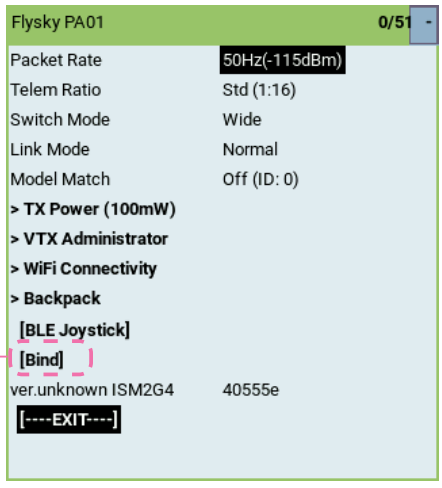
Receiver

Set the receiver corresponding to the current model.

You can change the parameters as follows:

On the main interface, press the scroll wheel briefly to enter the function menu, then select [Radio Settings] → [Tools] → [ExpressLRS] → Set Parameters.

Note: This transmitter comes with a default configuration .lua file. If you need to update or restore the .lua file, please visit: https://github.com/ExpressLRS/ExpressLRS/releases



微信公众号 Bilibili Website Facebook

7.3.5 External RF

Set the model channel range, radio frequency protocol, failsafe mode, refresh rate, and PPM frame rate of the external RF module.

EdgeTX supports the use of multiple RF modules. Each RF module supports different communication protocols and setup methods. For specific information, refer to the instruction manual of the corresponding RF module.

If another RF module is used, it should match a receiver that is compatible with the RF module. In this way, the binding can be completed.

In the setup of the internal RF or external RF, the delay can be properly reduced by setup the appropriate channel range, PPM frame and other parameters.

When you purchase the FRM303 RF module, set "External RF" to use it. FRM303 RF module, the data is stored in the RF module and supports up to 20 sets of models.

Module Status

Display the current connection status of RF module. "Ready" means the RF module is connected normally. "Not ready" means the RF module is connected abnormally.

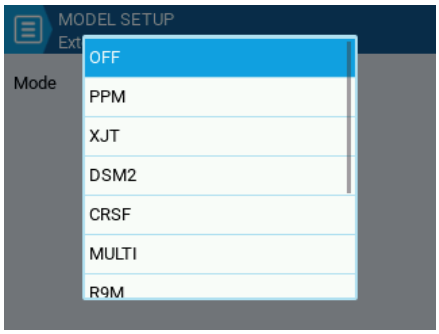
RF power

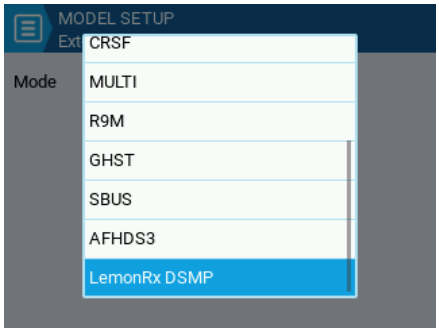
Set the output power.

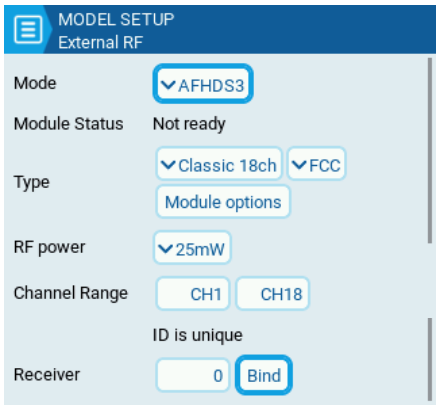
For other function items, refer to the Internal RF section.

Note: Please visit the Flysky website for updates on the FRM303.

⚠ Warning	• Do not use the USB Type-C cable to power on the RF module when the battery is not installed, to avoid loss of control due to unstable power supply!









7.3.6 Trainer

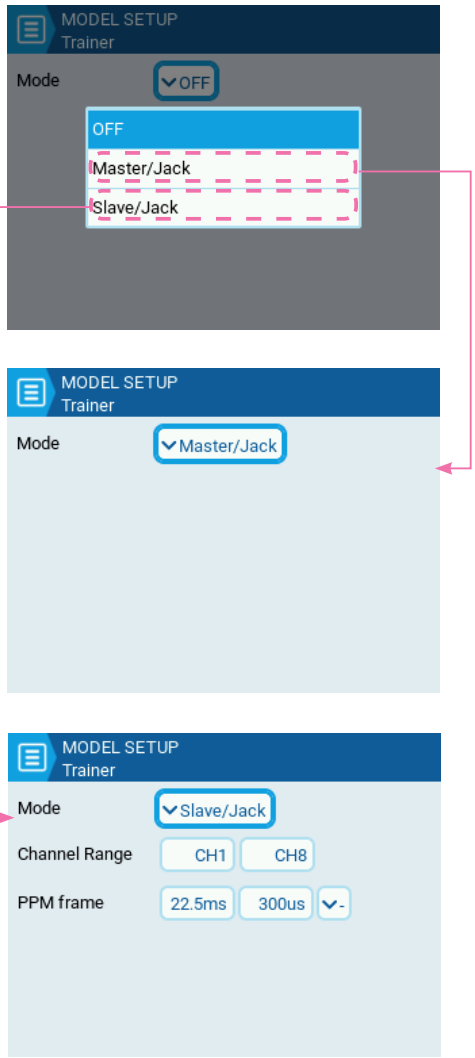
This function is used to set the master/slave mode of the trainer port. Two modes are available for selection.

When the "Master/Jack" is selected, the transmitter outputs trainer control signals to the model as the trainer mode, and student control signals are input from the audio jack.

When the "Slave/Jack" is selected, control signals of this transmitter are output to the trainer port from the audio jack. When the transmitter is connected to the simulator with a dongle, you need to select this option.

In this mode, you can set the channel range and PPM frame in the same way as the RF setup.

Set to '+' for standard PPM type or '-' for non-standard PPM type; when using head tracker PPM input, ensure the PPM signal type matches your setting.

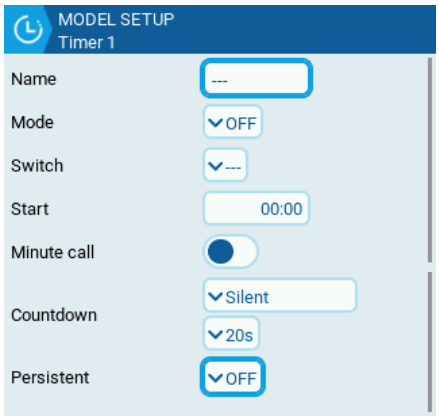


微信公众号 Bilibili Website Facebook

7.3.7 Timer1

Name

Name the timer with up to 8 characters.



Mode

Set the mode and switch for triggering the timer. You can set 6 groups of modes in total. A combination of different controls is used to control the start and stop of the timer. There are up-timer and down-timer.

Different timer modes can meet various flight condition requirements.

OFF Switch off the timer.

ON After the ON switch of the timer is triggered, it is not affected by other conditions. The timer starts the counting. When the ON switch returns to the initial position, the counting pauses.

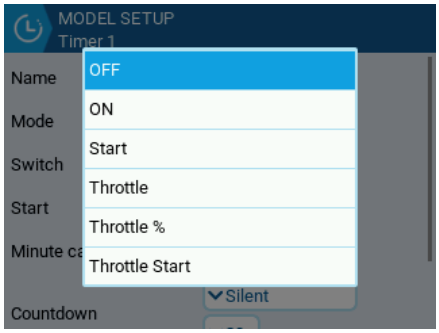
Start After the ON switch of the timer is triggered, it is not affected by other conditions. The timer starts the counting. When the ON switch returns to the initial position, the counting does not pause.

Throttle After the "ON" switch of the timer is triggered, when the throttle is not in the lowest position, the timer starts counting. When the ON switch of the timer returns to the initial position or the throttle returns to the lowest position, the counting pauses.

Throttle% After the "ON" switch of the timer is triggered, when the throttle is in the highest position, the timer starts counting, to calculate the time of full throttle. When the ON switch of the timer returns to the initial position or the throttle is not in the full-throttle position, the counting pauses.

Throttle Start After the "ON" switch of the timer is triggered, when the throttle is not in the lowest position, the timer starts counting. When the ON switch of the timer returns to the initial position, the counting pauses.

Note: The timer can be reset in "Reset"function.

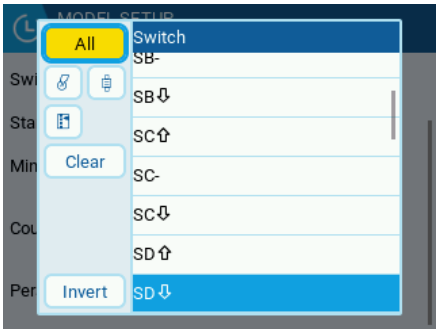


Switch

Position switch You can assign a toggle switch control or telemetry data control on the transmitter as the ON switch of the timer.

The " ↑ ", " ↓ " and "-" nearby the position switch indicate the highest, lowest and middle positions respectively. After selecting the target position of the position switch, you need to short press the scroll wheel for confirmation.

When the timer mode setup is completed, toggle the position switch. For example, "SD ↓ " means the timer is enabled when SD is in the lowest position.



Start

It is used to set the duration of the countdown. For example, if it is set to "01:00" to enable the timer, the timer starts counting down from 1 minute when the timing condition is triggered.

Minute Call

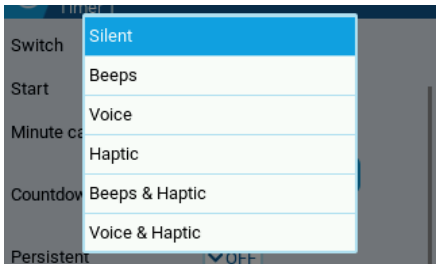
If this option is selected, the transmitter will broadcast once every one minute.

Countdown

It is used to set the transmitter broadcast mode and prompt time before the end of the countdown when the countdown times out.

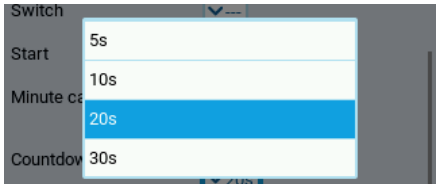
The countdown prompt is divided into four modes: "Silent", "Beeps", "Voice", and "Haptic".

Note: The "Voice" mode follows the system default voice settings when no special configurations are made.



Countdown Prompt Time

There are four modes: "5 seconds", "10 seconds", "20 seconds", and "30 seconds". When set to "10 seconds", the transmitter will activate countdown notifications 10 seconds before time expires, following the selected alert mode. If "Voice" mode is enabled, it will continuously vocalize "10, 9, 8, 7..." down to zero.



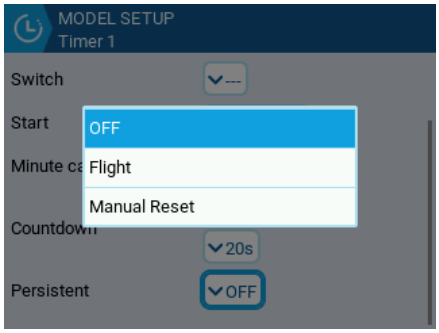
Persistent

It is used to set the current timing state of the timer, that is, the storage state by the shutdown or model change. There are three types: "OFF", "Flight", and "Manual Reset".

OFF The timer value is reset when switching models or when the radio is turned off / on.

Flight The timer value is NOT reset when switching models or when radio is turned off / on. The timer value is only reset when the Reset flight option is selected in the Reset telemetry menu.

Manual Reset The timer value is reset only when it is individually selected to be reset in the Reset telemetry menu.



7.3.8 Timer2

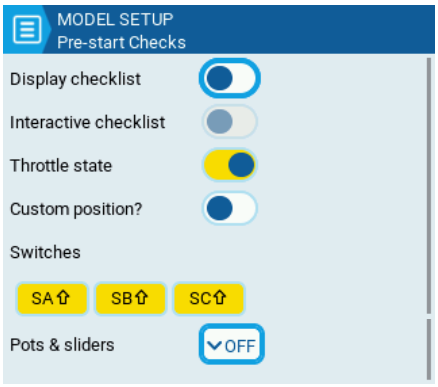
Refer to 7.3.7 Timer 1 on how to setting and using.

7.3.9 Timer3

Refer to 7.3.7 Timer 1 on how to setting and using.

微信公众号 Bilibili Website Facebook

7.3.10 Pre-start Checks



Display checklist

If a text file with the same name as the model is in the MODELS folder of the SD card, the transmitter will recognize it as a model check list. If this option is selected, the file will be displayed automatically.

Throttle State

When this item is selected, it will check whether the throttle is in the lowest position upon power-on. If the throttle is not in the lowest position, the system will give an alarm. If this item is not selected, the check is not performed.

Custom Position?

Turn on this function, you can customize the initial position of the throttle, and the system will be detected at power on with this value as the minimum throttle detection value.

Switch Position

It is used to set the initial position detection upon power-on. There are two states: one is to be detected yellow background, and the other not to be detected white background. Players can define the position of the switch for power-on detection as required.

Pots And Sliders (Non-returning middle dial)

The knob and slider are used to set whether to detect LS(S1) and RS(S2). There are three states: “OFF”, “AUTO”, and “ON”.

When “OFF” is selected, the positions of S1 and S2 are not detected upon power-on.

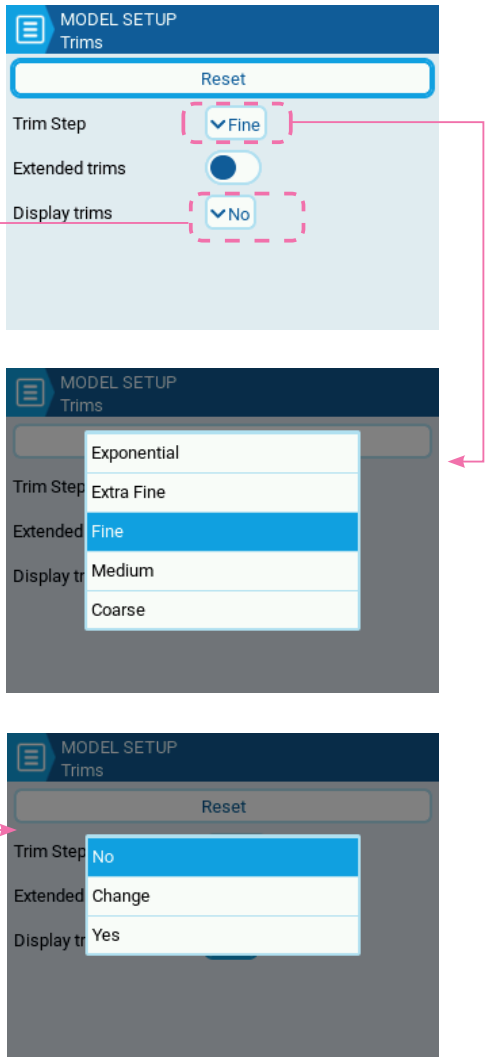
When “AUTO” is selected, it works automatically.

When “ON” is selected, it detects whether the S1 and S2 are in the initial position upon power-on.

7.3.11 Trims

With this function, you can set the step value and display mode of the trim.

If you select "Reset", all trims will return to the initial position.



Trim Step

Exponential To modify the trim step to exponential adjustment.

Extra Fine To modify the trim step value to 1.

Fine To modify the trim step value to 2.

Medium To modify the trim step value to 5.

Coarse To modify the trim step value to 8.

Extended Trims

After the "Expanded trims" is selected, it will expand the trim range. The default trim can be adjusted to a maximum of 25%. After it is selected, a maximum of 100% can be adjusted.

Display Trims

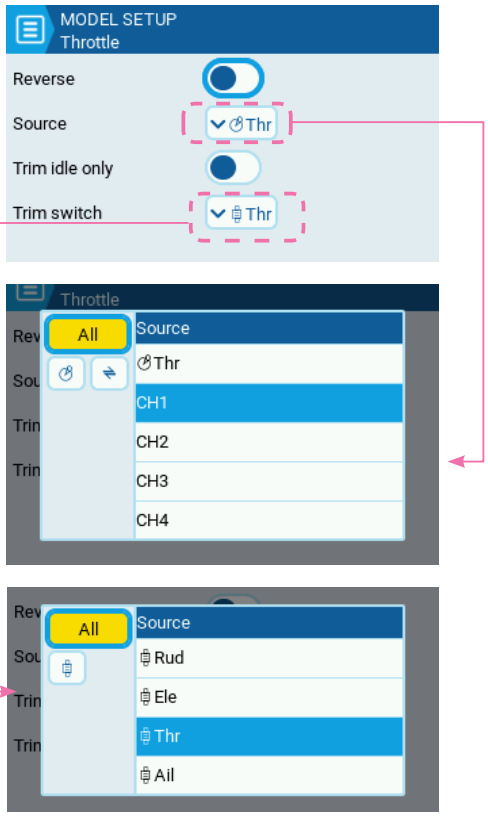
No The trim amount is displayed in the main interface. The current trim value will not be displayed when the trim value is changed.

Change Display the current trim value when the trim value is changed.

Yes Always display the current trim value.

微信公众号 Bilibili Website Facebook

7.3.12 Throttle



Reverse

The throttle will be reversed after this option is selected. It is recommended to debug in the "Mixes" menu or output setup when the throttle reverse is set.

Source

Set the input source of the throttle. It can be set to transmitter control or channel, generally set to "Thr".

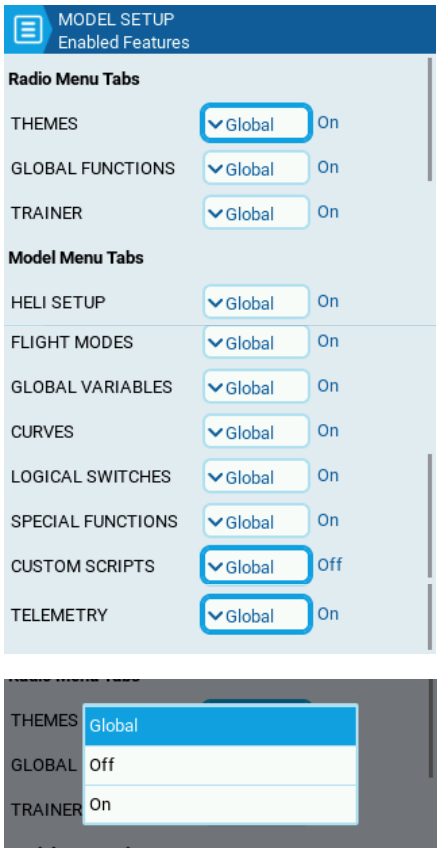
Trim Idle Only

When enabled, the throttle trim will only affect the bottom portion of the throttle band.

Trim Switch

Select the trim button to adjust the throttle, the default is "Thr".

7.3.13 Enabled Features



Configures which tabs are visible in the selected model's radio setup and model settings area.

Global

When selected, the tab will take the global value configured in the Enabled Features area in Radio Setup.

Off

When selected, this tab will not be visible when this model is loaded.

On

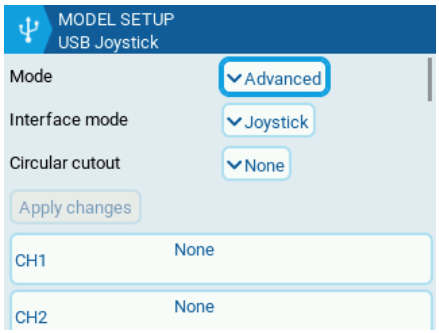
When selected, this tab will be visible when this model is loaded.

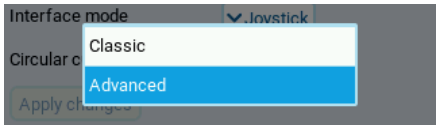
7.3.14 USB Joystick

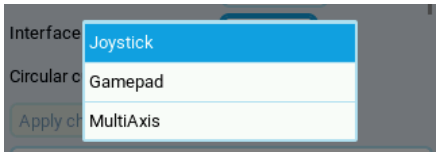
This function allows you to select the USB output function. It can be set to Classic or Advanced.

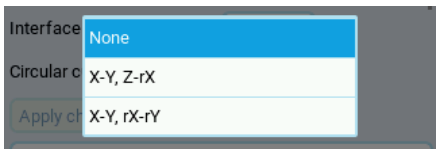
Classic

When the transmitter is connected to a computer, the transmitter acts as a standard HID device and is compatible with a wide range of emulators.









Advanced

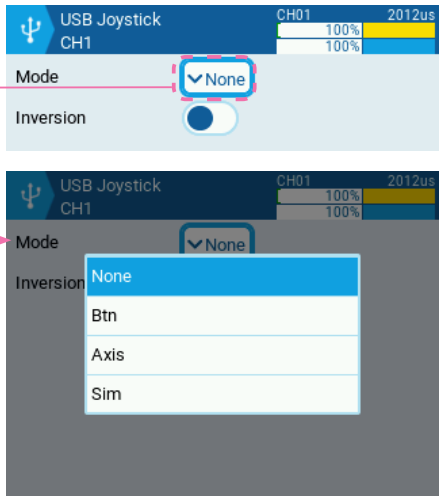
When set to Advanced, other devices can be connected to control the transmitter channel output. According to actual needs, you can set Interface mode and axis pairs mode can be set. For a specific channel, it can be set to "None", "Btn", "Axis" or "Sim" for specific channels.

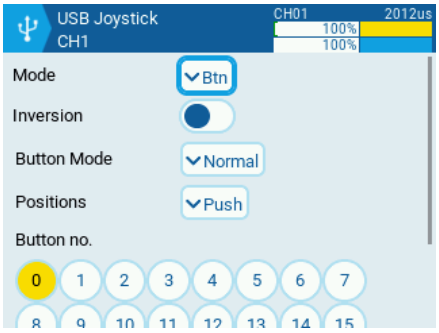
None Channel is not used.

Btn Channel is used to simulate a button.

Axis The channel is used to simulate an axis and will be mapped to one of the target device's default axes.

Sim The channel is used to simulate a common sim axis, such as Ail, Ele, Rud or Thr.







7.3.15 Customizable Switches

This function allows customization of illuminated button names, types, combo activation states, and multi-state LED colors (SW1~SW4 correspond to customizable Illuminated buttons 1~4).

Name

Set custom switch names (up to 3 characters supported).

Type

Set custom switch type: Options include 'None', 'Toggle', or '2POS'.

None: This switch is basically disabled.

Toggle: The switch state toggles between pressed and released positions.

2POS: The switch toggles its state with each press (OFF push ON push OFF ....).

Group

The switch group can be set as an input source for the "mix" function.

One switch of the group MUST stay on, in other words, the switches in the group cannot be all off. If you want this type of behavior, you should tick the check box at the right side of the screen.

Startup

Configure the switch state.

For switch not in group:

Last: Switch is set to the same state is was when the model was last used (it keeps old state).

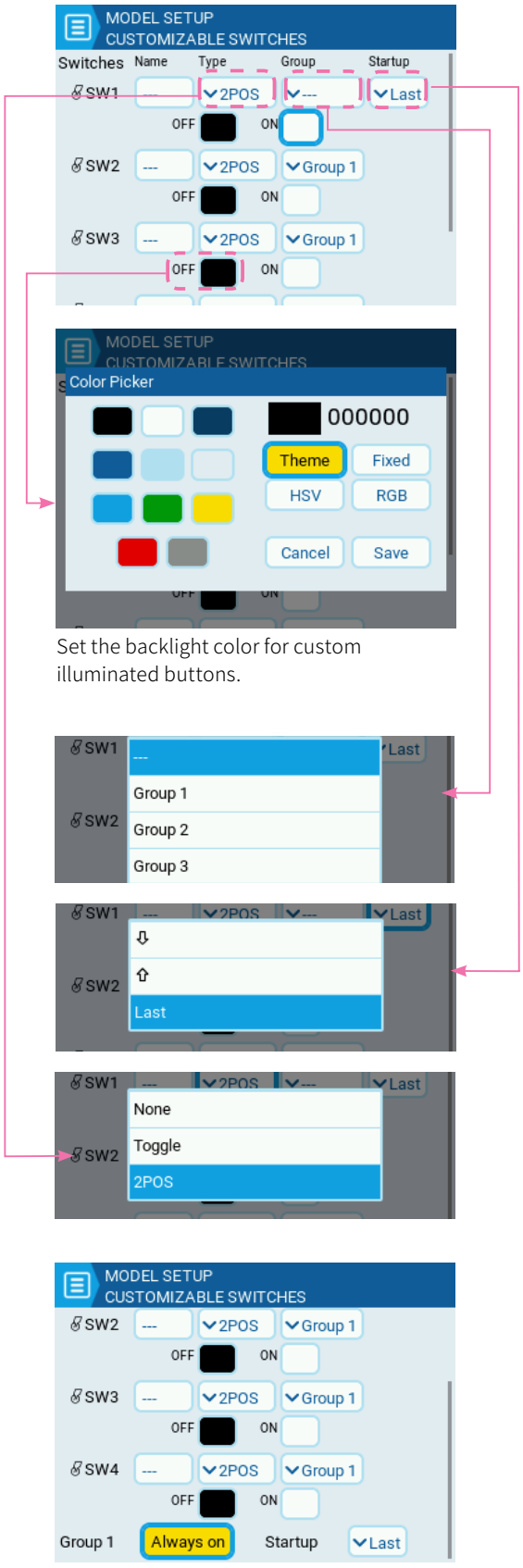
↑ : Switch is inactive.

↓ : Switch is active.

For groups:

Last: All switches in group are set to the same state is was when the model was last used (it keeps old state).

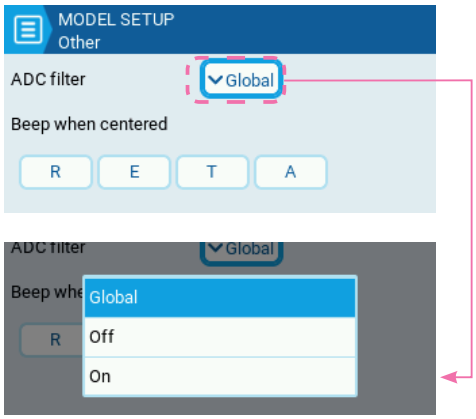
SWx: SWx is set to on, the others in groups are set to off.



Set the backlight color for custom illuminated buttons.

微信公众号 Bilibili Website Facebook

7.3.16 Other



ADC filter

Enables/disables the ADC filter for this model. The global option will take the value designated in the radio settings, which is on by default. Normally, this filter should be disabled for models with flight controllers.

Globel: When enabled, global functions programmed in the radio settings will apply to this model. When disabled, global functions will not apply to this model.

ON: This function is enabled for the current model.

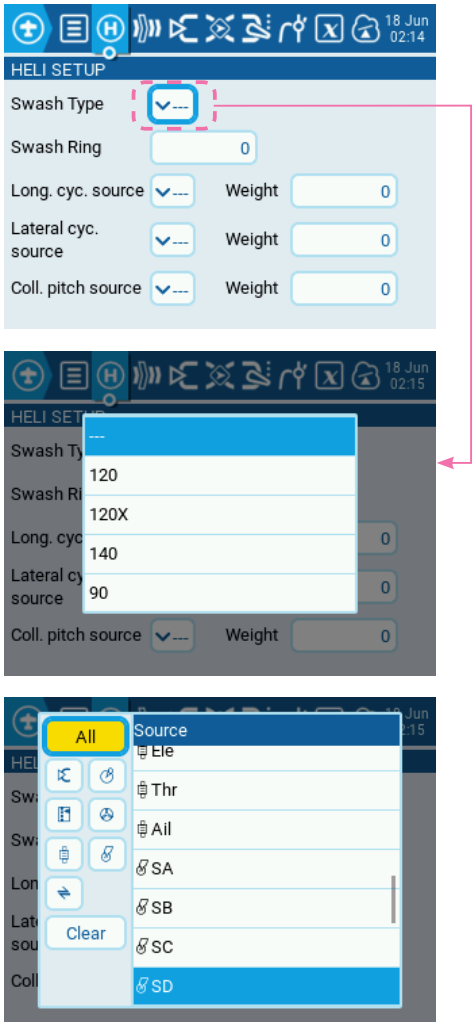
Off: This function is disabled for the current model.

Beep when centered

Set the stick midpoint detection of the channel. When the stick passes the midpoint, the transmitter will give a prompt through vibration. The dark background indicates that the middle position of the channel will be detected.

7.3.17 HELI Setup

Settings related to the helicopter modeling function. Normally CCPM helicopters need to be set up with this function.



Swash Type

Swash type for your model.

Swash Ring

Sets the swash ring limit only as needed.

Long. cyc. source/Lateral cyc.source/Coll. pitch source

Selects source input.

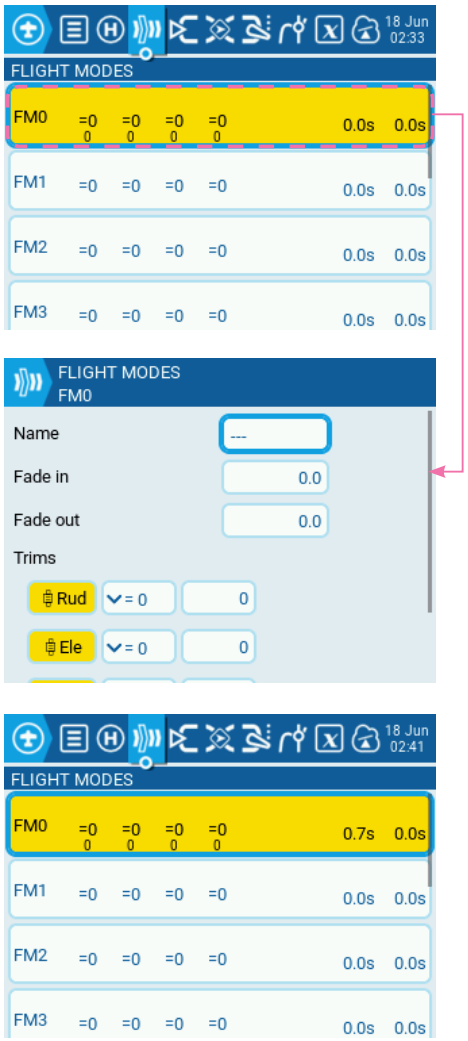
Weight Percentage value of the stick travel to use.

7.3.18 Flight Modes

To cope with different flight conditions or special flight tasks, you need to set different flight modes, to achieve some precise movements, or fast rolls. Therefore, the stick rate and travel requirements vary with flight modes.

With this function, you can set the name of each mode, ON switch, trim values of different modes, and the entry and exit time in the switching transition of each flight mode, respectively. Up to 9 flight modes can be set.

Take FM0 mode as an example.



Name

To set the name of the current flight mode. The name contains up to 10 characters.

Fade in

Gradually change the trim value when this flight mode is enabled. Specify the time until the value change is completed.

Fade out

Gradually change the trim value when this flight mode is disabled. Specify the time in seconds until the value change is completed.

Trim

To set the trim value.

微信公众号 Bilibili Website Facebook

7.3.19 Inputs

It is used to assign serial number, name, switch, curve, trim, and offset for each input. The serial number in the input setup does not represent the actual output channel serial number. The parameters in the input setup are set for each input source stick, toggle switch, knob, trim, button, and logic switch. The parameters set for each input source may be brought into the mixer that references the input for the calculation.

By setup each input to be referenced, it is more convenient for the use by the mixes in future.

"+": Used to select other input channels.

Function Settings:

Select the Input Settings function item to enter the editing interface, then configure parameters as needed.

Input name

To set the name of the channel, with up to 4 characters.

Line name

To set the name of the current default curve.

Source

It is used to set the input source. The source selection can be stick, knob, switch, button, trim, logic switch, etc.

Weight

Set the percentage of the input source. Adjust the rate output. The adjustable interval is between -100 and +100. When the value is less than 0, the input source will be reversed. The setup of the rate will affect the travel of the servo. The rate set in the mixes will have a superposition operation with the rate in the input setup, to prevent the output of the superposition operation from exceeding the maximum angle of the servo. The rate can be set in the input setup according to the travel of the servo to limit the maximum output of the servo.

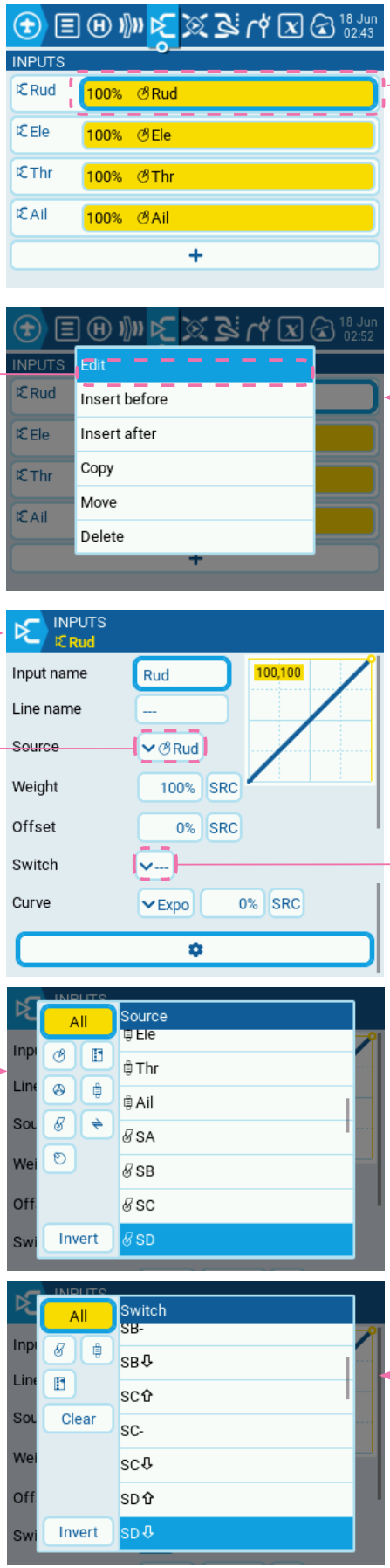
It is recommended to set the outputs when the servo is reversed.

Offset

Adjust the offset (upward and downward) of the curve. It affects the high travel, low travel and the neutral travel.

Switch

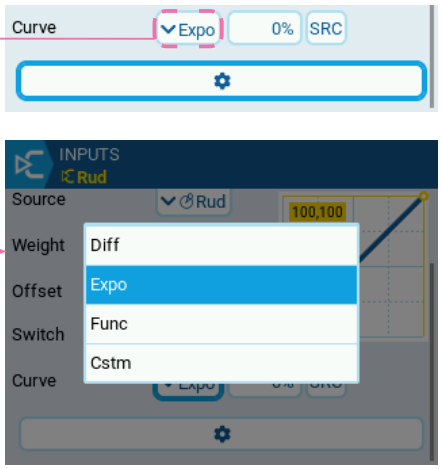
Enable and disable this input source. The switch can be a physical control and a logical switch. By default, it is enabled. In this case, it is selected as "---"



Curve

Set different output curves for various input sources. Different curves will have different manipulation feels. Users can set the appropriate curve according to their operating needs.

In this function, there are four modes: "Diff", "Expo", "Func" and "Cstm". A variety of output curves can be selected under different curve modes.



Diff Adjust the one-side curve without changing the linearity of the curve, but changing the high and low travel of the curve.

Expo Adjust the linearity and ratio of the curve. Adjust the two-sided curve at the same time. The larger the value is, the smoother the front-end curve change is. The back-end changes quickly. When the value is negative, it is opposite.

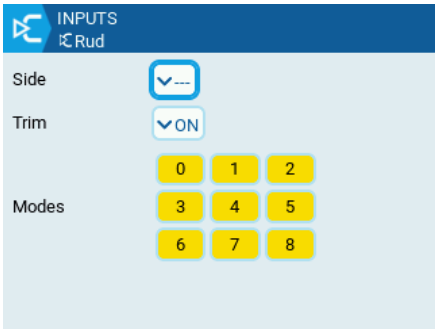
Func In this mode, a variety of algorithmic curves are preset. Users seldom use these curves.

Take "Func"--"f< 0" as an example. If the output source is < 0, the output = -100%. If the output source >0, the output = 0.

The above curve setup is affected by the rate and offset. The output will change according to the different setups of the ratio and offset. You can check the curve changes in the display bar to set it during the actual operations.

Cstm Select the curve in Curve Setup.

Select ⚙ to enter more setups.



Side The function is similar to Func. For example, if "x>0", then the low-side input is all "0".

Trim Set the trim button to use for this input setup.

Modes Select the flight mode that is effective for this input setup.

微信公众号 Bilibili Website Facebook

7.3.20 Mixes

The Mixes setup logic of EdgeTX is very flexible, that is, a variety of mixes input sources and output sources. There are three implementation modes: (1) a control → a channel; (2) a control → multiple channels; (3) multiple controls → a channel. Where in mode (3) is mostly applied to the setup of delta-wing airplanes.

To deeply understand the EdgeTX's mixer function, the above controls can be considered as "Input setup - Input source". In the "Input setup" menu, each line can be superimposed with more lines. There is one input setup, with multiple input sources.

The figure shows the basic setup of delta-wing airplane for input setup and mixes setup.

To understand the relationship between input setup and mixes correctly, the operator should understand the relationship by considering the action that his model needs to achieve. For example, for a delta wing, aileron roll and pitch are the actions achieved by the two wings, and the two wings are controlled by the servo. Under normal circumstances, the two servos are connected to the receiver CH1 and CH2 (The receiver's CH1 and CH2 output the transmitter's CH1 and CH2, corresponding to input source 1 and input source 2. However, for delta wings, CH1 and CH2 correspond to two different input sources respectively.) In the input setup, set the input source of input 1 and input 2 to Alie and Ele respectively (For the manual operation in USA, the receiver CH1 corresponds to Ali J1 stick, CH2 corresponds to Ele J2 stick).

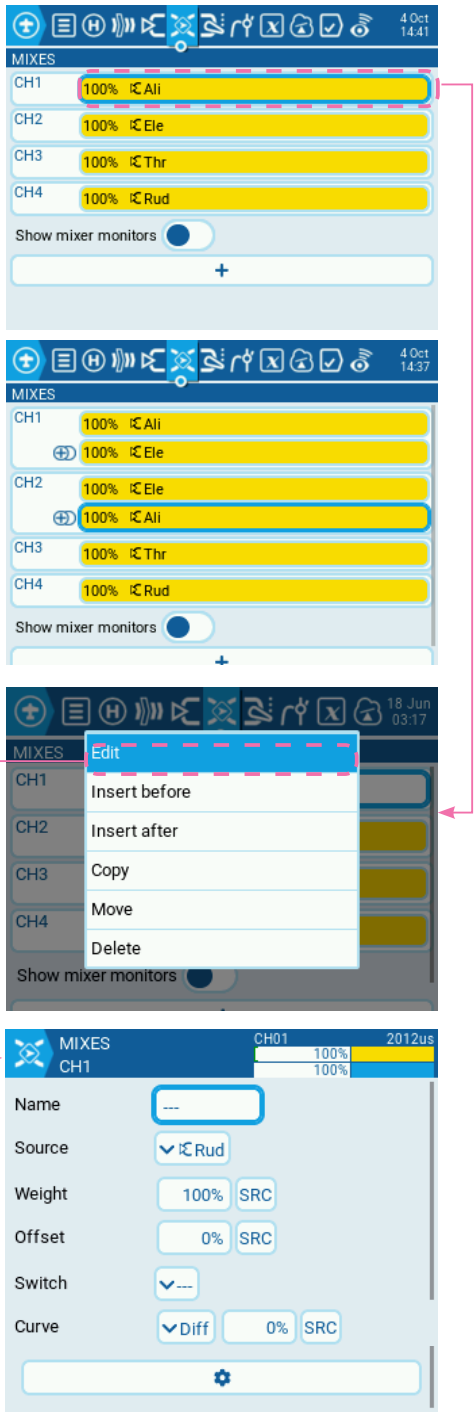
In the mixes setup, you can set which input sources are needed for each channel and the travel of the input sources according to the actions you need to achieve.

In the use of the EdgeTX's mixes, each mixes should be set with the consideration of the action that the model needs to achieve. It is helpful to better understand and realize the logic principle of each mixes.

The travel of each input source can be set in the mixes setup. The travel will have a superposition operation with the travel setup in the input setup. Therefore, pay special attention to the travel setup to prevent the travel from exceeding the limit of the servo in the debugging. Otherwise, the servo may be damaged.

Select "+" in the main interface to add a channel. You can also perform the channel setup.

On each main interface, you can perform the "Input setup" of the reference.



Edit

Set the selected input setup, such as name, effective flight mode, input source, curve, etc.

Insert Before

Add a new input setup before the selected entry.

Insert After

Add a new input setup after the selected entry. The main difference between the insertion before and after this entry is the "Multiplex".

Copy

Copy the selected input setup.

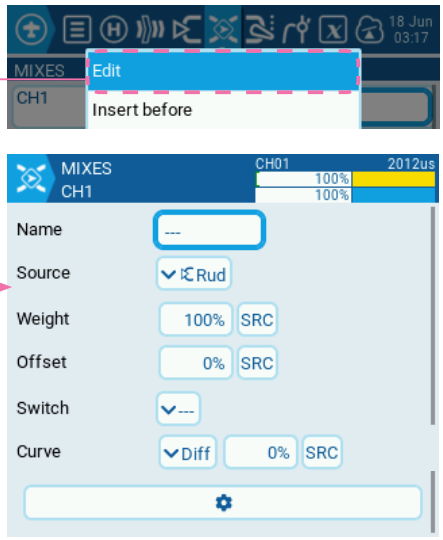
Move

Move the selected input setup.

Delete

Delete the selected input setup.

Select Edit to enter the setting interface.



Name It is used to set the mixes name. It is recommended to set the name of each mixer according to the model's action for better understanding and usage.

Source It is used to set the input source referenced by the mixes. The input source can directly refer to the input setup and various controls. The reference of the input setup as the input source can apply the parameters in the input setup to the mixer at the same time. However, the direct reference of a control does not necessarily cause the calling of the relevant setup of this control. It is recommended that the input source should reference the output setup to achieve more direct setup for each control.

Weight It is used to set the percentage of each input in the mixer and adjust the ratio output. The adjustment interval is from -100 to +100. When the value is less than 0, the input source will be reversed. This ratio will be superimposed with the ratio in the input setup.

Offset It is used to adjust the upward and downward offset of the curve, affecting the high travel, low travel, and neutral travel. The offset of the input setup and the mixes will be superimposed. In the debugging, take measures to prevent the superimposed travel from exceeding the maximum angle of the servo. Otherwise, the servo may be damaged.

Switch It is used to set a control to enable and disable the mixer. If there is no setup, "---" is displayed, indicating that the mixes is always effective.

Curve In the top display bar of this interface, you can check the current channel status.

Note: The usage is similar to the curve usage in the input setup.

微信公众号 Bilibili Website Facebook

Select ⚙ to enter more setups.

Modes This function allows you to set the flight mode applied to this curve. The yellow background indicates that the mode is applied, and the white background indicates that the mode is not applied. The applied mode will be available for query in the mixes interface.

Trim Specifies whether or not to include the trim values in this mix.

Warning Enable the warning tone of the mixes. By default, it is off. The highest value is "3". That is, after the mixes is enabled, the system will have a "beep" tone, indicating that the mixer has been enabled.

Delay up/dn prec Changes the precision for Delay up/dn between 0.0 and 0.00.

Delay up Implement the uplink delay function of the channel. The channel delay is a form of step. For example, if you set that the uplink delay is 2s and the stick output is 50%, the channel delay is from 0% to 50% in 2 seconds. Uplink refers to the travel interval of -100% → 0 → +100%.

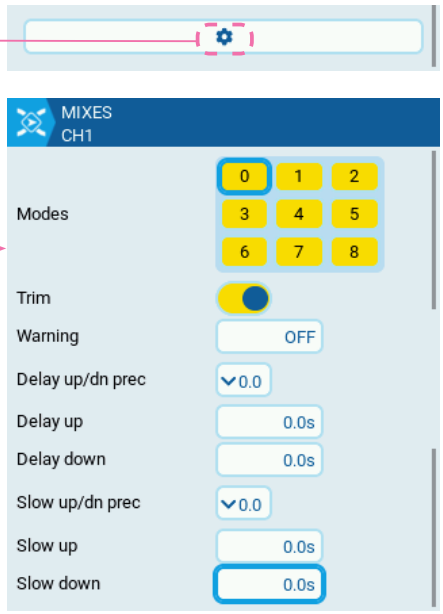
If the reversal is set, the output of uplink delay will also be reversed. The delay direction of uplink delay is not affected by the reverse setup. In the channel view interface, you can better distinguish them according to "Channel output" and "Mixe output".

After the uplink or downlink delay function is enabled, you can view the different icons on the channel status bar in the mixer main interface.

Delay down Implement the downlink delay function of the channel. The downlink refers to the travel interval of +100% → 0 → -100%.

Slow up Implement the uplink slow motion delay function of the channel. The channel delay mode is a form of slow motion. For example, if you set the 2s uplink slow motion and the stick output is 50%, the channel slowly increases from 0% to 50% in 2 seconds. Others are the same as the uplink delay.

Slow down Implement the downlink slow motion delay function of the channel.





7.3.21 Outputs

The function is where final adjustments to the control data are made (including subtrims, curves, endpoint, and center values) before finally sending the control data to the RF module.

Add All Trims to Subtrims

When selected, adds the current trim value to the subtrim value for each configured output. The trim value is then reset to zero.

Extended Limits

When enabled, it increases the minimum(-150) and maximum(150) range for the output values.

Select any channel. Four options are displayed.

Edit

Set relevant parameters for the selected channel, such as channel name, midpoint, reverse, etc.

Name It is used to name the current channel, with containing up to 6 characters.

Subtrim It is used to set the midpoint of the current channel. The maximum value is +100 and the minimum value is -100.

When the mixer of the channel references one or more input setups, the value of the midpoint will be involved in the calculation of different curves and different superposition methods.

Min It is used to set the minimum output value of the channel. This setup will limit the final channel output value after various curves and superimpositions are set.

Max It is used to set the maximum output value of the channel. This setup will limit the final channel output value after various curves and superimpositions are set.

Inverted It is used to set the output reversal of the channel.

Curve The selected curve is the used curve for this channel.

PPM Center Sets the midpoint of the channel. This value will be superimposed with the value of the trim button.

Subtrim Mode Defines how the subtrim value affects the min/max output values. Two options: Center Only or Symmetrical.

Reset

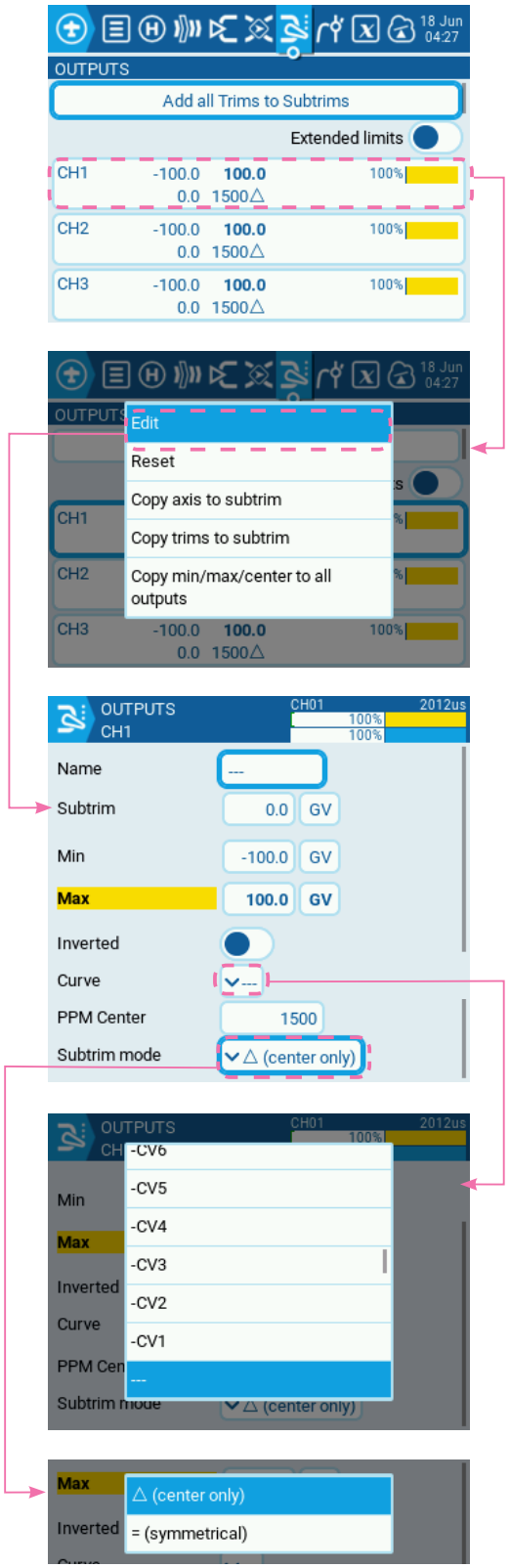
Reset the selected channel to the system default parameters.

Copy Stciks to Subtrim

Adds the current value of the stick deflection as the subtrim value.

Copy Trims to Subtrim

Adds the current trim value to the subtrim value. The trim value is not changed.



微信公众号 Bilibili Website Facebook

7.3.22 Curves

Up to 32 different curves can be set in the EdgeTX, including "Input setup", "Mixes", "Input setup", etc.

Edit

Opens the curve configuration page.

Preset

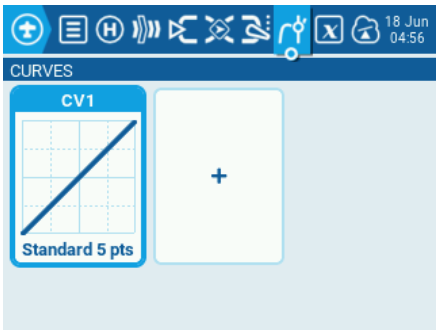
Allows you to set the curve to one of the preset slope values (-45 to 45 degrees in 15 degree increments). The curve will have 5 points and smoothing is not enabled by default.

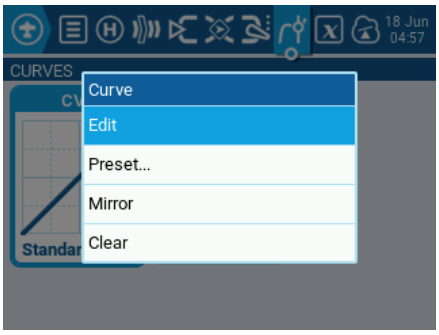
Mirror

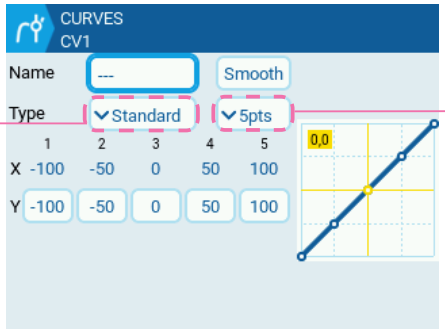
Mirrors the selected curve.

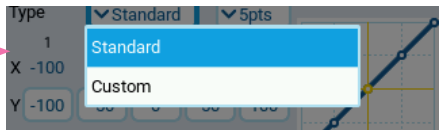
Clear

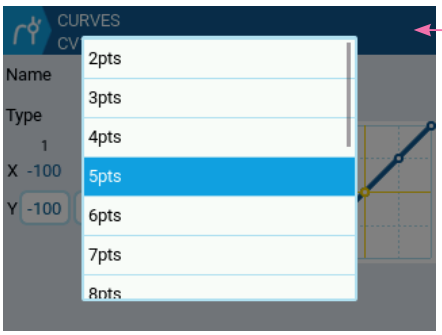
Clears all curve values from the selected curve.











Choose any curve and select "Edit" to set the curve.

Name To set the name of the curve.

Type You can set two types: "Standard" and "Custom".

Smooth When the smooth effect is enabled, the turning point of the curve will be smooth.

In the "CURVES", you can set up to 17 points. The value of each point can be set individually. When the set points exceed the maximum number of points that can be displayed on the screen, slide the screen frame to switch.

7.3.23 Global Variables

Global variables are variables whose values are shared across all the configuration menus of a model.

Select any variable to display the options "Edit" and "Clear".

Edit

Opens the global variable setting menu for the selected global variable.

Clear

Clears the global variable values for all flight modes for the selected global variable.

Select "Edit" to enter the setting interface. You can set the value of the global variable for each flight mode, or you can inherit the value from another selected flight mode. It contains the following configuration options:

Name Name for the global variable.

Unit Allows you to add a % label to the displayed values when selected. It does not affect how the values are calculated.

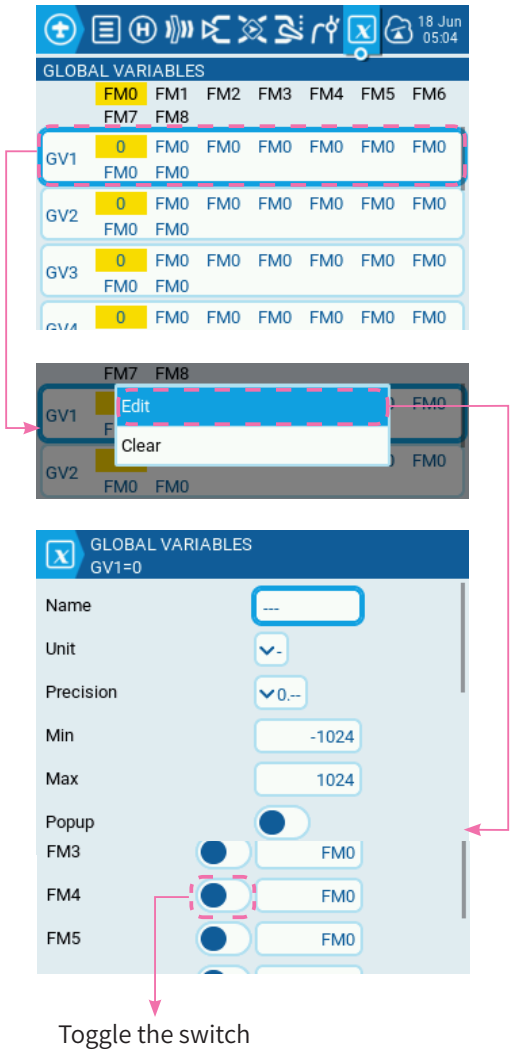
Precision Allows you to select the number precision options whole numbers and decimal.

Min Defines the minimum that is allowed for the global variable.

Max Defines the maximum value that is allowed for the global variable.

FM0 The value of the global variable on Flight Mode 0.

FM1 -> FM8 Toggle switch disabled: Global variable value for the selected flight mode is inherited from the flight mode; Toggle switch enabled (highlighted): The global variable value for the selected flight mode is manually defined in the text field.



微信公众号 Bilibili Website Facebook

7.3.24 Logical Switches

The EdgeTX logic switch is a kind of programmable virtual switch, unlike the usual logic switch. The input sources selected are for various physical control switches. The input source of the EdgeTX logic switch can be physical control switches, telemetry values height, temperature, current, voltage, speed, RSSI, etc., other logic switches, input values of controls, global variables, timers, etc.

Select "+" to set an unused logic switch.

Select any of the logic switches to display the options Edit, Copy, Paste and Clear.



Edit

Opens the Logical Switches configuration page for the selected logical switch.

Copy

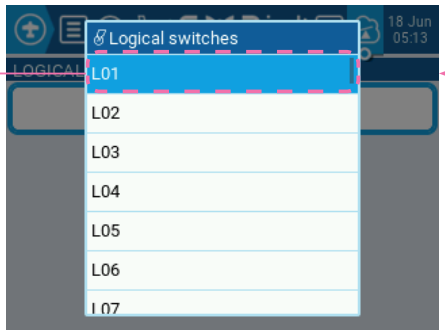
Copies the selected logical switch.

Paste

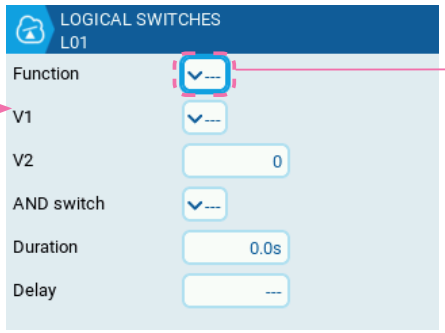
Pastes a copied logical switch onto the selected logical switch.

Clear

Deletes all configuration options for the selected logical switch.



Select "Edit" to enter the setting interface.



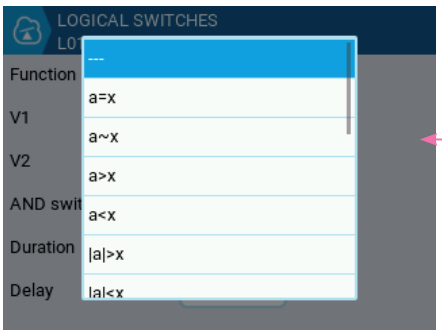
Function Configure the desired logic function.

Type 1: Variables And Constants

There are 6 options: a=x; a~x; a>x; a< x; |a|>x; |a|< x;

"a" is the judgment input source of V1, and "X" is a set constant.

Example: Select "a>x", set V1 input source to throttle stick, x=50. That is, when the throttle stick output is greater than 50, the logic switch will be triggered.



Type 2: Logical Operations

AND AND operation. V1 and V2 input sources are in the AND relationship. The logical switch is enabled if both outputs are true.

Example: Select V1 input source "SA lowest" and V2 input source "SC highest". When SA and SC are in the lowest and highest position respectively, the logical switch is enabled.

OR OR operation.

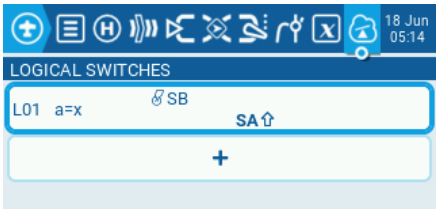
XOR XOR operation.

Type 3: Variables And Variables

a=b The logical switch is enabled when the value of V1 input source is equal to V2 input source.

a>b The logical switch is enabled when the value of V1 input source is greater than V2 input source.

a<b The logical switch is enabled when the value of V1 input source is less than V2 input source.



Other options are as follows:

"AND" Used to set the general switch of the logic switch. By default, it is set to "---", indicating that the general switch is ON. You can also set other input sources as the switch.

"Duration" Used to set the effective time that the logic switch is ON. The maximum value is 25s.

"Delay" Used to set the delay time for triggering the logic switch to turn on. The maximum value is 25s.

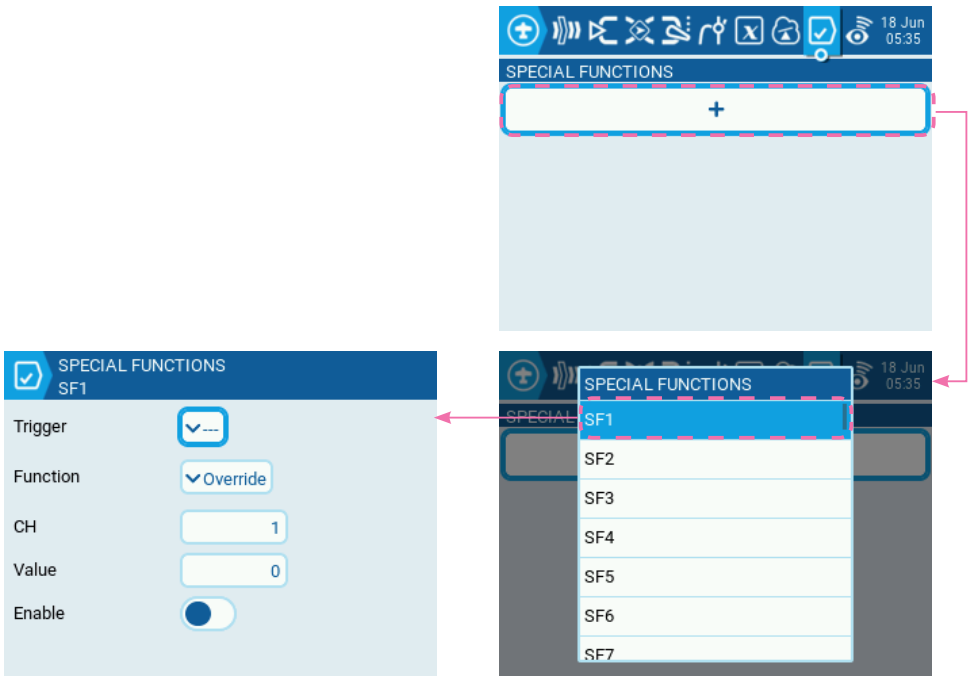
7.3.25 Special Functions

This system can implement some special functions, such as locking channel value, reset, volume, playing audio files, etc. More complex functions can be implemented after some lua scripts are used.

EdgeTX provides 64 kinds of special function setup for each model. The effective switch of special functions can also be a variety of input sources, similar to input setup, and mixer input setup. You may also choose a logic switch as the input setup.

Switch

It is the switch used to assign special functions to take effect.



微信公众号 Bilibili Website Facebook

Function

Used to trigger the corresponding functions after the special function switch is enabled and effective. Some of the functions are listed below:

"Override" Makes the specified channel value output a fixed value.

" Trainer" Enables trainer mode.

"Reset" Reset the specified timer or transfer-back parameter.

"Set" Sets the specified timer to the specified value.

"Adjust" Changes the value of the specified global variable.

"Volume" Used to adjust the volume of the transmitter.

"SetFailsafe" Sets the custom failsafe values for the selected module (Internal/External) to the current stick position when activated.

"Play Sound" Plays the specified tone file after triggering the special function switch to take effect.

"Play Track" Plays the specified track in SD card after triggering the special function switch to take effect.

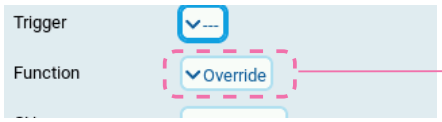
"Play Value" Plays the current output audio value of the specified channel.

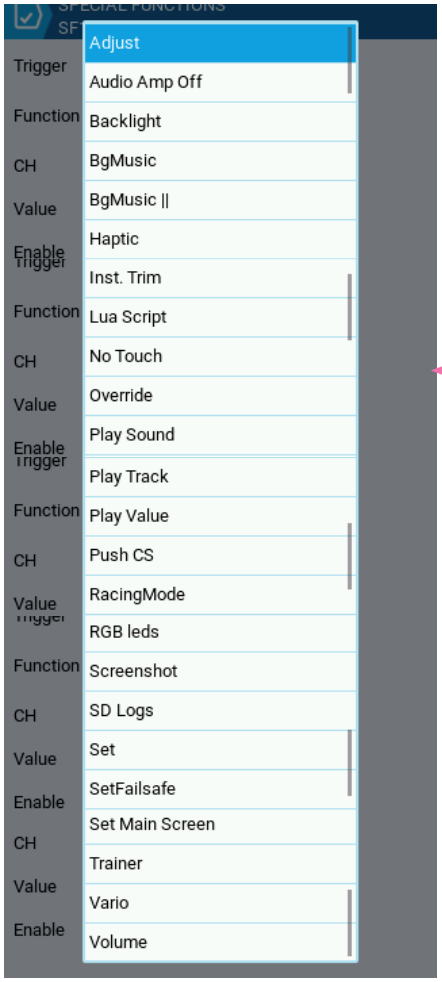
"Lua Script" Execute the Lua script defined in the 'value' field, which must be located in the "/SCRIPTS/FUNCTIONS/" directory on the SD card.

"Vario" Enables the variometer beeping sound for the ascent and descent of the model.

"SD Logs" Creates a log .csv file of the radio and telemetry values in the LOGS folder on the SD Card.

"Screenshot" Shoots the current screen and store it to SD card under the SCREENSHOTS folder.





Value

Select sthe input source, audio file, and lock output value for the value playback.

Cycle

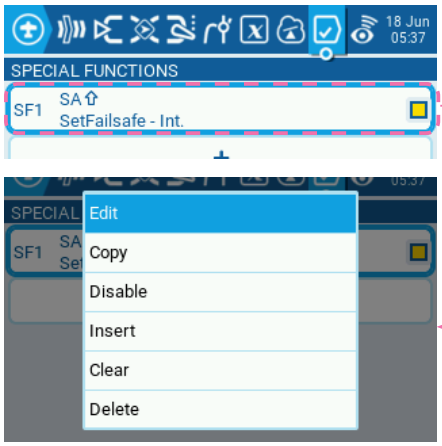
Sets the number of cycles."!1x--do not change"; "1x" -- trigger only once; "1s" -- trigger every 1s.

Reset

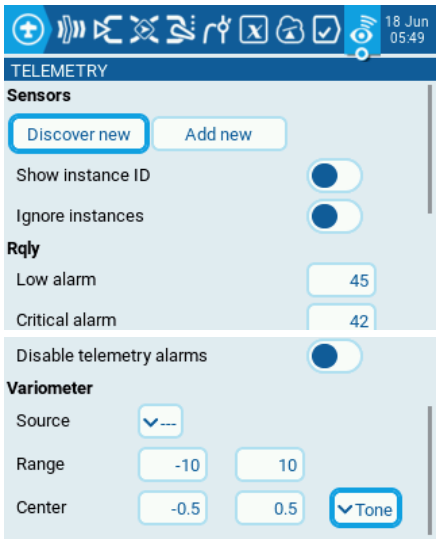
Selects the function to be reset, such as "Timer 1", "Timer 2", etc.

Enable

Sets the special function to take effect.



7.3.26 Telemetry



Sets the relevant functions for RSSI signal strength, sensor telemetry data and altitude change rate. The transfer-back function of the EdgeTX is very powerful. It can connect multiple sensors of the same type. This function can be used to edit the functions of the sensor. In EdgeTX, all telemetry values can be used as a single sensor data for editing related functions.

When the FRM303 RF module is connected, you can view the temperature, external power supply voltage and power information in the telemetry interface(Select "Discover new" to view the related infornation).

Displays the current sensor number, the name of the transfer-back telemetry data, detection digit, and ID.

The following is the sensor telemetry data in the connection to the Flysky receiver:

RSNR Receiver signal-to-noise ratio.

A1 Displays the current input operating voltage of the receiver.

RQly Receiver link quality (valid packets)

RSSI Displays the current signal strength value.

Tmp1 Displays the current temperature sensor value.

Alt Displays the current altitude value of the barometric pressure sensor.

Pres Displays the current barometric pressure value of the barometric pressure sensor.

微信公众号 Bilibili Website Facebook

Select the sensor value, then select "Edit" to edit the function of the sensor.

Name Name of the sensor - up to 4 characters.

Type Options are custom or calculated. Custom sensors are defined by the hardware. Calculated sensors are a sensors whose value is calculated using other sensors values.

ID This number indicates what type of sensore it is. It contains two parts. The first part is the ID number which defines the sensor type. The second part is the instance number for the hardware. If multiple sensors of the same type are configures, the instance numbers must be unique.

Unit Selects the appropriate calculation unit for the sensors. For example, V voltage, A current, m length or height, mAh battery capacity, etc.

Precison "0.--": Displays the accuracy to a single digit; "0.0-" shows the accuracy to one decimal point; "0.00" shows the accuracy to two decimal points.

Ratio Specifies the ratio value to multiply with the sensor value as needed by some sensors.

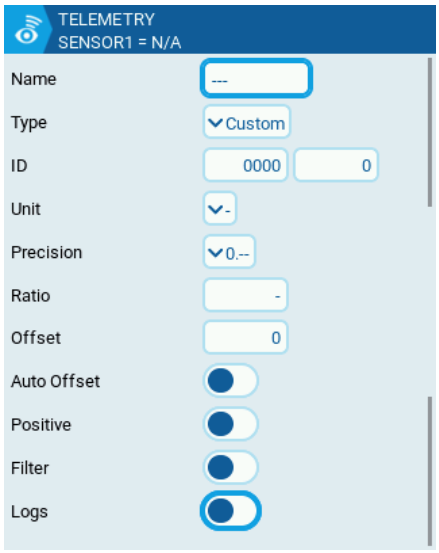
Offset Specifies the offset value to add to the sensor value.

Auto offset value When selected, the first received value is used as offset. You can use the Reset telemetry option to reset the offset on already configured sensors.

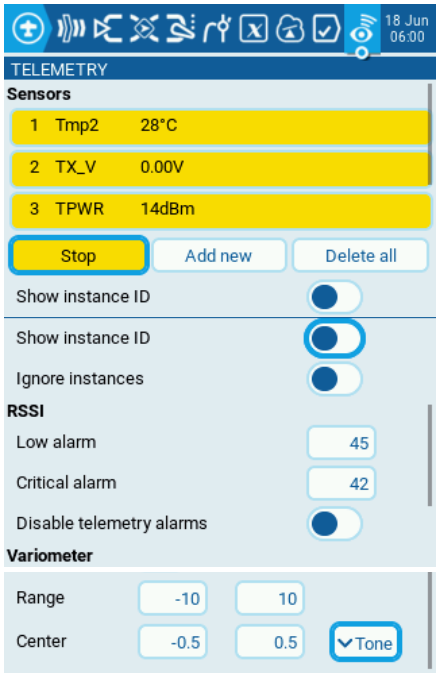
Positive Displays all detected negative values as positive values. Displays zero when the sensor value becomes a negative number.

Filter When selected, the sensor value becomes a rolling average of the last 5 received values.

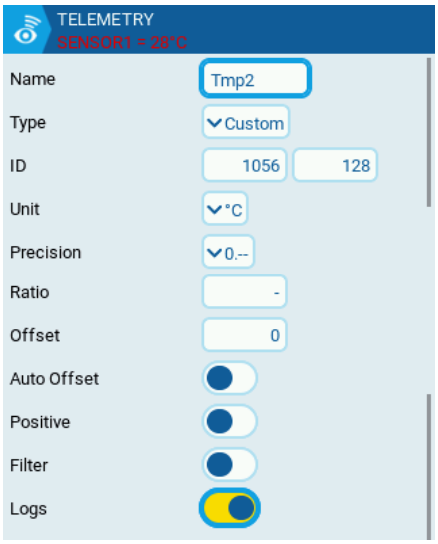
Logs Enable the record function. The detection data will be recorded on the SD card. The logging needs to be activated by the special function.



连接 FRM303 后界面



If type is set to Calculated,



Formula After selecting the type of "Operation", you can select different calculation formulas to calculate the detection values of multiple sensors.

Add Calculate the sum of the selected multiple detection values.

Average The average of the selected multiple detection values is calculated.

Minimum The minimum value of the selected multiple detection values is displayed.

Maximum The maximum value of the selected multiple detection values is displayed.

Multiply Multiplication of the selected multiple detection values.

Totalize Calculates the cumulative value of one sensor.

Consumpt Calculates the power consumption (mAh) by cumulatively add the values of current sensor.

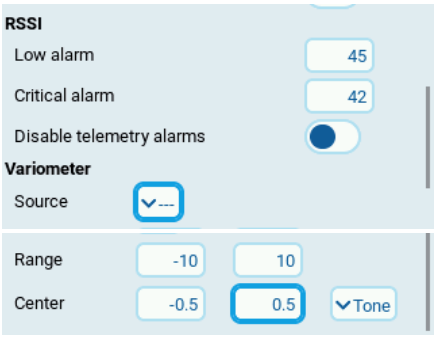
Distance Calculates the distance between the receiver and the transmitter using GPS sensor and altimeter values.

Input source 1 Select the transfer-back data for calculation and display.

Input source 2 Select the transfer-back data for calculation and display. Up to 4 input sources are allowed.

Persistent When selected the sensor values will be saved when switching between models or powering down the transmitter.

RSSI



"Low alarm" and "Critical alarm" Set the alarm threshold value of the signal strength.

"Disable telemetry alarm" Set whether to enable RSSI signal strength alarm.

Variometer

EdgeTX can alert the user of these altitude changes by providing a rising/lower pitched tone. Use the Variometer menu to set the actual frequency and volume of the tone to be played.

Source Sets the sensor to use as the variometer.

Range Sets the ascent/descent range that will trigger the change in the Variometer beeping pitch. If climb/descent rate is within the range specified here, beeping pitch will change according to that value.

Center Specifies the range for ignoring changes in climb/descent rates. When the climb/descent rate is within the range specified here, the beeping pitch will not change.

Tone/Silent Sets whether to beep when climb/descent rate is within the range set by Center.

微信公众号 Bilibili Website Facebook

7.4 Radio Settings

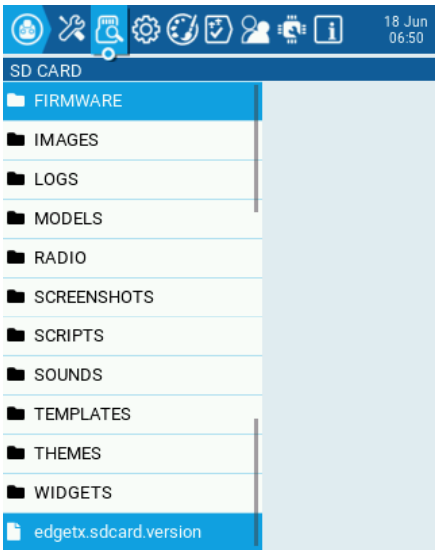
7.4.1 Tools

The Tools page in Radio Settings is where you can select Lua script-based tools for execution. Lua scripts that are located on the SD card in the Tools folder will be listed here.



7.4.2 SD Card

Used to browse SD card directory. It can perform the functions of viewing text, executing lua files, playing audio, copying, renaming and deleting SD card files.



FIRMWARE Transmitter firmware upgrade directory. The downloaded transmitter firmware is in this folder for upgrade.

IMAGES Model picture folder. Images are in the JPEG format in RGB color.

LOGS Log folder.

MODELS This folder contains model information.

RADIO This folder contains the radio.yml file. This file contains all the radio configuration data. If this file is corrupted or missing, the radio will go into Emergency Mode and will create a new radio.yml file with the default settings.

SCEENSHOTS The screenshot images will be saved under this path when the screenshot function is enabled.

SCRIPTS lua Lua script folder.

SOUNDS Voice folder. The recommended audio format: audio sampling rate 8 kHz, 16 kHZ or 32 kHZ wav audio with bit depth 8 or 16.

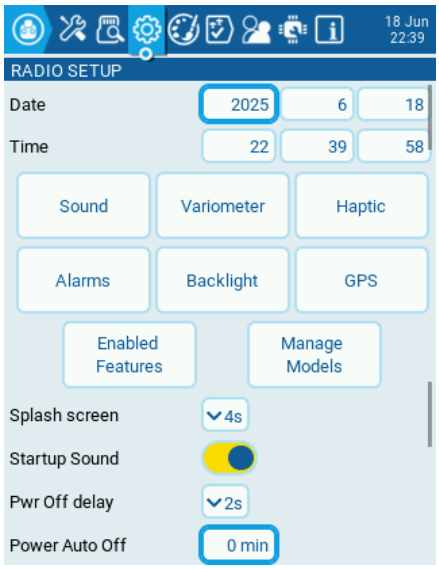
TEMPLATES This is where model template files are stored.

THEMES This folder contains the theme packs for EdgeTX. The EdgeTX SD card comes with several theme packs automatically installed.

WIDGETS This is where the widget files are stored.

7.4.3 Radio Setup

Note: Edge TX : 2.11.3-6 Version Update Features



Date & Time

Display the current system date and time. The values can be modified. By default, the date and time are displayed in the upper left corner of the main interface.

Sound

Set the trigger conditions, volume, tone, and broadcast duration of each sound in the system.

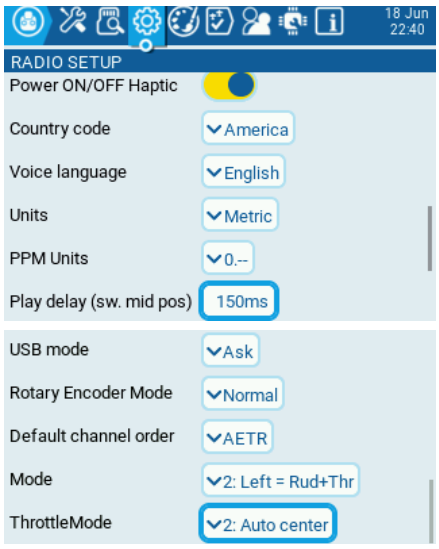
Mode Set the trigger conditions for the effective sound.

Quiet All sounds are turned off.

Alarm Only the warning tone is allowed.

Nokey Disable the touch tone only.

All Enable all prompt tones.



Volume Set the primary volume level change the prompt volume, voice volume and background voice volume at the same time.

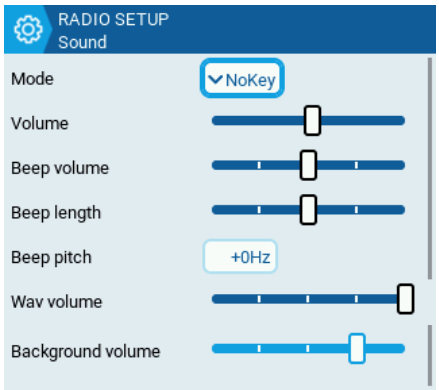
Beep volume Set the prompt voice volume.

Beep length Sets the prompt voice duration.

Beep pitch Sets the prompt pitch. The range is 0-300 Hz.

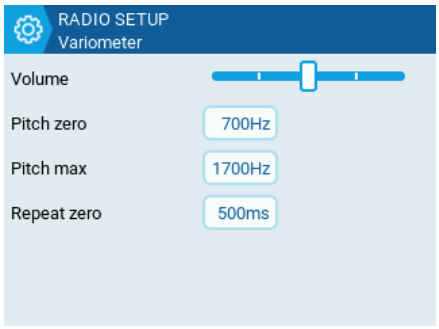
Wav volume Sets the voice volume.

Background volume Sets the background music volume.



微信公众号 Bilibili Website Facebook

Variometer



Uses the receiver with the transfer-back function, with GPS or barometer sensor information. Select "Discover new" on the "Model Settings" - "TELEMETRY" page. Select "Alt" or other altitude information as the input source for the altitude change rate. Calibrate the range to use the altitude change rate function correctly.

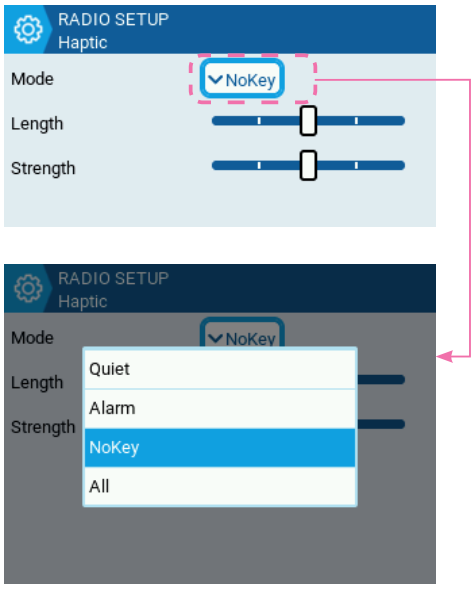
Volume Sets the volume level of the altitude change rate.

Pitch zero Range: 300 Hz-1100 Hz.

Pitch Max Range: 900 Hz-2500 Hz.

Repeat zero 200 ms-1000 ms

Haptic



Mode

--"Quiet" All vibrations are off.

--"Alarm" Only the warning vibration is available.

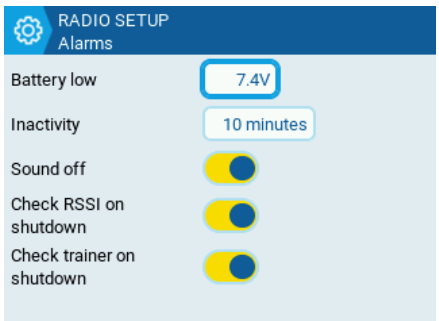
--"Nokey" Disables the touch vibration only.

--"All" Enables all vibrations.

Length Sets a single vibration feedback duration.

Strength Sets the vibration strength.

Alarm



Battery low Sets the low voltage alarm value. The range is 3.0v-12v.

Inactivity If the transmitter is not operated for a long period, the system gives an alarm. The range is 0-250 minutes. When it is set to 0, the alarm is disabled.

Sound off If the Sound off is selected, the alarm sound is disabled.

Check RSSI on shutdown Checks whether the transmitter is connected to an aircraft before the transmitter is powered off. The prerequisite is that the aircraft is enabled with the transfer-back function to correctly transfer back the RSSI.

Check trainer on shutdown Check for the presence of a trainer signal output or input when powering off the transmitter.

Backlight

To set the backlignt parameters of the screen.

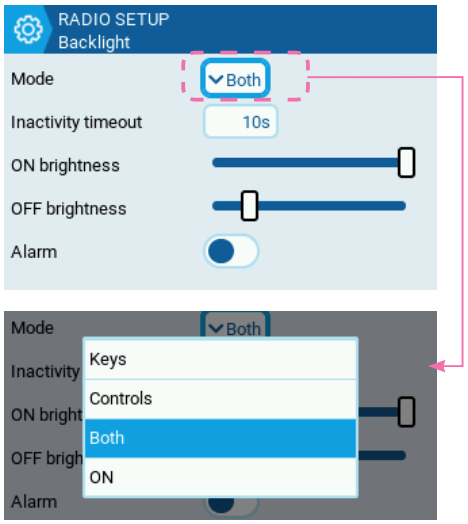
Mode Sets how to enable the backlight.

Inactivity timeout The length in seconds that the backlight is on.

ON brightness To disable the backlight.

OFF brightness To enable the backlight always.

Alarm The backlight turns on when there are alarms or warnings.



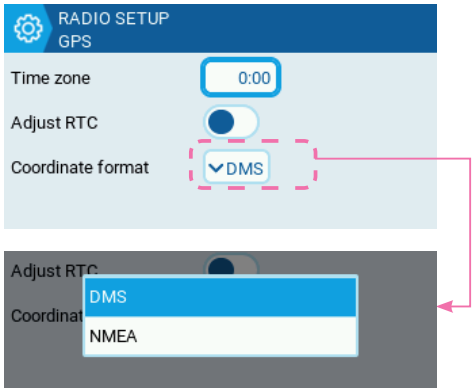
GPS

The GPS configuration settings are only for when a GPS has been installed on the transmitter, not the model's GPS.

Time zone Sets time zone accordingly.

Adjust RTC Adjusts the transmitter's real-time clock to match the time determined by the GPS.

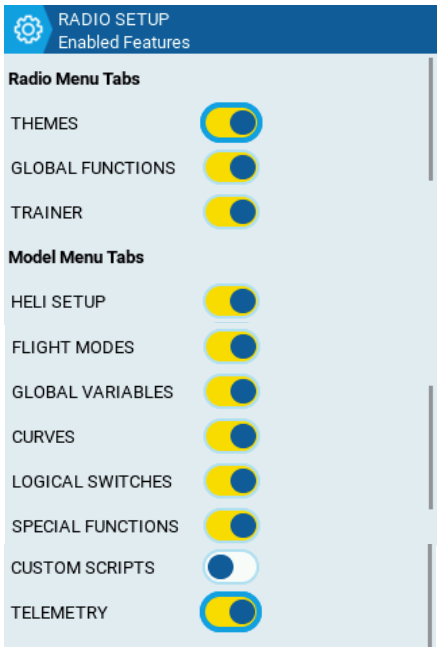
Coodinate formate The GPS coordinate format that will be displayed.



Enabled Features

Allows you to configure the Global settings for which tabs are visible in the Radio Setup and Model Settings area of EdgeTX.

Note: Turning off a tab only hides the tab and does not change the items already configured in that tab.



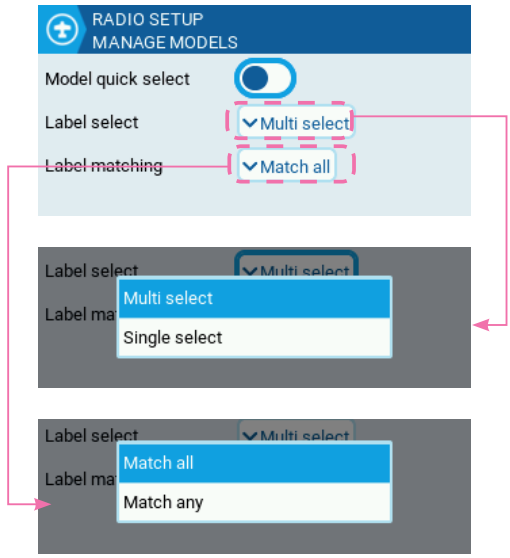
微信公众号 Bilibili Website Facebook

Manage Models

Configure the operation mode for the [Manage Models] function.

Model quick select When set to OFF, switching models under the "Manage Models" function requires selecting "Select Model" from the menu for activation; conversely, selecting a model in the "Manage Models" interface immediately switches models when set to ON.

Label select Configure the "Manage Models" function by selecting one or more filter criteria in the left label bar; the system will automatically hide models that do not match the selected labels. It can be set as "Multi select" or "Single select". When set to Multiple Choice, matching conditions can be configured.



Splash screen

Set the duration for the startup screen display.

Startup Sound

Enable/Disable the startup sound effect.

Power Off Delay

Sets the time to wait for the transmitter to shut down after pressing and holding the shutdown button. The range is 0-4s.

Power Auto Off

When enabled, the device will automatically power off if the following conditions persist for the set duration: no RF signal transmission and inactive trainer mode.

Power ON/OFF Haptic

When enabled, the device will trigger vibration alerts in the following two scenarios:

- Power-on: Vibration indicates the power button can be released.
- Power-off: Vibration signals the transmitter is shutting down.

Country Code

It needs to match the geographic location to keep the radio frequency transmission parameters to comply with the regulatory requirements.

Voice Language

Selects the language for broadcasting. This list contains all supported languages. Ensure that the voice package for the language is loaded onto the SD card located in a subfolder of the SOUNDS directory.

Units

Selects the metric or imperial unit for the telemetry value.

PPM Uints

Set the display precision for PPM values.

Play Delay(SW. mid pos)

Delayed broadcast range: -150 ms-850 ms.

USB Mode

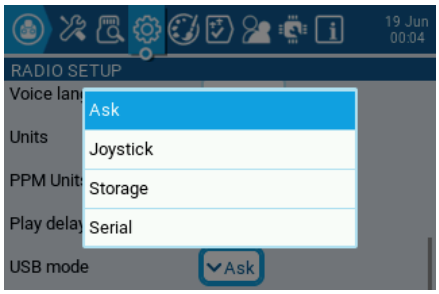
Sets the computer-transmitter recognition mode after the connection via the USB cable.

--"Ask" After the connection via USB, a pop-up window is displayed, indicating to select the joystick, U disk or serial mode.

--"Joystick" Connects to the simulator.

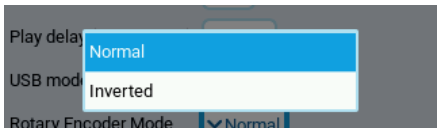
--"Storage" Reads and modifies files in the SD card.

--"Serial" Used to upgrade the firmware.



Rotary Encoder Mode

Set the scroll wheel mode: Normal or Reverse.



Default Channel Order

Sets the stick mode of each model. If it is selected as AETR, the channel is CH1-Ail, CH2-Ele, CH3-Thr, or CH4-Rud.

Mode

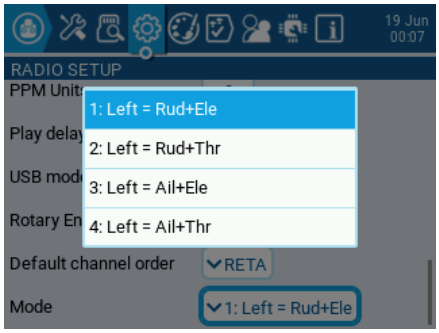
Sets the channel assignment of the left and right sticks.

1：left=Rud+Ele It indicates that the left stick corresponds to yaw and pitch, respectively.

2：left=Rud+Thr It indicates that the left stick corresponds to yaw and throttle, respectively.

3：left=Ail+Ele It indicates the left stick corresponds to aileron roll and pitch, respectively.

4: left=Ail+Thr It indicates the left stick corresponds to aileron roll and throttle, respectively.



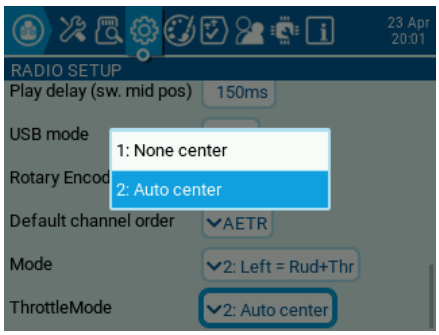
Note: When setting the throttle controlled by the right gimbal or the throttle controlled by the left gimbal, you should first set the "Default Channel Mode" and "Mode".It is not recommended to change through "Input" and "Mixes", as the setting process is more complicated.

For example, if you want to set the throttle controlled by the left gimbal, you should choose the default "Default Channel Mode" as "AETR", "Mode" as "2: left=Rud+Thr", and "Mode" as "2: left=Rud+Thr". is "2: left=Rud+Thr".

Throttle Mode

Sets the throttle stick self-centering:

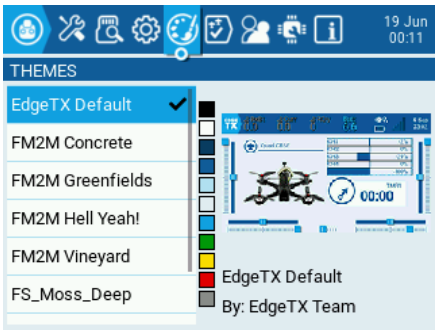
- Set to "Auto Center", the dead zone setting takes effect on the throttle channel. The transmitter will not check the throttle channel status at power-on.
- Set to "None Center" , the throttle channel has no dead zone. The transmitter will check the throttle channel status at power-on (enabling the "Pre-start Checks >Throttle State" function).



微信公众号 Bilibili Website Facebook

7.4.4 Themes

Allows you to apply different colored themes to your transmitter.



7.4.5 Global Functions

Global Functions GF1 to GF64 are allowed to customize functions that apply to all models, such as specific switches, potentiometers, sliders, etc. Thus, it is unnecessary to set the same function on "Model Settings" in the "SPECIAL FUNCTIONS" page. For specific setup, see the "SPECIAL FUNCTIONS".



7.4.6 Trainer

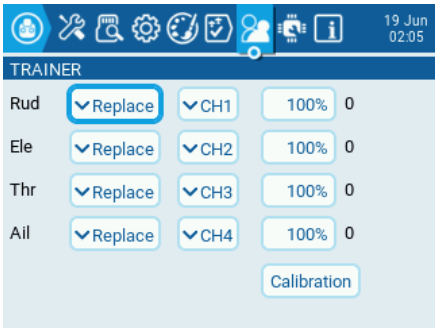
It is used to select the trainer control or student control for the output computation mode after the trainer mode is enabled. Assign the trainer control output channel corresponding to the student control input channel, to calibrate the stick deviation of both.

There are three options for the output computation mode: "OFF", "=(Replace)" and "+(Add)".

"=(Replace)" The student control command completely replaces the trainer control command in this channel.

"+(Add)" The trainer control command and the student control commands are added together in this channel.

"OFF" The channel will not be controlled by student control.

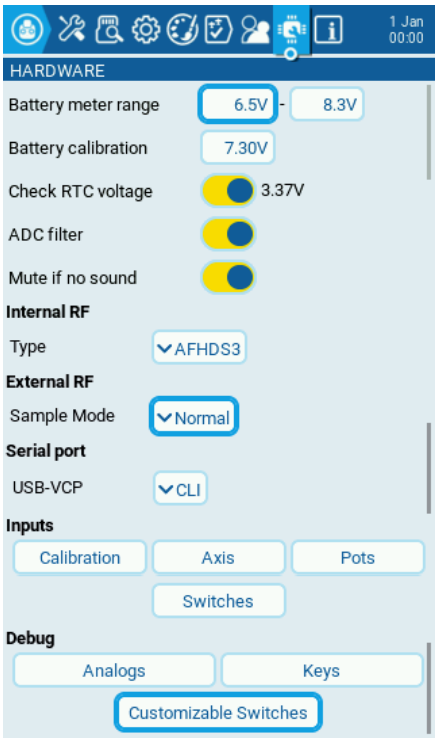


Calibration

It is used to calibrate the stick deviation between the trainer control and the student control. By using this function, you can put all the sticks of the trainer control and the student control in the middle position. To complete this, select the "Calibrate" button.

1. The process of using the trainer function:"Model Settings" -- "Trainer" -- "Mode": Select the appropriate connection mode for trainer control and student control.
2. "Model Settings" -- "Special function": Enable the trainer function and assign the trainer function switch.
3. "Radio Settings"--"Trainer": Set the relevant input and output parameters for trainer control and student control.

7.4.7 Hardware



Voltage

Battery meter range Set the voltage test range of the transmitter battery. This transmitter operates using a 2S 18650 battery, therefore the recommended voltage range should be set to 6.0V~8.4V.

Battery calibration Calibrate the voltage detection value of the transmitter battery. The detection value of the transmitter may differ from the battery voltage. You can calibrate it by using this function.

Note: Calibrate according to the voltage readings from a multimeter.

Check RTC voltage Set whether to enable the detection of time driven battery voltage.

ADC fliter Enables or disables the ADC Filter. This filter can also be enabled/disabled per model in the model settings.

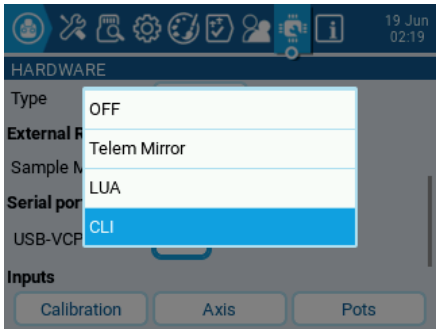
Mute if no sound When enabled, puts the transmitter in mute mode until a sound needs to be played. This prevents interference noise from high-powered TX modules from coming out of the transmitter speakers.

Internal RF

Type Displays the internal RF type.

External RF

The default setting of Normal should be used by most users.



Serial Port

Displays a list of available auxiliary serial ports that can be configured and used.

Telem Mirror The same telemetry data that goes to the external module bay is sent to the serial port.

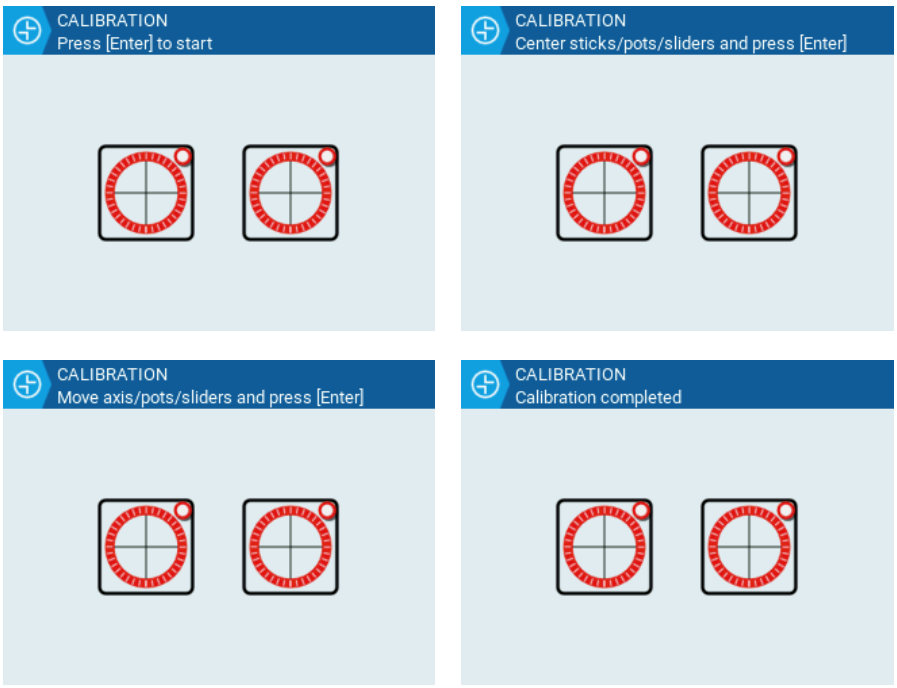
LUA Send/receive data to/from Lua script.

CLI Send commands to the radio via the command line.

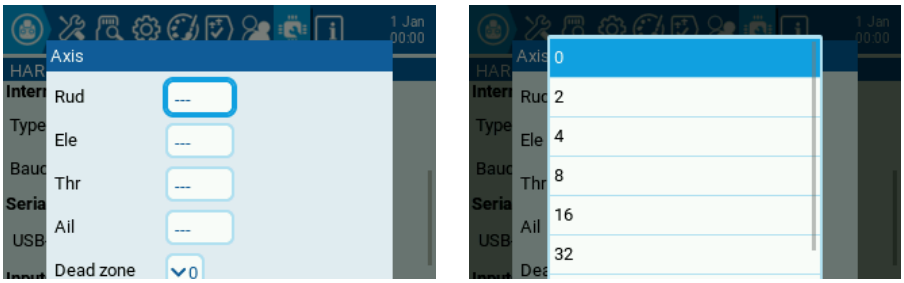
微信公众号 Bilibili Website Facebook

Inputs

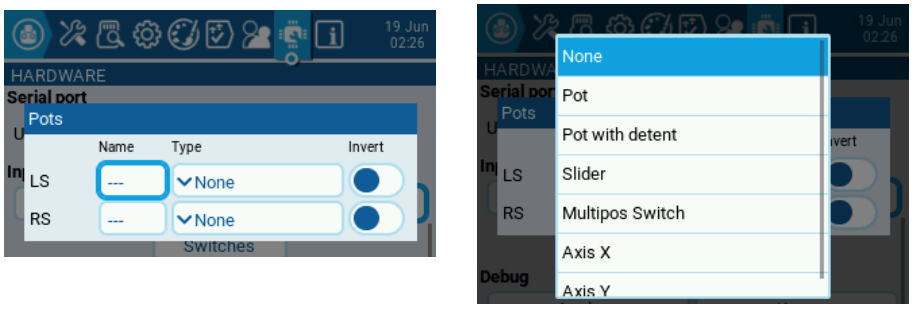
Calibration You can calibrate the travel of the stick and knob. After the transmitter is used for a period of time, if there is drift or deviation of the stick or knob, you can calibrate it by this function.Refer to the pictures below for the steps of calibration.



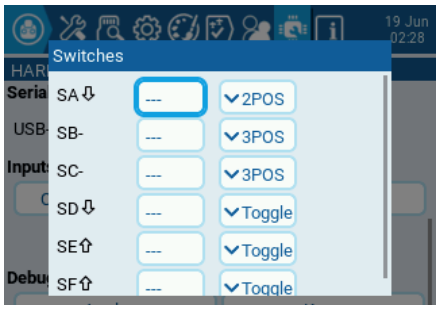
Axis To modify the default name of CH1-CH4 and set the dead zone.



Pots To modify the default names of LR and RS, as well as the knob setup type.



Switches To modify the default name of each switch, and set the switch types such as two-way switch, three-way switch, and kickback switch.

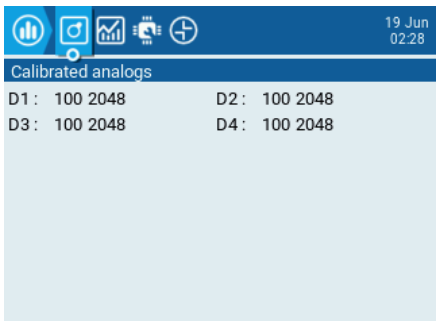


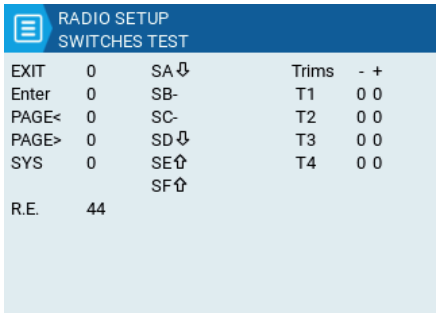
Debug

Allows for testing and debugging of the analog controls and keys.

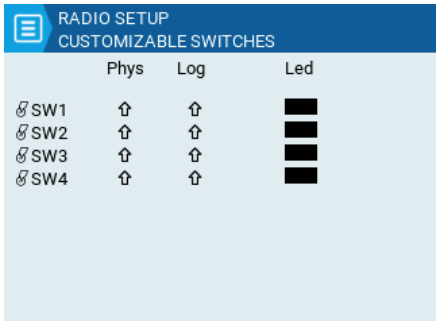
Debug Analogs Shows you the data for your analog control and the screen on your transmitter.

Debug Keys Shows you the digital data for your keys or switches, ect.





Customizable Switches Display the configuration details of customizable switches.

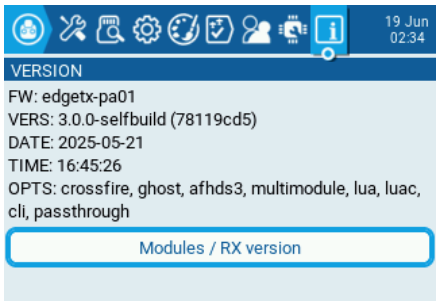


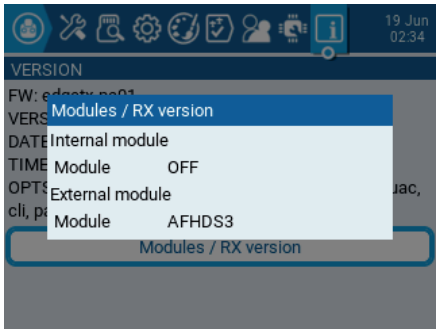
微信公众号 Bilibili Website Facebook

7.4.8 Version

Displays information about the current firmware version, including firmware number, firmware update time, supported wireless protocols, etc.

Modules / RX Version It gives you information about the activated RX modules for the currently selected model.





7.5 Screen Settings

With this function, you can select the display and layout parameters of the "Top bar" widget and "Main view" widget in the main interface.

In the Main View menu, select the layout box to select different layouts, such as "1+2", "1+3", "2+2", "2x2", etc. Users can choose different layouts according to their needs.

In the Main View of EdgeTX, you can set multiple interfaces as required by selecting "+" in the top box. After the setting is completed, you can switch to other interfaces by sliding the screen in the main interface. You can set up to ten sub-interfaces.

Select the "Set" widget to enter the setup interface. Select on a layout box. A pop-up window is displayed, indicating to select a widget. That is, you can select different widgets, such as "Models current model", "Outputs current output of each channel", "Timer", etc.

Top Bar

Selects whether to enable the top bar display in the main interface.

Flight Mode

Selects whether the flight mode can be displayed.

Slider

Selects whether the main interface displays the position of the slider S1/S2.

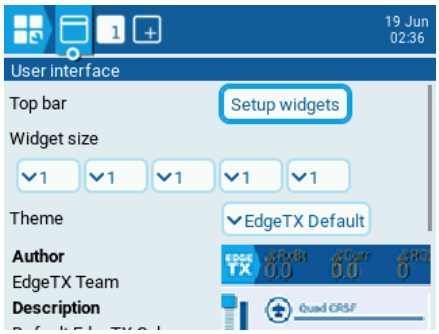
Note: The setting will only be displayed after configuring the "Type" under [Radio Settings] > [Hardware] > [Pots].

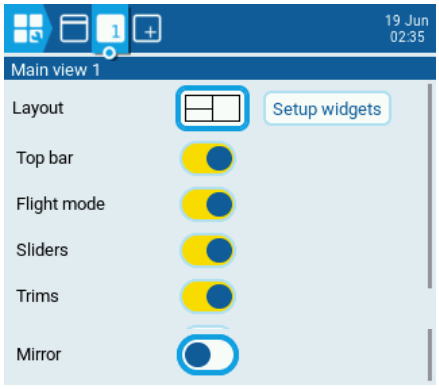
Trim

Selects whether the main interface displays the trim position.

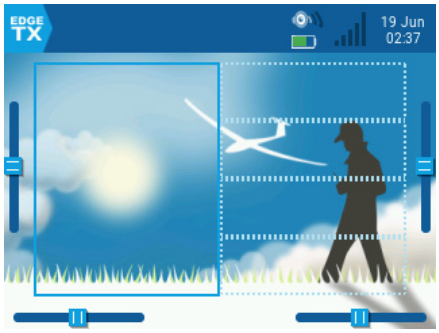
Mirroring

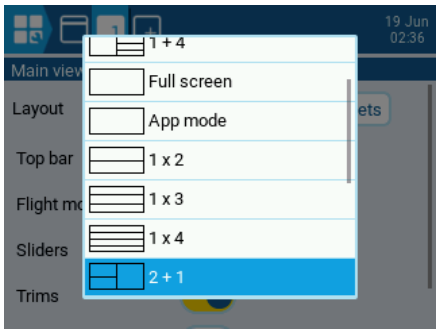
Toggles whether to mirror the selected widget layout.







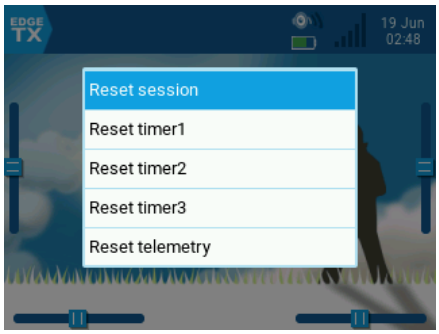




微信公众号 Bilibili Website Facebook

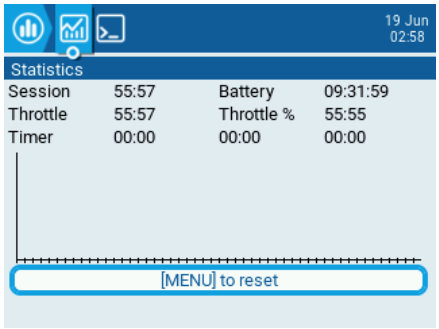
7.6 Reset Telemetry

You can reset the flight record, timer and transfer-back parameters. For the reset function, you can assign a switch as a reset switch by choosing "Model Settings"--"Special Function" or "Global Function".



7.7 Statistics

Presents you with statistics regarding radio usage. Except for Battery, all data is reset once the radio is powered off.



Session

The amount of time that the transmitter has been turned on.

Battery

The amount of time that the radio has been on since the last reset.

Throttle

The amount of time that the throttle has was above the 0% stick position.

Throttle %

The amount of time that the throttle has was above the 50% stick position.

Timer

The current values of Timer 1, Timer 2, Timer 3.

7.8 About EdgeTX

To view EdgeTX related information by scaning the code in this function.

8. Product Specifications

This section describes the specifications for PA01.

8.1 Transmitter Specifications

Product Model	PA01
Compatible Receiver	FTr16s, TMr and other AFHDS 3 protocol receivers ; PR01, PR02 and other ELRS receivers
Compatible RC Model	Educational, Competition, and Recreational Models
Number of Channels	18-channel is for Internal RF, and 32-channel is for External RF
RF	2.4GHz ISM
Maximum Power	< 20dBm (e.i.r.p.) （EU）
RF Protocol	AFHDS 3/ELRS
Resolution	4096
Data Connector	USB Type-C, 3.5mm Audio Jack, 3.5mm Trainer Connector(DSC)
Low Voltage Alarm	<7.2V
Antenna	Single antenna (external folding antenna)
Input Power	7.4V 2*18650 LiPo
Display	2.4 inch 240*320 full dot color non-touch IPS screen
Firmware Update	Supported
Temperature Range	-10°C ~ +60°C
Humidity Range	20% ~ 95%
Color	Black translucent
Dimensions	174.9*131.4*67.5mm
Weight	Transmitter weight: 343g (without battery); Charging base weight: 67g (without battery)
Charging Jack	Yes (USB Type-C Port)
Certifications	CE, FCC ID: 2A2UNPA0100

微信公众号 Bilibili Website Facebook

9. Package Contents

The accessories included are different in different versions, please consult your dealer for details.

10. Certifications

10.1 DoC

Hereby, [ShenZhen FLYSKY Technology Co., Ltd.] declares that the Radio Equipment [PA01] is in compliance with RED 2014/53/EU.
The full text of the EU DoC is available at the following internet address: www.flyskytech.com/info_detail/10.html

10.2 CE Warning

The ce warns that the installation of the antenna used in this transmitter must be kept in distance from all the personnel and shall not be used or used with any other transmitter. The end user and the installer must provide antenna installation instructions and transmitter operating conditions to meet the requirements for rf exposure compliance.

10.3 FCC Statement

This equipment has been tested and found to comply with the limits for a Class B digital device pursuant to part 15 of the FCC rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or televison reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:
-- Reorient or relocate the receiving antenna.
-- Increase the separation between the equipment and receiver.
-- Connect the equipment into an outlet on a circuit different from that to which the receiver is connected.
-- Consult the dealer or an experienced radio/TV technician for help.

To assure continued compliance, any changes or modifications not expressly approved by the party responsible for compliance could void the user's authority to operate this equipment. (Example use only shielded interface cables when connecting to computer or peripheral devices).

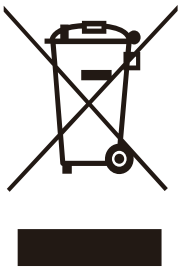
This equipment complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions:
(1) This device may not cause harmful interference.
(2) This device must accept any interference received, including interference that may cause undesired operation.

Caution!
The manufacturer is not responsible for any radio or TV interference caused by unauthorized modifications to this equipment. Such modifications could void the user authority to operate the equipment.
1. Move all your channels to the desired position.
2. Select [All channels] and then [Yes] in the confirmation box.

微信公众号 Bilibili Website Facebook

10.4 Environmentally friendly disposal

Old electrical appliances must not be disposed of together with the residual waste, but have to be disposed of separately. The disposal at the communal collecting point via private persons is for free. The owner of old appliances is responsible to bring the appliances to these collecting points or to similar collection points. With this little personal effort, you contribute to recycle valuable raw materials and the treatment of toxic substances.



CAUTION
RISK OF EXPLOSION IF BATTERY IS REPLACED BY AN INCORRECT TYPE.
DISPOSE OF USED BATTERIES ACCORDING TO THE INSTRUCTIONS

CAUTION
- replacement of a battery with an incorrect type that can defeat a safeguard (for example, in the case of some lithium battery types);
- disposal of a battery into fire or a hot oven, or mechanically crushing or cutting of a battery, that can result in an explosion;
- leaving a battery in an extremely high temperature surrounding environment that can result in an explosion or the leakage of flammable liquid or gas; and
- a battery subjected to extremely low air pressure that may result in an explosion or the leakage of flammable liquid or gas.

Figures and illustrations in this manual are provided for reference only and may differ from actual product appearance. Product design and specificatiions may be changed without notice.

www.flysky-cn.com

Copyright ©2025-2026 Flysky Technology Co., Ltd.

Release Date:2026-04-27

微信公众号 Bilibili Website Facebook

FCC ID: 2A2UNPA0100

Manufacturer: ShenZhen FLYSKY Technology Co., Ltd.
Address:16F, Huafeng Building, No. 6006 Shennan Road, Futian District, Shenzhen, Guangdong, China